CELESTIAL CROSSING EVIDENCE IN THE D101 OCXO SIGNAL

A Foundational Educational and Archival Reference on Repeated Crossing States, Direct, Antipodal, and Orthogonal Celestial Geometry, Earth-Moon Barycentric Motion, Palindromic Variance Envelopes, Chirp Trajectories, CWT and FFT Derivatives, Two-Dimensional Recurrence, and the Volume-Filling Cell3 Plane System

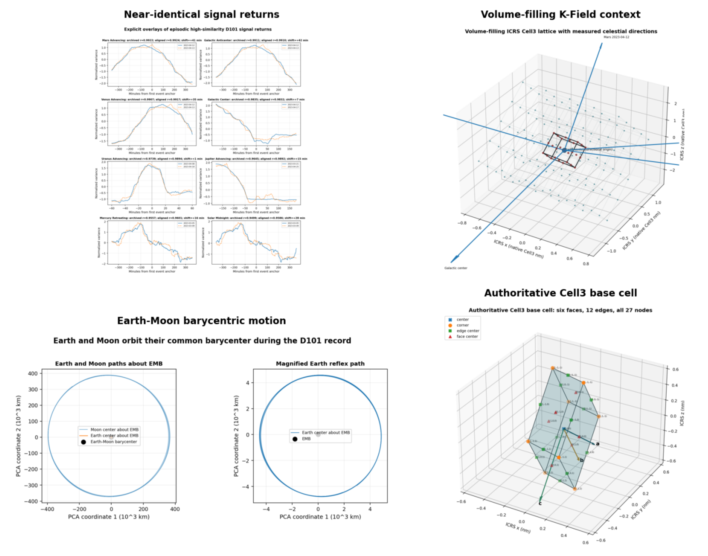

Figure 1. Data-derived title composite. Upper left: the 78.269-day D101 archive. Upper right: the volume-filling ICRS Cell3 context. Lower left: the actual Earth-Moon barycentric path. Lower right: the authoritative 27-node Cell3 base cell.

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Archival publication epoch: 1786043894 | Generated UTC: 2026-08-06T19:18:14Z
Machine-readable companion: K-Field_Celestial_Crossing_Evidence_in_the_D101_OCXO_Signal_1786043894.json
Rendered with Python ReportLab using Helvetica, Helvetica-Bold, Times-Roman, and Courier only.

Proprietary Notice and Document Contract

This paper is a standalone educational and archival engineering reference. It reconstructs the complete explicit crossing evidence in the D101 OCXO pulse-width archive, including direct transits, antipodal transits, and the two 90-degree orthogonal Earth-rotation positions. Every numerical figure is generated from the actual D101 analysis ledgers, the original D101 archive, or the authoritative machine-readable Cell3 geometry. No illustrative synthetic signal is used as experimental evidence.

Central result: the D101 signal repeatedly enters distinct variance, chirp, spectral-polarity, harmonic-drift, and multi-event recurrence states at defined celestial crossing phases. The states differ between direct, antipodal, and orthogonal positions. Their detailed waveform changes from event to event, but their event-centered signal classes recur across the archive.

Table 1. Document contract and scope

Document item	Archival specification
Source archive	d101_b.zip; 6,762,475 one-second observations
Primary measured variable	pulse_w, the OCXO differential pulse-width time series
Crossing classes	Solar, lunar, CMB, Galactic, seven planets, antipodes, and two 90-degree orthogonal positions
Signal methods	Moving mean, moving standard deviation, palindrome correlation, CWT, CWT derivatives, FFT, FFT derivatives, 2-D FFT, chirp ridge/notch analysis
Coordinate correction	Exact flattening of the laboratory latitude cone and target-relative hour-angle registration
Spatial reference	Authoritative 27-node Cell3 direct and reciprocal plane geometry
Earth-Moon datum	Earth-Moon barycenter base trajectory plus lunar-opposed Earth reflex and geodetic laboratory rotation
Machine-readable authority	K-Field_Celestial_Crossing_Evidence_in_the_D101_OCXO_Signal_1786043894.json

Abstract

The D101 archive contains 6,762,475 nominal one-second pulse-width observations collected from March 2 through May 22, 2023. The investigation tested whether signal structure is organized around celestial crossing geometry rather than around UTC alone. The tested geometry includes solar upper and lower culmination, lunar upper and lower culmination, lunar horizon crossings, the advancing and retreating directions of the Earth-CMB vector, the Galactic-center axis and its antipode, upper and antipodal transits of Mercury through Neptune, and the two Earth-rotation positions located 90 degrees from every crossing.

The signal was examined through moving-standard-deviation envelopes, event-centered mirror symmetry, recurrent waveform templates, raw and angularly normalized chirp detection, carrier-harmonic drift, continuous-wavelet-transform derivatives, FFT derivatives, and two-dimensional FFTs across event number and crossing phase. The analysis was repeated after analytically removing the laboratory-latitude cone traced by the local upward vector.

The results establish five connected signal layers. Broad palindromic variance envelopes occur around selected crossings, especially CMB retreating, Galactic anticenter, Moonset, Mercury retreating, Saturn transits, Uranus retreating, solar transit, solar midnight, and lunar lower culmination. Chirp events occur in both fast 2-512-second and long 2-minute-4-hour bands. The strongest controlled angularly normalized chirps occur at Uranus advancing and Mercury advancing. Opposite 90-degree quadratures exhibit a systematic ridge/notch inversion at long periods: all 24 tested parent crossing classes have negative direct correlation between their two orthogonal spectral templates. Two-dimensional FFT results show alternating, episodic, or multi-event recurrence with characteristic intervals of approximately two to eight crossings rather than one invariant daily waveform. Latitude unwinding leaves fixed celestial axes unchanged in scalar coherence, as analytically required, but materially sharpens lunar horizon-crossing structure and converts temporal chirps into physically interpretable angular-wavelength trajectories.

The combined evidence defines a crossing-conditioned signal architecture containing palindromic envelopes, chirp trajectories, spectral polarity reversals, harmonic carrier modulation, and recurrence over multiple celestial crossings. The strongest repeated profiles return with archived correlations above 0.99 after separations from one day to several weeks. Their explicit ICRS positions frequently approach projected Cell3 node rays, while their episodic selection is governed by recursive translational phase.

The Earth-Moon barycenter is the smooth translational datum for this phase. The Earth center moves opposite the lunar vector by approximately 4,407-4,932 km during the archive, and the laboratory adds a daily geodetic displacement of approximately 6,369 km. The observed crossing phase is therefore the sum of EMB lattice phase, Earth-reflex phase, and laboratory-rotation phase. This decomposition explains why lunar x/y/z covariance can encode the detector's own position around the barycenter and why angularly identical transits do not reproduce uniformly every day.

The archive therefore provides a foundational signal and geometric reference for experiments that map Earth rotation, celestial direction, Earth-Moon barycentric motion, and rectilinear K-Field plane structure into oscillator-frequency observables.

Keywords: D101; OCXO; K-Field; celestial crossing; transit; antipodal transit; orthogonal transit; palindrome; chirp; continuous wavelet transform; two-dimensional FFT; latitude-cone unwinding; CMB; Galactic center; Cell3.

Reader Guide

Chapter 1 begins with the explicit repeated signals so that observation precedes interpretation. Chapter 2 establishes the measurement record. Chapters 3 through 5 construct the authoritative Cell3 base cell, the volume-filling ICRS lattice, and the celestial-vector crossing taxonomy. Chapters 6 and 7 establish the Earth-Moon barycentric datum and analytically unwind the laboratory latitude cone. Chapter 8 identifies the explicit ICRS anchor geometry. Chapters 9 and 10 define the signal operators, controls, and the absence of one invariant daily waveform. Chapters 11 through 14 explain palindromes, chirps, CWT derivatives, FFT derivatives, and two-dimensional recurrence. Chapters 15 through 18 examine solar, lunar, CMB, Galactic, planetary, and orthogonal crossings. Chapters 19 through 21 derive the integrated symmetry model, state falsifiable predictions, and conclude. The appendices preserve the complete numerical ledgers, source hashes, figure inventory, and machine-readable record.

Throughout the paper, empirical observations, derived quantities, geometric implications, and mechanism-level interpretations are kept distinct. A measured event-centered correlation is reported as an observation. A wavelength conversion is reported as a derivation. A connection to a rectilinear spatial structure is reported as a geometric interpretation. This separation lets a student reproduce each layer independently.

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PART I. BEGIN WITH THE OBSERVED SIGNAL

1. The Repeated Signal Events

1.1 Observation before interpretation

The primary empirical result is that D101 signal profiles recorded on different dates can return to nearly the same structured state. The recurrence is visible in broad moving-standard-deviation envelopes and includes internal peaks, valleys, shoulders, and inflection points. It is not merely a return of average amplitude. The matched events are separated by intervals ranging from approximately one day to more than two months.

Each pair in the opening atlas is generated directly from the one-minute D101 feature matrix. The archived event correlation is reproduced in the panel title. A common lag of at most 60 minutes is permitted because the prior event search showed that the strongest state can occur near, rather than exactly at, the nominal meridian anchor. The displayed alignment shift is recorded explicitly and is not hidden in the plot.

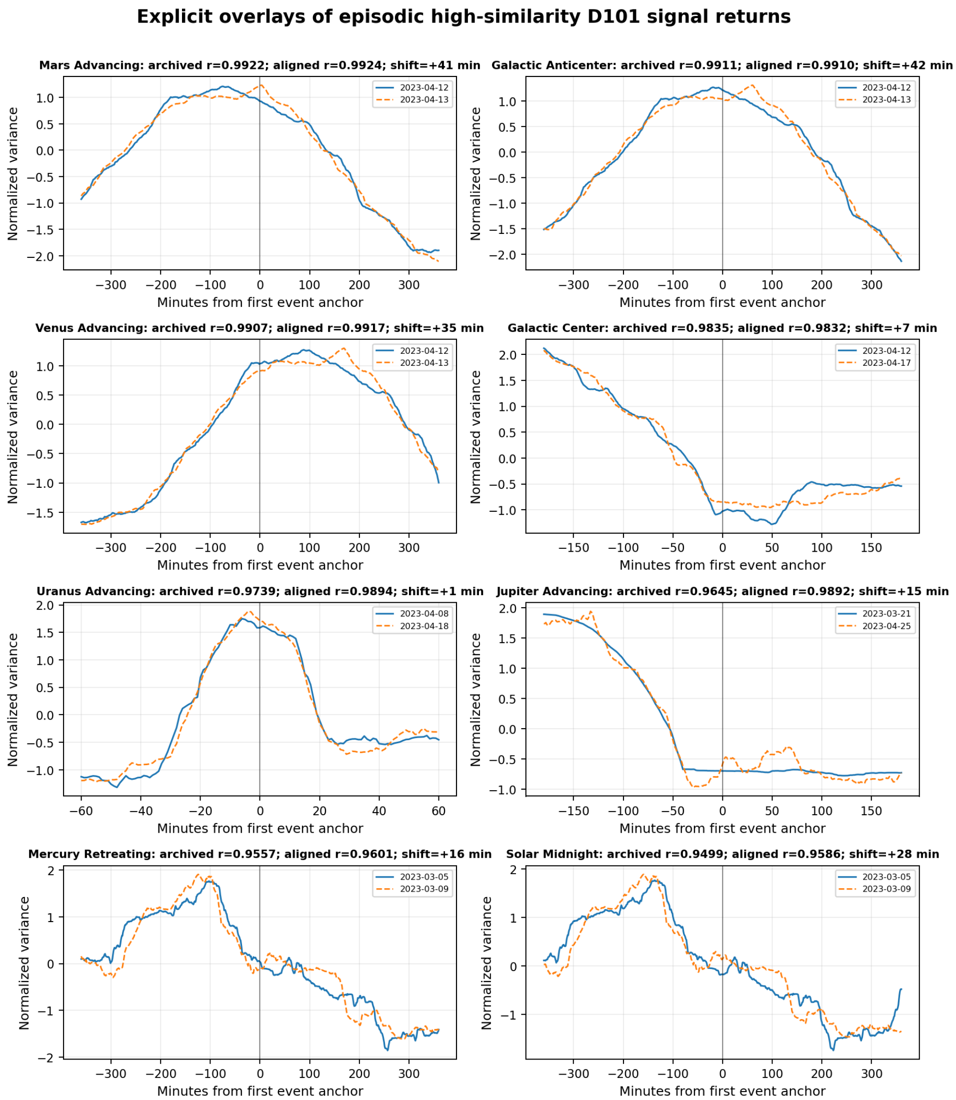

Figure 2. Explicit overlays of eight episodic D101 repeat states.

Reading guide. Read each panel as two separately recorded variance profiles. The solid and dashed lines were acquired on the dates printed in the legend. The correlations remain high over broad multi-hour structures even when the events are separated by weeks.

1.2 Residual structure

A profile-pair correlation near one can be produced only when the ordered shape is reproduced. The residual panels therefore provide an essential second view. They show where the two traces differ after normalization and alignment. The residuals are small relative to the common structure, but they are not zero; the repeated state is a highly similar physical configuration rather than a bit-for-bit duplicated data block.

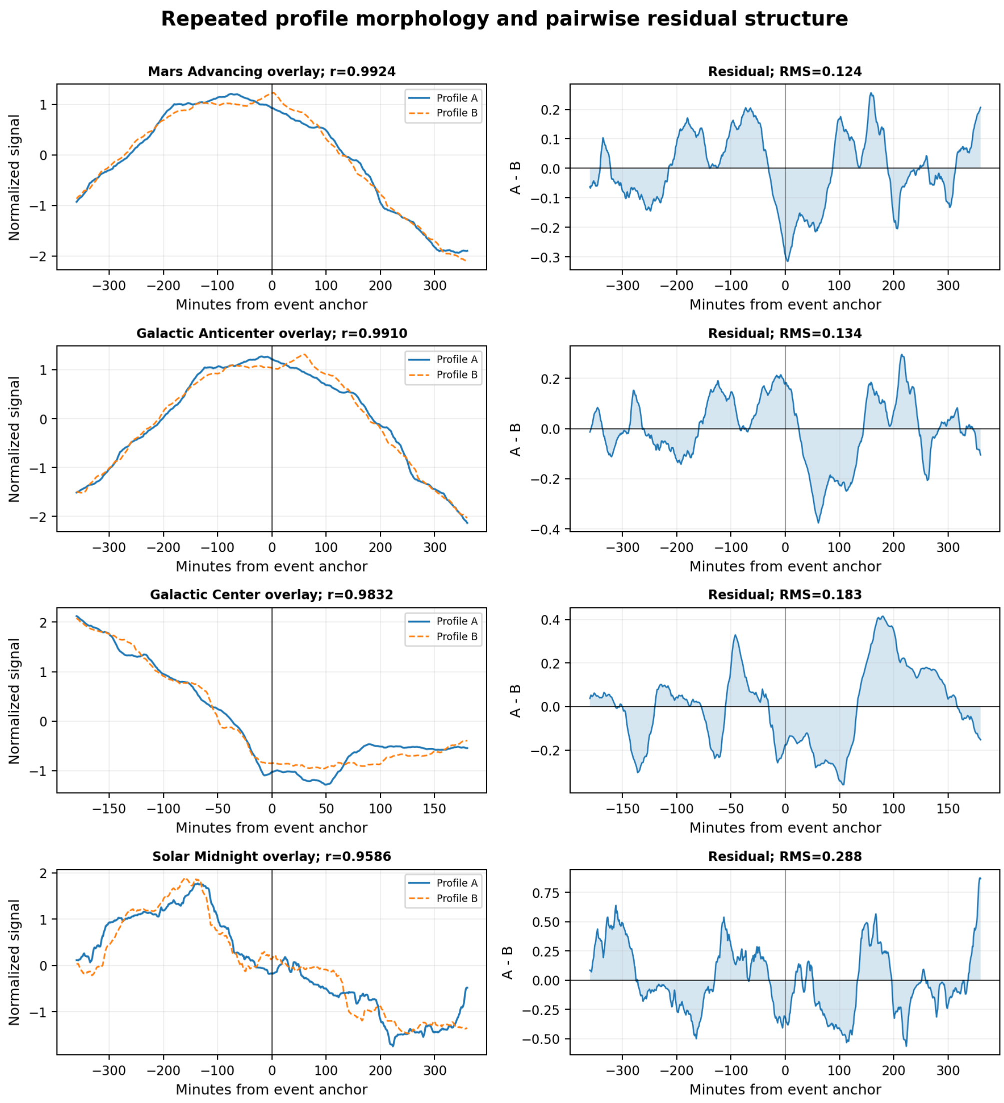

Figure 3. Four repeated profiles and their pairwise residuals.

Reading guide. The left column shows the matching signals. The right column isolates the remaining difference. A low residual across both broad and local features demonstrates recurrence of morphology rather than simple baseline agreement.

1.3 Returns over days, weeks, and months

The complete high-similarity ledger contains 205 profile pairs with archived correlation at or above 0.90, representing 253 unique anchors and 17 event classes. High-similarity pairs occur at adjacent-day separation, at intervals of several days, and at intervals approaching the full archive length. There is no single preferred separation that reduces the phenomenon to an ordinary daily instrumental cycle.

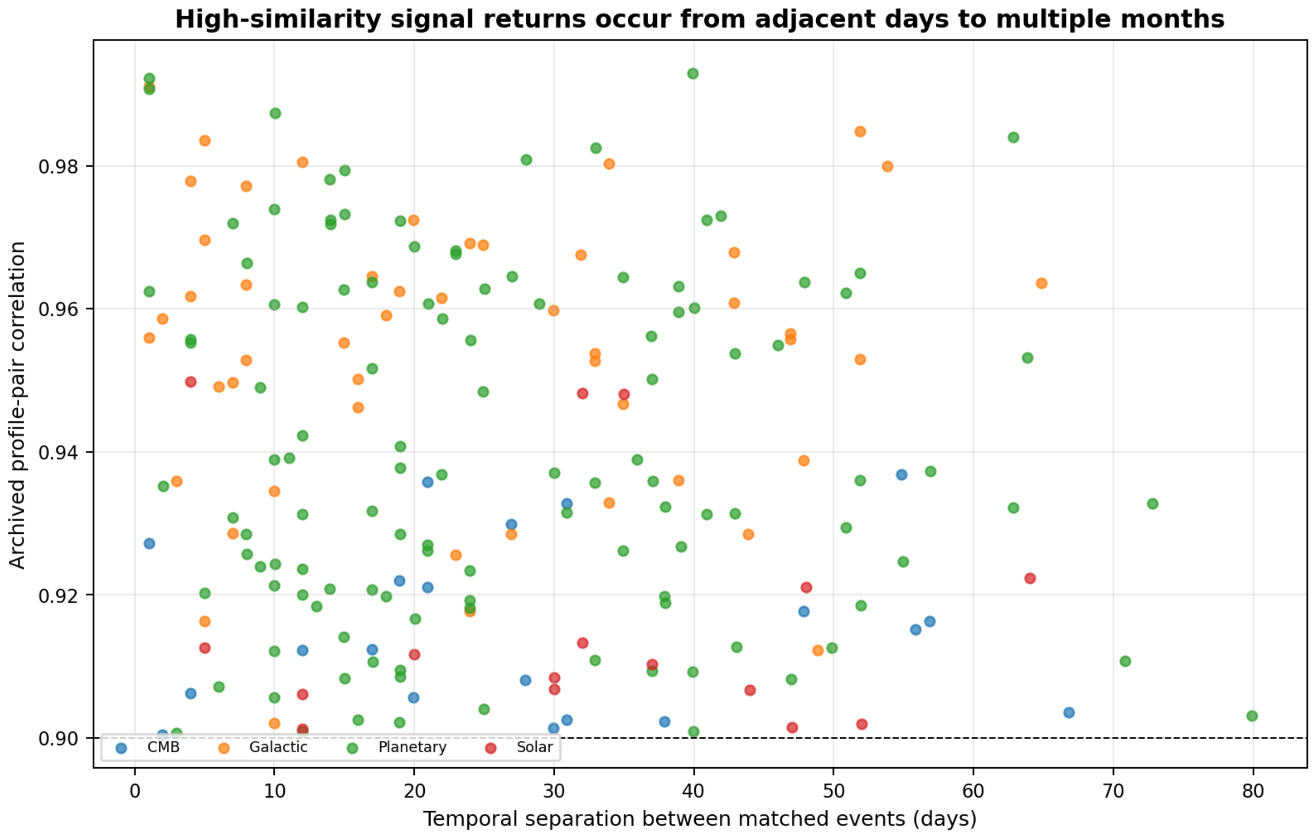

Figure 4. Correlation of every high-similarity pair versus temporal separation.

Reading guide. Solar, lunar, CMB, Galactic, and planetary labels all contain episodic returns. The horizontal threshold identifies the complete r >= 0.90 selection used in the geometric reconstruction.

1.4 The April 12-13 common-state return

The strongest cross-label example occurs on April 12 and 13, 2023. Venus advancing, Galactic anticenter, and Mars advancing each reproduce with correlation near 0.99 during the same approximately three-hour celestial sector on both dates. These are three independently defined event labels, yet they return together. The most economical interpretation is that the labels intersect one global K-Field state rather than representing three unrelated planet-specific processes.

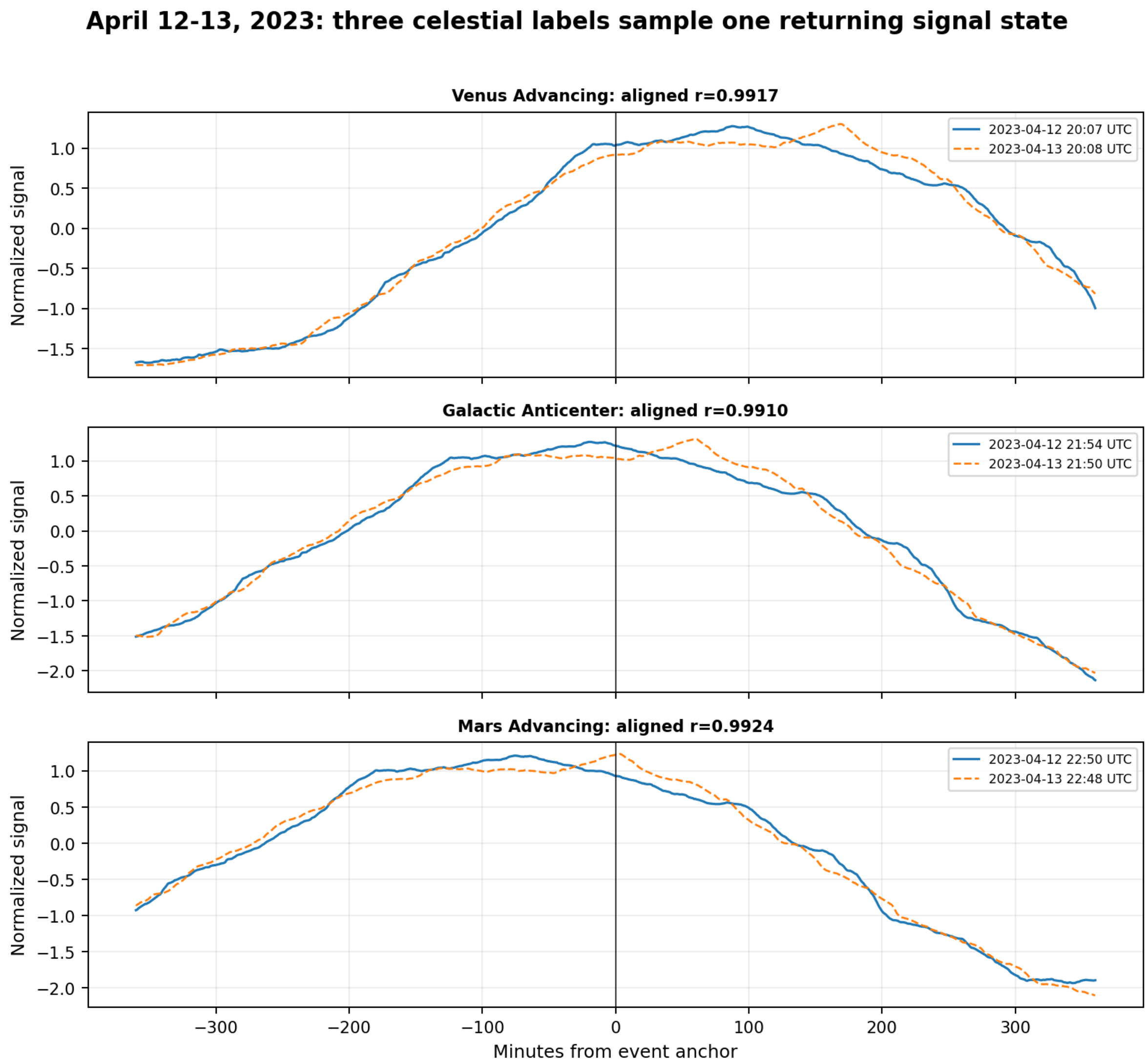

Figure 5. The April 12-13 Venus, Galactic-anticenter, and Mars common-state cluster.

Reading guide. The three panels are aligned independently. Their shared return demonstrates that several celestial labels can sample the same underlying spatial and rotational configuration.

1.5 Why ordinary daily recurrence is insufficient

Archive-wide lag correlation is weak at the sidereal, solar, and lunar day. The strongest correlations between 1,320 and 1,540 minutes are only about 0.008-0.026. Exact sidereal-day correlations range approximately from -0.021 to +0.011; solar-day correlations from -0.023 to +0.010; and lunar-day correlations from -0.033 to +0.017. The signal therefore does not reproduce as one invariant waveform every day.

The observational requirement is stricter: a celestial orientation must coincide with an appropriate translational phase of the observer inside the volume-filling K-Field structure. The remainder of the paper constructs that geometry and shows how the Earth-Moon barycenter, Earth reflex motion, and laboratory rotation determine the precise return condition.

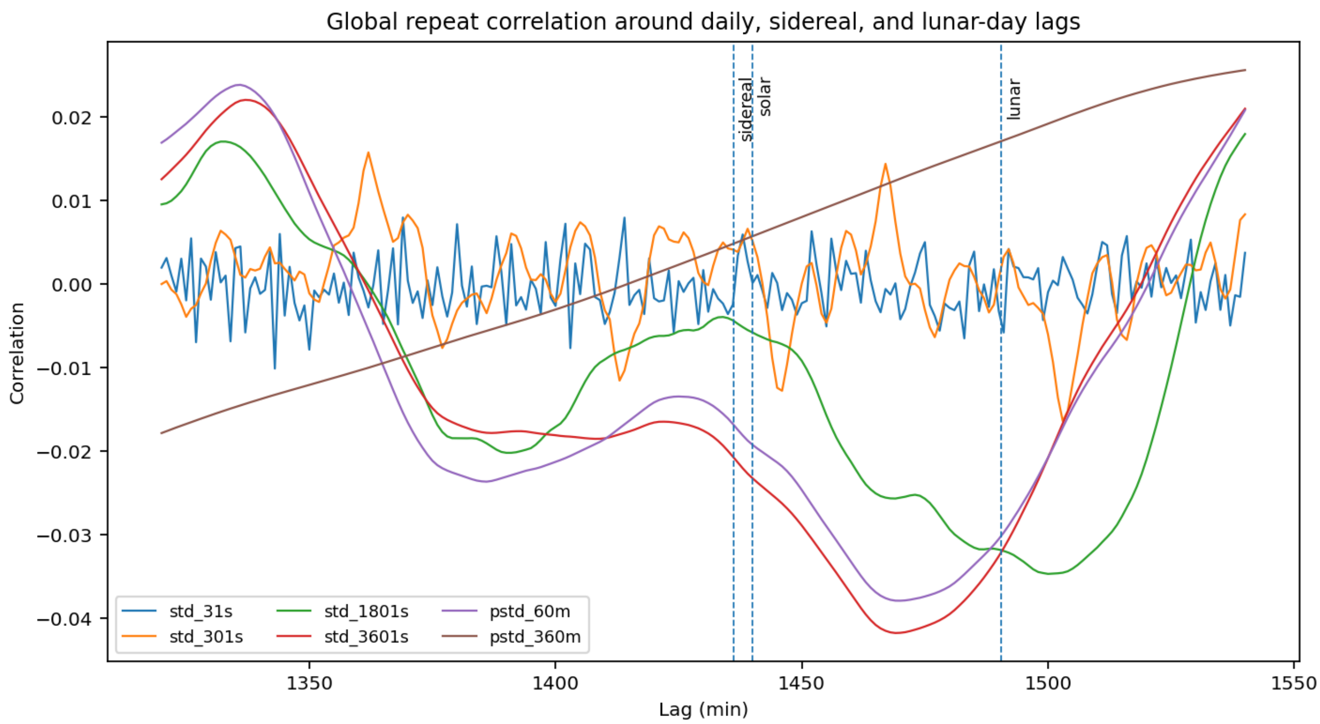

Figure 6. Daily, sidereal, and lunar lag comparison.

Reading guide. The absence of a strong archive-wide daily peak separates episodic state returns from a conventional daily environmental cycle.

2. D101 Instrument Record and Archive Reconstruction

2.1 Measured signal

D101 records the differential behavior of two nominally identical 20 MHz oven-controlled crystal oscillators mounted orthogonally in a common plane and thermal environment. Their outputs are heterodyned, and the resulting pulse-width record measures the evolving fractional-frequency differential. The measured column used throughout this paper is pulse_w.

Pulse width is treated as a time-series observable rather than as an absolute frequency calibration. Slow drift, short carrier modulation, transient variance, and crossing-centered structure coexist in the same record. The analysis therefore separates the signal into multiple time scales instead of reducing it to one averaged daily statistic.

Table 2. D101 archive census

Archive quantity	Value
Total source rows	6,762,475
Nominal rows	6,752,500
Fault-state rows	9,975
Start UTC	2023-03-02 21:58:22+00:00
Finish UTC	2023-05-22 17:17:10+00:00
Observed duration	78.269375 days
Duplicate timestamps	0
Gaps greater than 1 s	206
Largest gap	217,747 s
Nominal pulse median	3e+06
Nominal pulse standard deviation	18064.243
Temperature range	36.3750 to 46.9375 C

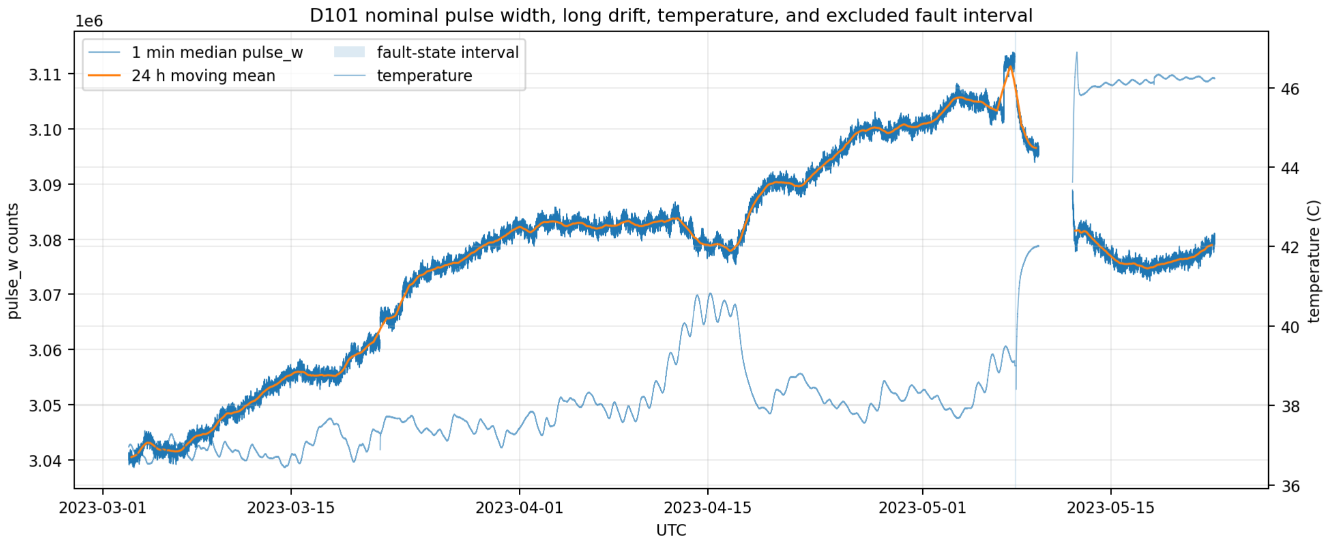

Figure 7. Reconstructed D101 archive timeline.

Reading guide. The chronological record is assembled from concatenated run blocks. The nominal and fault-state intervals are separated before spectral and variance analysis.

2.2 Segment structure and exclusions

The archive contains five principal acquisition segments and 206 timing gaps. The largest gap is 217,747 seconds, approximately 2.52 days. No duplicate timestamps were found. Event-centered profiles are accepted only when the required pre-event and post-event interval is sufficiently complete; this is why usable event counts vary slightly among channels and spans.

Table 3. Acquisition segments

Segment	Start UTC	Finish UTC	Rows	Fault rows
1	2023-03-02 21:58:22+00:00	2023-03-21 14:58:33+00:00	1,616,412	0
2	2023-03-21 15:00:16+00:00	2023-04-23 20:59:09+00:00	2,872,244	0
3	2023-04-23 21:09:04+00:00	2023-05-07 22:58:37+00:00	1,216,156	9,975
4	2023-05-07 23:00:19+00:00	2023-05-09 15:02:02+00:00	144,103	0
5	2023-05-12 03:31:09+00:00	2023-05-22 17:17:10+00:00	913,560	0

The May 7 high-count interval is a separate counter state. Approximately 9,838 samples occupy a 23-cycle accumulation state and 137 samples occupy a 128-cycle state. It is retained as an instrument-state observation but excluded from nominal spectra, moving statistics, transit templates, and chirp matrices.

2.3 Carrier ladder and directly observable frequency range

The dominant embedded carrier occurs at approximately 0.0372543 Hz, corresponding to a period of 26.8425 seconds. Strong components occur at approximately two, four, and eight times the base frequency, and a broader integer ladder extends through at least the thirteenth harmonic. Coherent fractional motion of several harmonics is stronger evidence of true carrier modulation than motion of one isolated spectral bin.

```
f0 = 0.0372543 Hz
T0 = 1/f0 = 26.8425 s
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The raw cadence is exactly one sample per second. The Nyquist frequency is therefore 0.5 Hz. The directly observable period range used in the chirp analysis is 2 seconds through 4 hours. Frequencies between 0.5 and 2 Hz cannot be uniquely reconstructed from this archive because they alias into the sampled band.

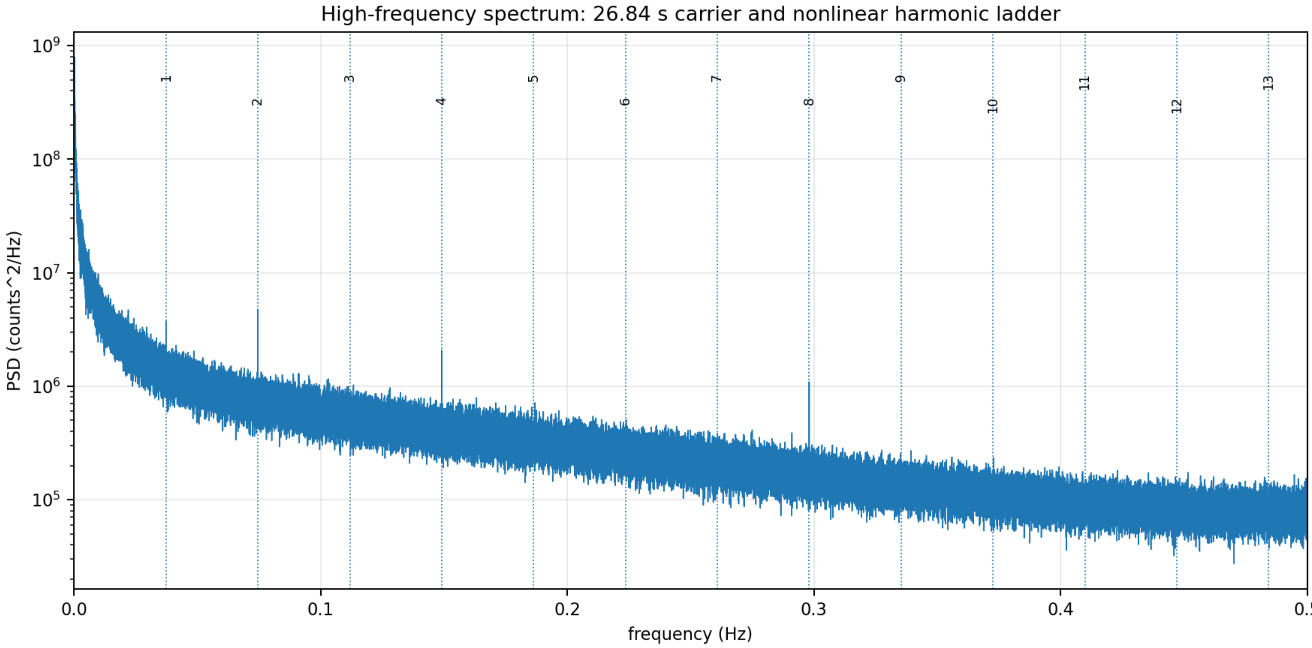

Figure 8. High-frequency power spectral density and carrier ladder.

Reading guide. The base carrier and power-of-two harmonics are embedded within the nominal OCXO signal. The figure is generated from the one-second archive after removal of the separate counter state.

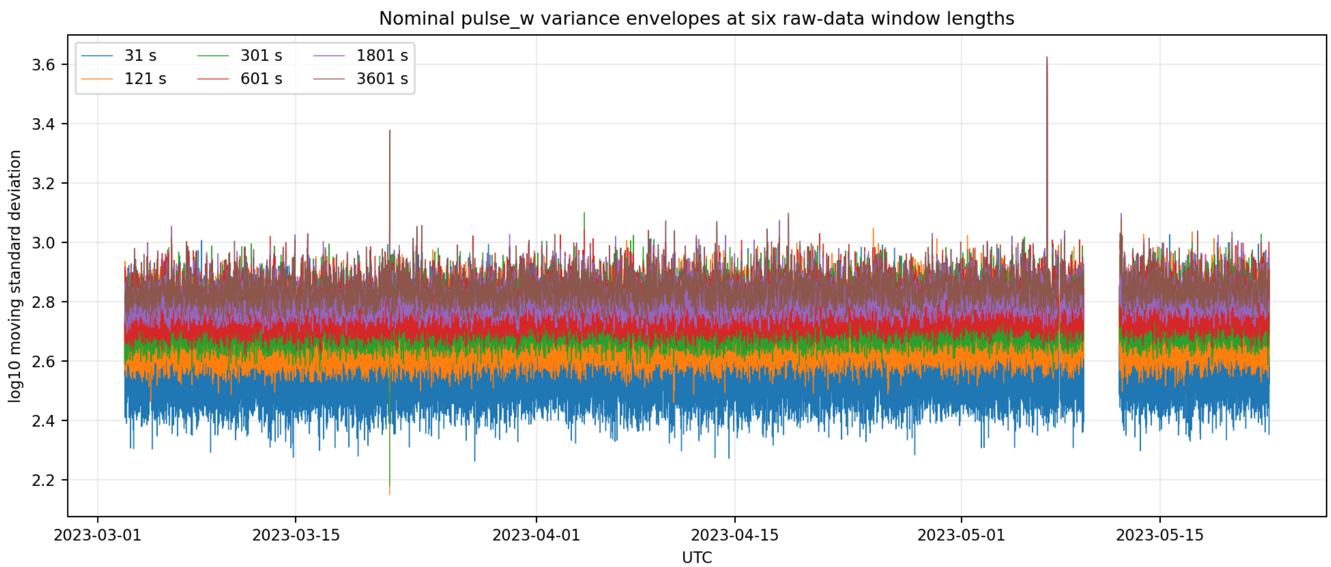

Figure 9. Multiscale moving-standard-deviation representation.

Reading guide. Short windows expose carrier and compact event modulation; long windows expose broad crossing-centered variance envelopes and slow background drift.

PART II. CONSTRUCT THE SPATIAL CONTEXT

3. Authoritative Cell3 Base Cell, Planes, and Complete Node Structure

3.1 Role of the Cell3 reference

The D101 crossing analysis measures temporal and angular signal organization. The Cell3 geometry supplies the declared rectilinear spatial reference used to interpret directed crossings as motion through plane families. The Cell3 basis is not estimated from the D101 archive in this paper; it is imported from the authoritative K-Field machine-readable geometry and retained exactly. This separation preserves dimensional correctness and prevents the crossing signal from being used to redefine its own spatial reference.

3.2 Direct basis and metric

Table 4. Authoritative Cell3 direct basis

Vector	ICRS components (nm)	Length (nm)	RA	Dec
a	0.327501242056, -0.054897007884, -0.037487502895	0.334179679059	350.484327659 deg	-6.440864866 deg
b	0.035330775844, 0.207319850440, -0.547401085389	0.586410890412	80.328748765 deg	-68.983472217 deg
c	-0.010129317627, -0.389787702366, -0.689734408367	0.792319765042	268.511403628 deg	-60.519741294 deg

Table 5. Cell3 metric invariants

Metric quantity	Value
det(B)	-0.117914891911749 nm^3
Physical volume	0.117914891911749 nm^3
Angle a-b	83.933483012249 deg
Angle a-c	80.448130018322 deg
Angle b-c	50.363231813718 deg
Area ab	0.194869165460674 nm^2
Area ac	0.261106235790182 nm^2
Area bc	0.357809539128446 nm^2

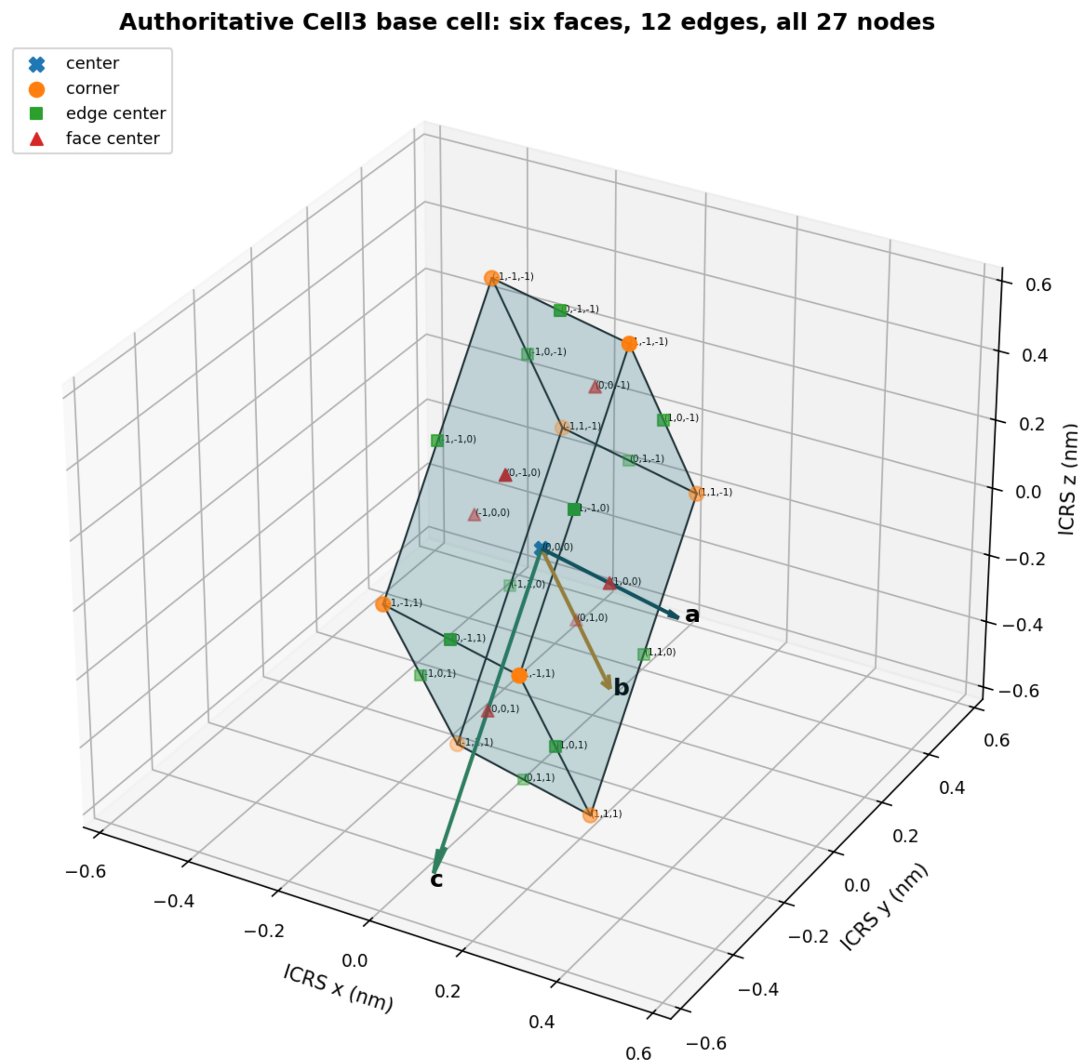

Figure 10. Dimensionally correct Cell3 base cell with all nodes.

Reading guide. The centered primitive cell is reconstructed from the authoritative ICRS basis. All six boundary faces, 12 corner edges, 27 half-step nodes, integer labels, and basis vectors are shown in nanometres.

3.3 Complete node definition

```
r(i,j,k) = (1/2) B [i,j,k]^T,
where i,j,k are in {-1,0,+1}
```

The node census is eight corners, twelve edge centers, six face centers, and one body center. The complete ledger is given below. The lambda index is i + j + k and identifies the seven-sheet (111) family.

Table 6. Complete 27-node Cell3 registry

Node ID	Type	Indices	ICRS coordinate nm	lambda	Sheet offset nm
N(-1,-1,-1)	corner	(-1, -1, -1)	-0.176351350, 0.118682430, 0.637311498	-3	-0.453818064111
N(-1,-1,+0)	edge center	(-1, -1, 0)	-0.181416009, -0.076211421, 0.292444294	-2	-0.302545376074
N(-1,+0,-1)	edge center	(-1, 0, -1)	-0.158685962, 0.222342355, 0.363610956	-2	-0.302545376074
N(+0,-1,-1)	edge center	(0, -1, -1)	-0.012600729, 0.091233926, 0.618567747	-2	-0.302545376074
N(-1,-1,+1)	corner	(-1, -1, 1)	-0.186480668, -0.271105272, -0.052422910	-1	-0.151272688037
N(-1,+0,+0)	face center	(-1, 0, 0)	-0.163750621, 0.027448504, 0.018743751	-1	-0.151272688037

Node ID	Type	Indices	ICRS coordinate nm	lambda	Sheet offset nm
N(-1,+1,-1)	corner	(-1, 1, -1)	-0.141020574, 0.326002280, 0.089910413	-1	-0.151272688037
N(+0,-1,+0)	face center	(0, -1, 0)	-0.017665388, -0.103659925, 0.273700543	-1	-0.151272688037
N(+0,+0,-1)	face center	(0, 0, -1)	0.005064659, 0.194893851, 0.344867204	-1	-0.151272688037
N(+1,-1,-1)	corner	(1, -1, -1)	0.151149892, 0.063785422, 0.599823995	-1	-0.151272688037
N(-1,+0,+1)	edge center	(-1, 0, 1)	-0.168815280, -0.167445347, -0.326123453	0	0.000000000000
N(-1,+1,+0)	edge center	(-1, 1, 0)	-0.146085233, 0.131108429, -0.254956791	0	0.000000000000
N(+0,-1,+1)	edge center	(0, -1, 1)	-0.022730047, -0.298553776, -0.071166661	0	0.000000000000
N(+0,+0,+0)	center	(0, 0, 0)	0.000000000, 0.000000000, 0.000000000	0	0.000000000000
N(+0,+1,-1)	edge center	(0, 1, -1)	0.022730047, 0.298553776, 0.071166661	0	0.000000000000
N(+1,-1,+0)	edge center	(1, -1, 0)	0.146085233, -0.131108429, 0.254956791	0	0.000000000000
N(+1,+0,-1)	edge center	(1, 0, -1)	0.168815280, 0.167445347, 0.326123453	0	0.000000000000
N(-1,+1,+1)	corner	(-1, 1, 1)	-0.151149892, -0.063785422, -0.599823995	1	0.151272688037
N(+0,+0,+1)	face center	(0, 0, 1)	-0.005064659, -0.194893851, -0.344867204	1	0.151272688037
N(+0,+1,+0)	face center	(0, 1, 0)	0.017665388, 0.103659925, -0.273700543	1	0.151272688037
N(+1,-1,+1)	corner	(1, -1, 1)	0.141020574, -0.326002280, -0.089910413	1	0.151272688037
N(+1,+0,+0)	face center	(1, 0, 0)	0.163750621, -0.027448504, -0.018743751	1	0.151272688037
N(+1,+1,-1)	corner	(1, 1, -1)	0.186480668, 0.271105272, 0.052422910	1	0.151272688037
N(+0,+1,+1)	edge center	(0, 1, 1)	0.012600729, -0.091233926, -0.618567747	2	0.302545376074
N(+1,+0,+1)	edge center	(1, 0, 1)	0.158685962, -0.222342355, -0.363610956	2	0.302545376074
N(+1,+1,+0)	edge center	(1, 1, 0)	0.181416009, 0.076211421, -0.292444294	2	0.302545376074
N(+1,+1,+1)	corner	(1, 1, 1)	0.176351350, -0.118682430, -0.637311498	3	0.453818064111

3.4 Central face planes

The ab, ac, and bc central planes are defined by fixing k = 0, j = 0, and i = 0 respectively. Each contains nine half-step nodes. They are the central members of the three primitive face-plane families and form a natural coordinate skeleton for rectilinear crossing analysis.

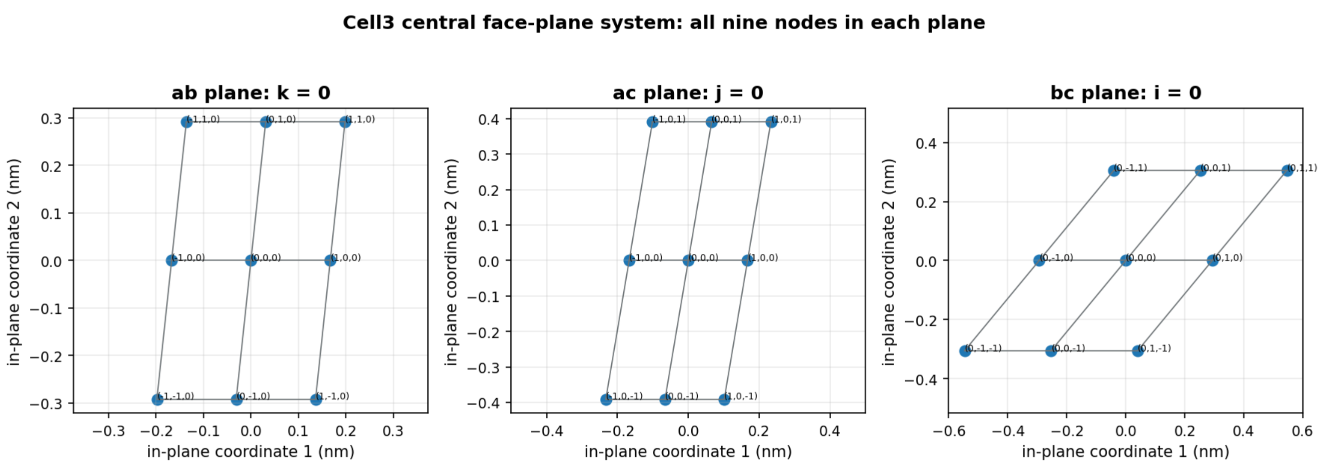

Figure 11. Cell3 central face planes and all nine nodes per plane.

Reading guide. Each panel uses an orthonormal in-plane coordinate basis derived from the exact Cell3 vectors. Node labels retain the authoritative integer address.

3.5 Reciprocal plane atlas

Table 7. Authoritative reciprocal plane families

Family	hkl	Spacing nm	RA deg	Dec deg	Unit normal ICRS
(100)	(1, 0, 0)	0.329546529696	355.201792089	1.869279271	0.995965190, -0.083602163, 0.032619291
(010)	(0, 1, 0)	0.451597379721	84.132672287	-29.408562676	0.089052594, 0.866576769, -0.491033948
(001)	(0, 0, 1)	0.605097741518	258.000576203	-21.000692601	-0.194092201, -0.913177169, -0.358379235

Family	hkl	Spacing nm	RA deg	Dec deg	Unit normal ICRS
(110)	(1, 1, 0)	0.266175054578	27.349970065	-15.252460507	0.856930360, 0.443241629, -0.263072646
(101)	(1, 0, 1)	0.306442673082	326.873754622	-8.694351580	0.827844954, -0.540205857, -0.151163370
(011)	(0, 1, 1)	0.576944779286	106.780626621	-75.703939557	-0.071291375, 0.236417305, -0.969032712
(111)	(1, 1, 1)	0.302545376075	3.082265054	-28.568337610	0.876976866, 0.047223124, -0.478206600

3.6 Seven physical sheets

The (111) family partitions the 27 nodes into seven indexed sheets with populations 1, 3, 6, 7, 6, 3, and 1. The adjacent half-step sheet spacing is 0.151272688037 nm, while the indexed reciprocal spacing is 0.302545376075 nm. The sheet normal has right ascension 3.082265 degrees and declination -28.568338 degrees.

Table 8. Complete seven-sheet membership

lambda	Population	Offset nm	Polarity	Node indices
-3	1	-0.453818064111	-1	(-1, -1, -1)
-2	3	-0.302545376074	1	(-1, -1, 0); (-1, 0, -1); (0, -1, -1)
-1	6	-0.151272688037	-1	(-1, -1, 1); (-1, 0, 0); (-1, 1, -1); (0, -1, 0); (0, 0, -1); (1, -1, -1)
0	7	0.000000000000	1	(-1, 0, 1); (-1, 1, 0); (0, -1, 1); (0, 0, 0); (0, 1, -1); (1, -1, 0); (1, 0, -1)
1	6	0.151272688037	-1	(-1, 1, 1); (0, 0, 1); (0, 1, 0); (1, -1, 1); (1, 0, 0); (1, 1, -1)
2	3	0.302545376074	1	(0, 1, 1); (1, 0, 1); (1, 1, 0)
3	1	0.453818064111	-1	(1, 1, 1)

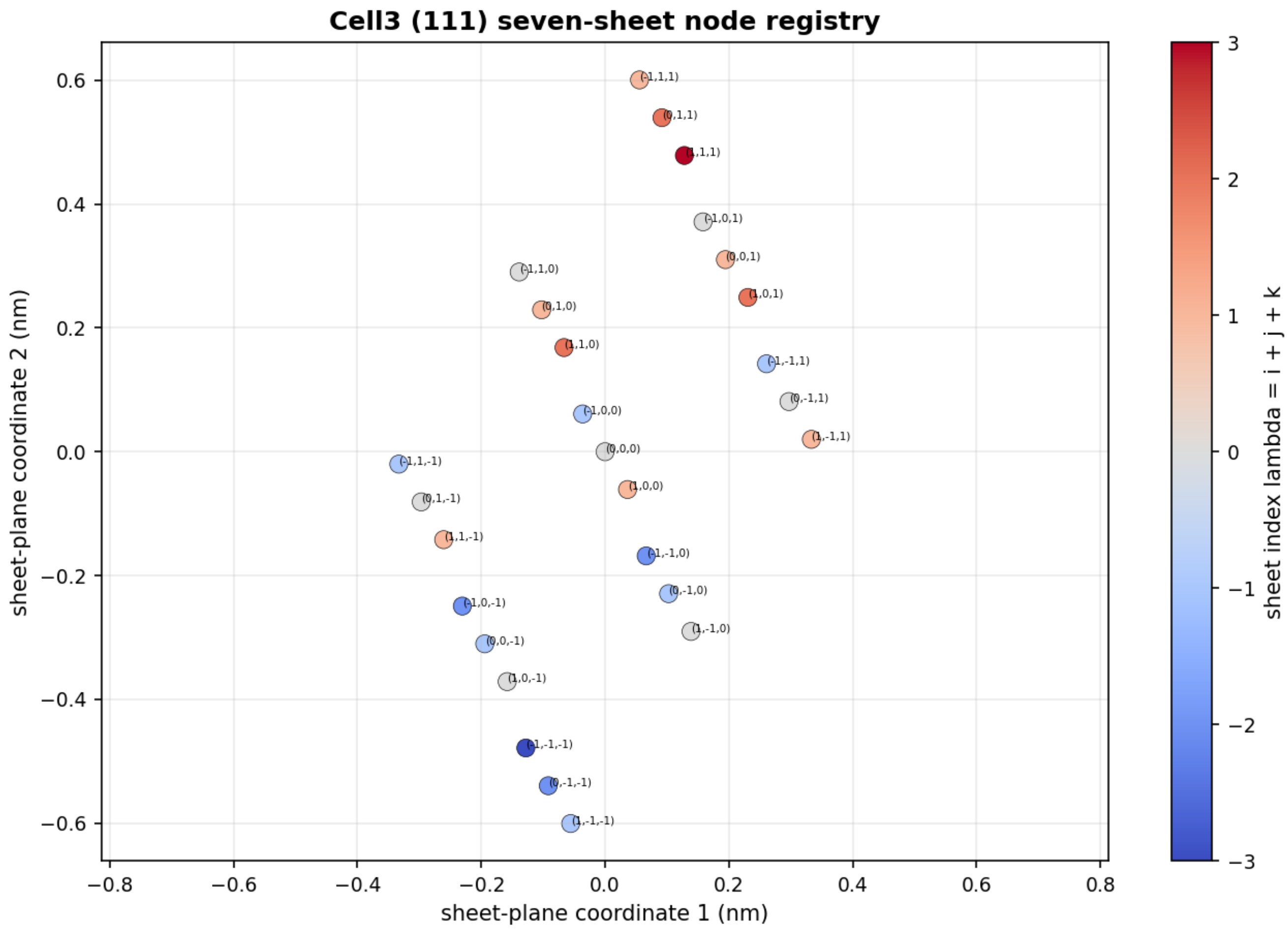

Figure 12. Cell3 seven-sheet projection and node membership.

Reading guide. The projection shows every node in the (111) sheet plane. Color is the exact integer sheet index lambda = i + j + k.

4. The Volume-Filling, Correctly Oriented K-Field Structure

4.1 From one Cell3 to a filled spatial volume

The authoritative Cell3 is a primitive oblique cell, not an isolated object. Translating its direct basis through all integer triples produces a volume-filling lattice:

```
L = { B n : n in Z^3 }.
```

The repeated reciprocal planes extend across that volume. A moving observer therefore encounters a sequence of boundaries whose orientations are fixed by the ICRS basis and whose spacings change with the declared recursive order. The base cell supplies the local node and face registry; the integer translations supply the global volume.

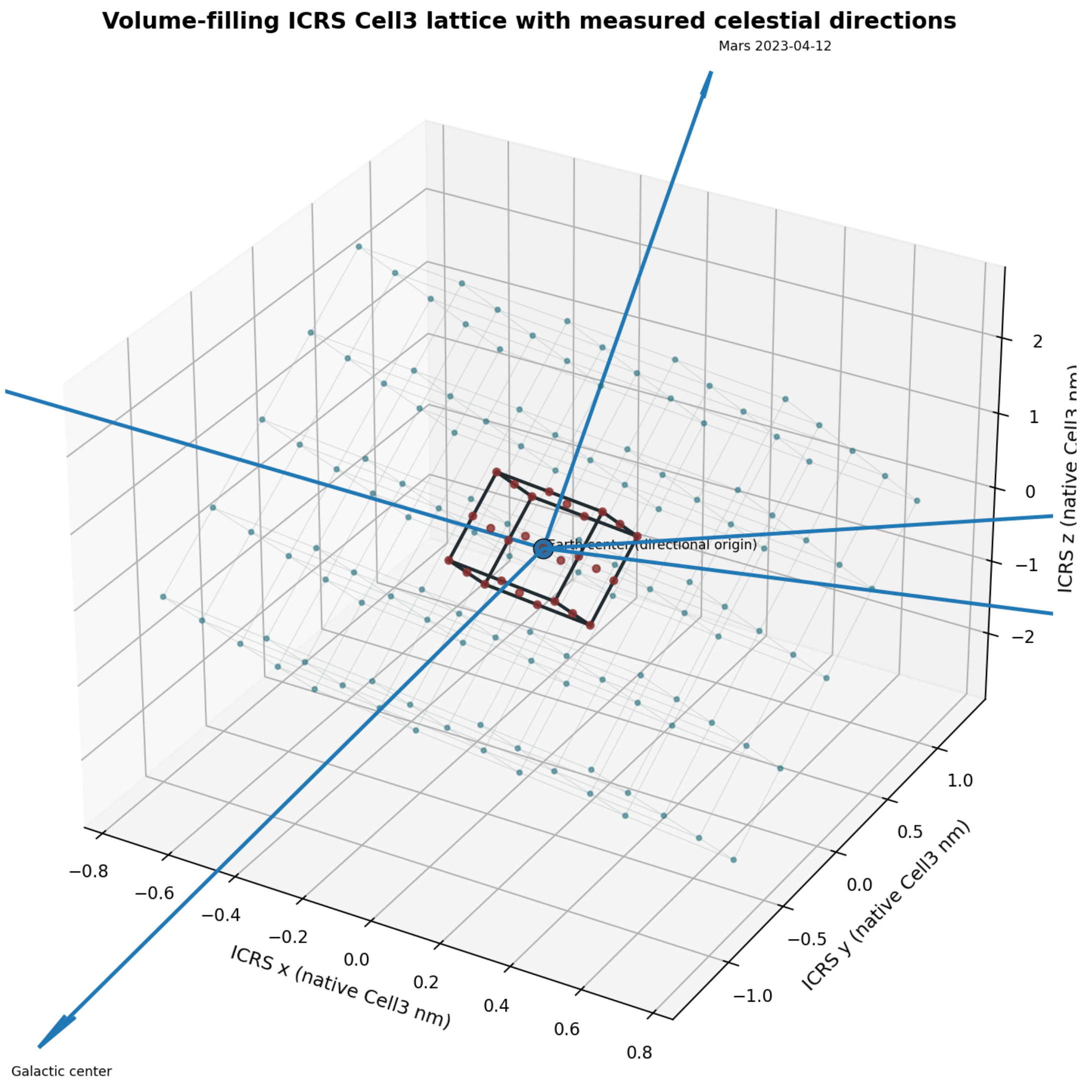

Figure 13. Volume-filling Cell3 lattice with Sun, Galactic-center, CMB, Mars, and Uranus directions.

Reading guide. The central cell and repeated translations are dimensionally correct in native nanometres. Celestial arrows are normalized in length solely for visibility; their ICRS directions come from the measured archive and declared fixed vectors.

4.2 The observer crossing a boundary

A boundary event is local. Near one plane, the complete three-dimensional problem can be reduced to a normal coordinate and two in-plane coordinates. The Earth-Moon barycenter establishes the smooth translational path. Earth reflex motion displaces the Earth center opposite the Moon. The daily laboratory vector displaces the detector from the Earth center. The actual detector crosses when the sum reaches the plane.

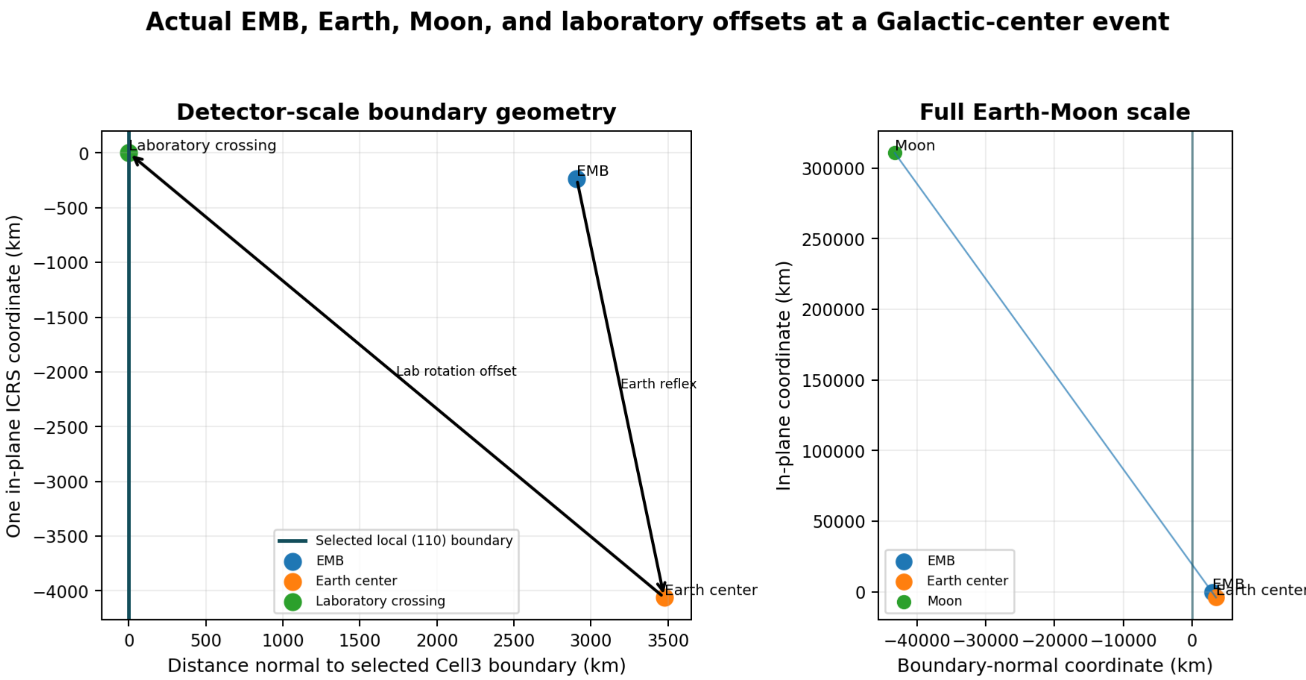

Figure 14. Actual local boundary geometry at a high-similarity Galactic-center anchor.

Reading guide. The selected Cell3 (110) plane is shown as the zero normal coordinate. The EMB, Earth center, laboratory, and Moon are placed from the actual D101 ephemeris and geodetic vectors at the event time.

4.3 Celestial vectors as orientation gates

The CMB velocity direction, Galactic-center axis, solar direction, and planetary directions do not define different lattices. They define different orientation gates through the same fixed ICRS plane system. A transit, antipodal transit, or orthogonal position selects a directed projection of the observer and instrument axes onto that plane structure. The observed signal can therefore differ by event label even when several labels occupy the same global state.

5. Earth and Celestial Vectors Inside the K-Field Volume

5.1 Hour angle as the event coordinate

A celestial transit is not fundamentally a clock time. It is a geometric condition defined by the relation between the local meridian and a celestial direction. For a target with right ascension alpha(t), the target-relative Earth-rotation phase is the hour angle H(t).

```
H(t) = theta_LST(t) - alpha(t)
```

Upper culmination occurs at H = 0 degrees. The antipodal or lower culmination occurs at H = 180 degrees. The two orthogonal positions occur at H = -90 degrees and H = +90 degrees. When the target is fixed in ICRS, alpha is constant. For the Sun, Moon, and planets, alpha changes with time and must be included in event registration.

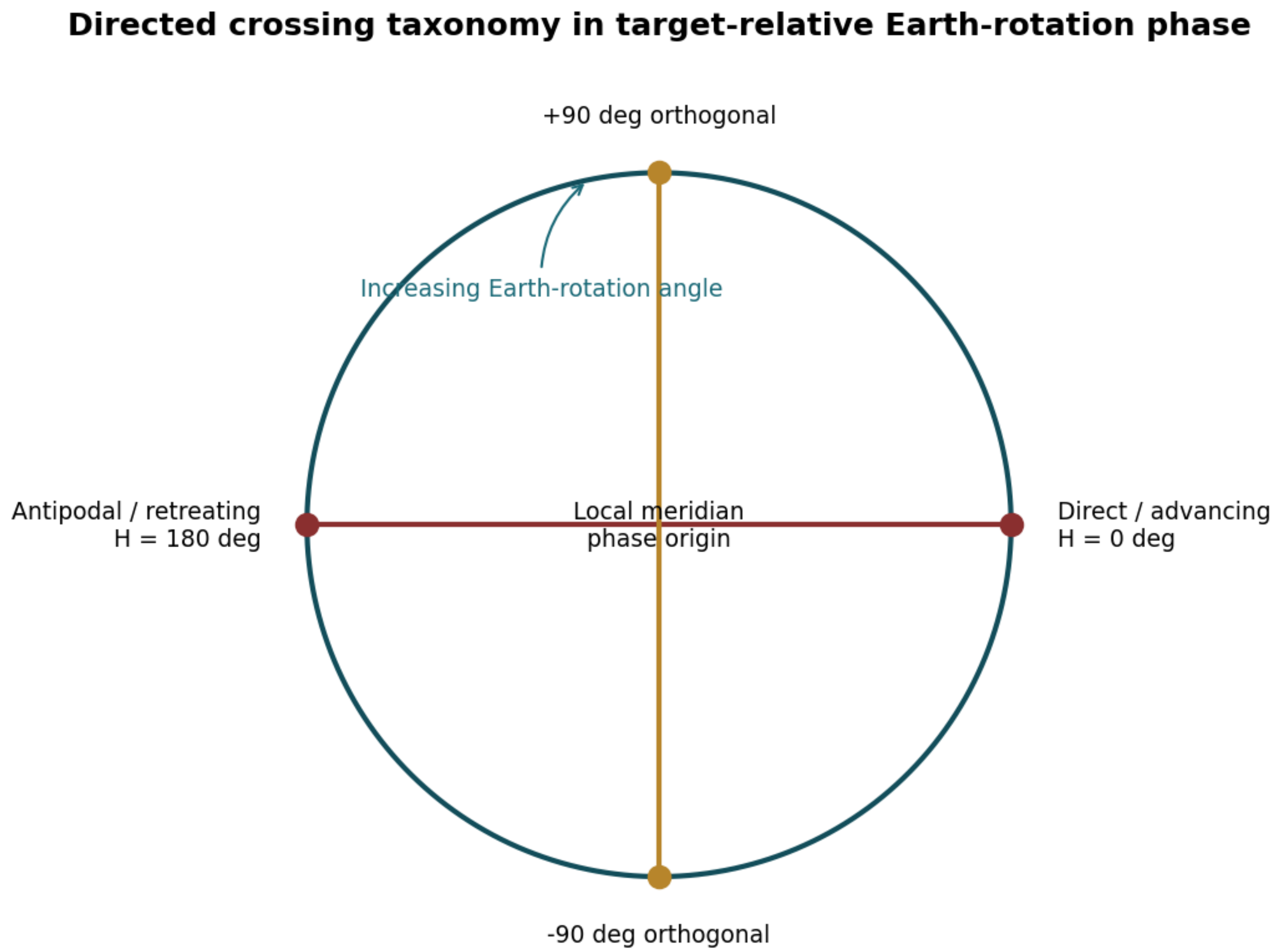

Figure 15. Direct, antipodal, and orthogonal event geometry.

Reading guide. The four positions form a complete directed Earth-rotation phase cycle. Advancing and retreating sides are separated by 180 degrees; orthogonal sides are separated by 90 degrees.

5.2 Fixed directions

Table 9. Fixed ICRS crossing axes

Direction	Right ascension	Declination	Role
CMB advancing	189.285391 deg	+16.626278 deg	Supplied Earth-CMB velocity direction
CMB retreating	9.285391 deg	-16.626278 deg	Antipode of supplied CMB direction
Galactic center	266.405100 deg	-28.936175 deg	Milky Way center direction
Galactic anticenter	86.405100 deg	+28.936175 deg	Antipode of Galactic center

The CMB and Galactic axes are fixed in the ICRS frame over the duration of the experiment. Their event ordering is therefore controlled almost entirely by sidereal rotation. This makes them essential reference cases for distinguishing target-motion corrections from pure latitude scaling.

5.3 Solar, lunar, and planetary directions

Solar and lunar positions were generated from the ephemeris at one-minute cadence. Planetary upper and antipodal crossings were evaluated for Mercury, Venus, Mars, Jupiter, Saturn, Uranus, and Neptune. Earth was excluded because a geocentric direction toward Earth is undefined. Lunar rise and set are horizon crossings rather than meridian culminations; they were retained as independent event classes because prior D101 work identified strong lunar-coordinate covariance.

5.4 Orthogonal duplication

Because the antipodal event is already 180 degrees from the direct event, direct +90 degrees and antipodal -90 degrees describe the same physical sector apart from small target motion during the half-day interval. Duplicate labels are therefore useful for validating the event computation but must not be counted as independent detections. The solar evening quadrature and Saturn quadrature results provide explicit examples of this equivalence.

PART III. ESTABLISH THE EARTH-MOON BARYCENTRIC DATUM

6. Earth-Moon Barycentric Motion as the Translational Datum

6.1 Why the Earth-Moon barycenter is required

The smooth translational datum is the Earth-Moon barycenter, abbreviated EMB. The Earth center is not the center of the Earth-Moon system. It executes a monthly reflex path opposite the geocentric lunar vector. The laboratory adds a daily rotational path about the Earth center. Treating the Earth-center heliocentric position as one undifferentiated coordinate retains the numerical displacement but hides the phase mechanism that produces lunar covariance and episodic lattice closure.

```
r_obs(t) = r_EMB(t) - mu r_Moon/Earth(t) + r_lab/Earth(t)
```

Here mu = m_Moon/(m_Earth + m_Moon) = 0.01215058561. During the D101 archive the Earth-center displacement from the EMB ranged from approximately 4,407 to 4,932 km, with mean magnitude about 4,675 km. The geocentric laboratory radius was approximately 6,369 km. Monthly reflex and daily rotation are therefore comparable observer-offset motions.

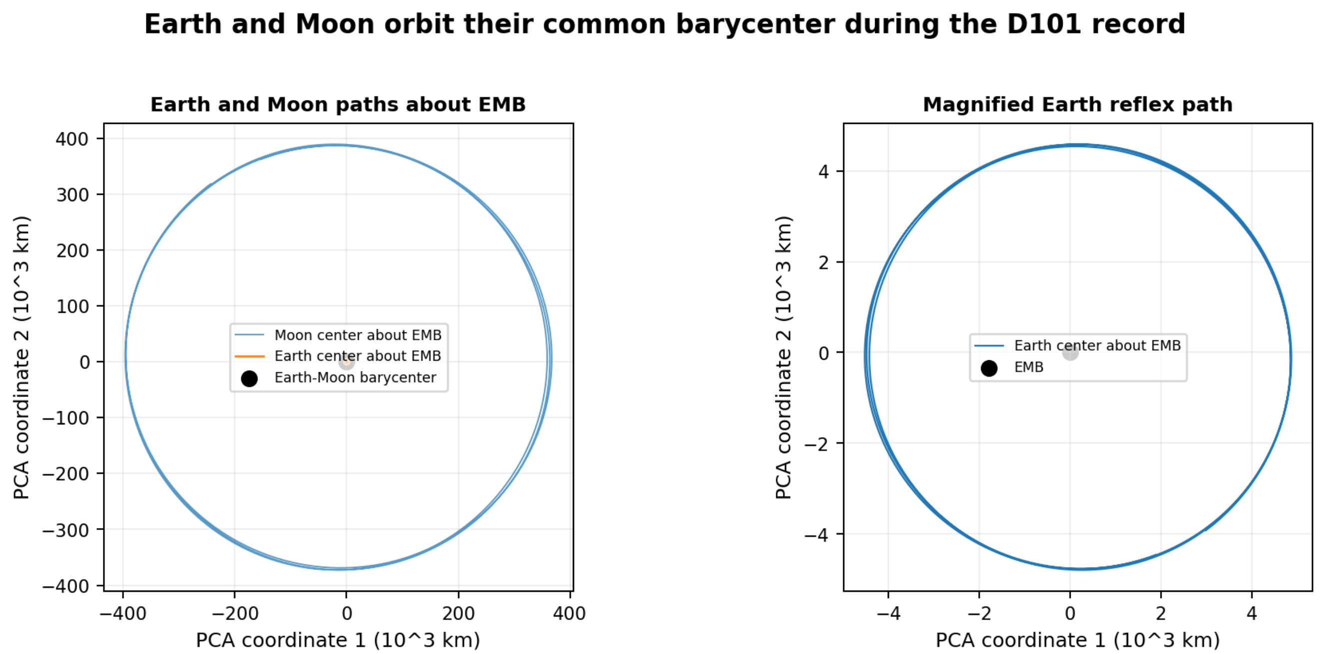

Figure 16. Actual Earth and Moon paths about their common barycenter.

Reading guide. The left panel places the Earth and Moon on the same dimensionally scaled barycentric axes. The right panel magnifies the Earth reflex path, which is exactly opposite the Moon and smaller by the Earth-Moon mass fraction.

6.2 Nested detector motion

The detector trajectory is a nested curve. The EMB translates through the K-Field. The Earth center moves around the EMB over the lunar cycle. The laboratory rotates around the Earth center every sidereal day. The observed path is consequently neither a straight line nor a single circular motion, even over a short archive.

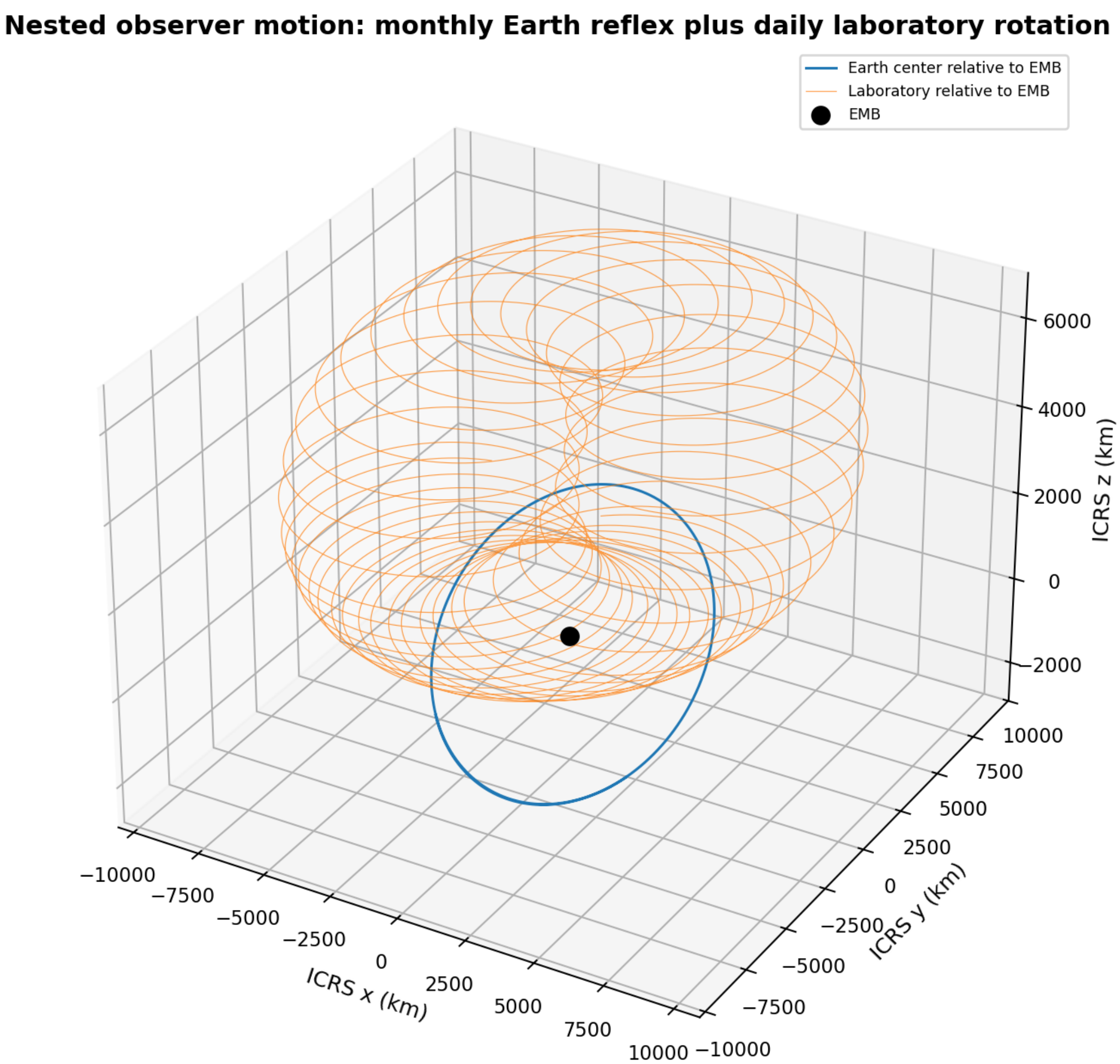

Figure 17. Actual nested observer motion during the central portion of the archive.

Reading guide. The smooth Earth-center reflex path and the daily laboratory loops are shown in ICRS kilometres. The daily loops migrate as the Earth center moves around the EMB.

6.3 Phase modulation in recursive Cell3 units

For reciprocal plane h and recursive order N, the complete detector phase is

```
Phi_obs(h,N,t) = Phi_EMB(h,N,t) + deltaPhi_reflex(h,N,t) + deltaPhi_rot(h,N,t).
```

The monthly Earth-reflex term and daily laboratory term are both large at low recursive order and decrease rapidly with increasing order. Their median peak-to-peak amplitudes are approximately 15 cycles at order 16, 1.7 cycles at order 17, 0.19 cycles at order 18, 0.021 cycles at order 19, 0.0023 cycles at order 20, and 0.00026 cycles at order 21. Order 19 is especially important because its offset amplitude is comparable with many measured episodic closure residuals.

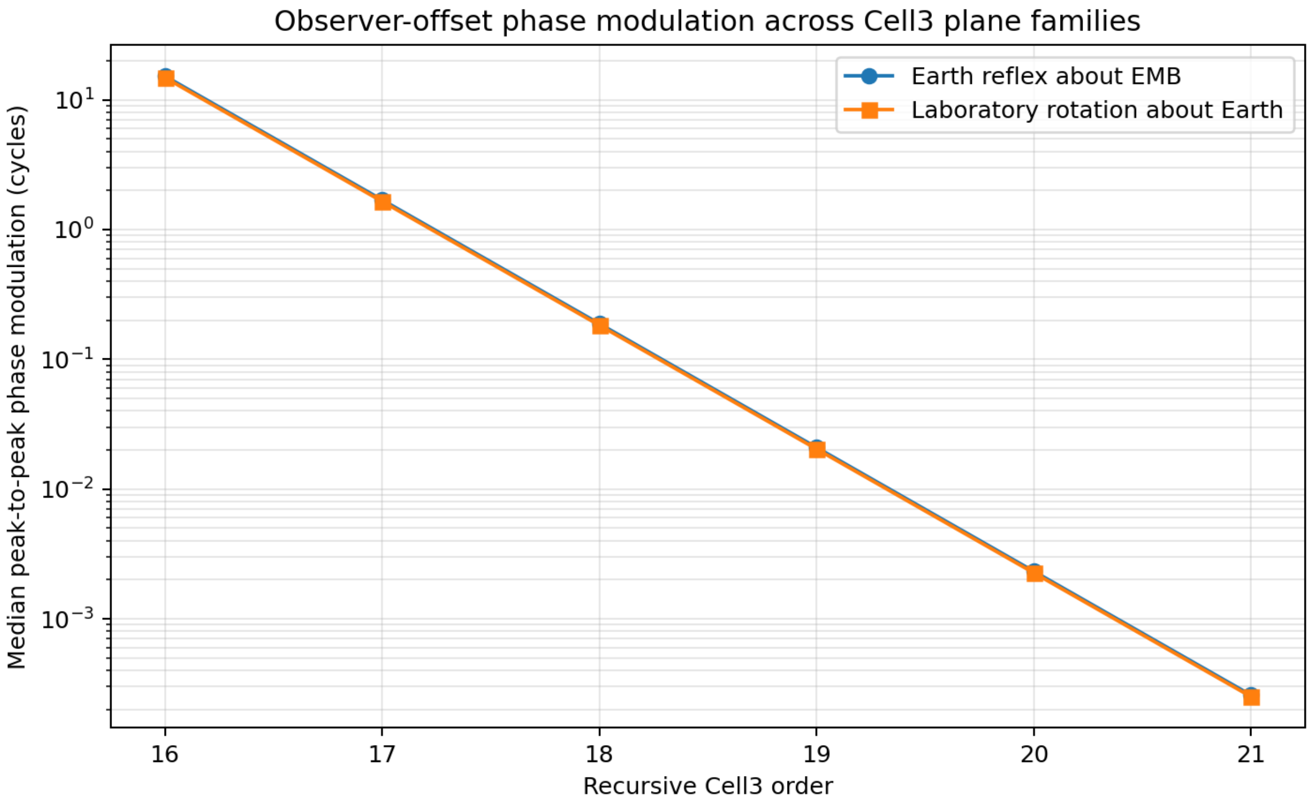

Figure 18. Earth-reflex and laboratory phase modulation versus recursive Cell3 order.

Reading guide. The monthly and daily observer offsets are expressed directly in plane cycles. Their comparable magnitudes explain why both must be retained when locating a terrestrial detector crossing.

6.4 Barycentric returns and compensated detector returns

Two recurrence mechanisms are observed. In a barycentric return, the EMB base phase itself approaches the same lattice state and removal of Earth reflex improves closure. In a compensated detector return, the base phase is less exact but the known Earth-reflex or laboratory offset completes the detector closure. Neither component may be discarded; the decomposition identifies which part of the geometry produced the return.

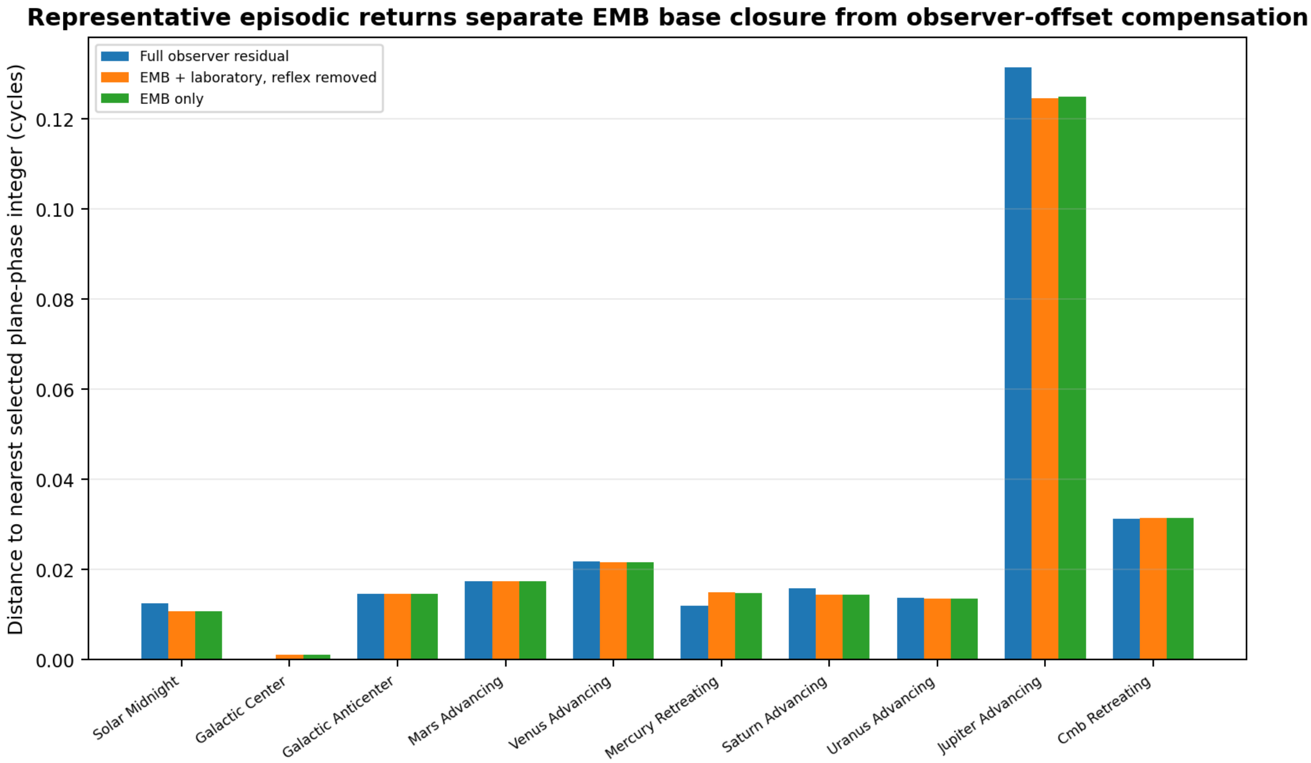

Figure 19. Full-observer, EMB-plus-laboratory, and EMB-only closure residuals.

Reading guide. Solar midnight and Saturn examples improve when reflex is removed, whereas the Galactic-center example is sharpened by reflex compensation. The event response therefore contains both barycentric and compensated returns.

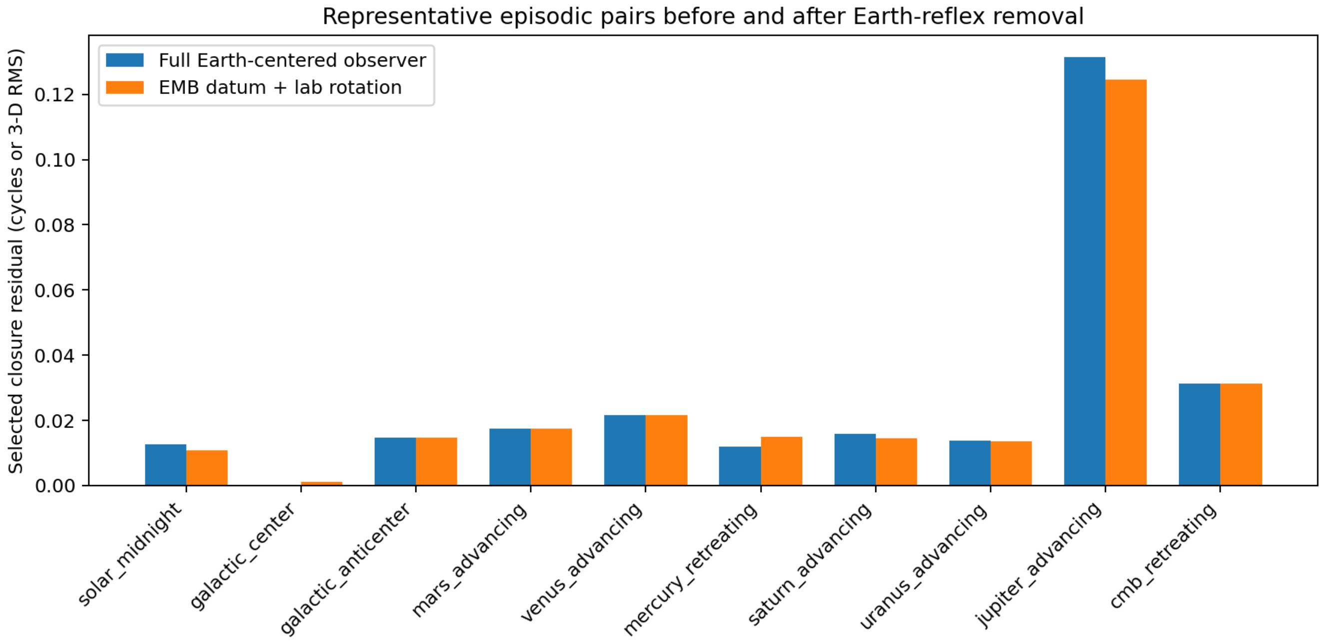

Figure 20. Representative pair closure before and after barycentric decomposition.

Reading guide. Each bar is calculated from the actual pair displacement. The comparison shows which high-similarity states are base-phase returns and which are completed by observer-offset modulation.

6.5 Revised interpretation of lunar covariance

The Moon coordinates are not only external astronomical covariates. They directly encode the position of the Earth center relative to the EMB:

```
r_Earth/EMB = -mu [x_Moon, y_Moon, z_Moon]^T.
```

A regression between D101 variance and lunar ICRS x, y, z can therefore arise because the detector is moving through the K-Field lattice in opposition to the Moon. Coefficient reversals are expected when the active Cell3 plane family or recursive order changes. This explains why strong local lunar covariance can coexist with a nonstationary global transfer function.

7. Analytical Unwinding of the Laboratory Latitude Cone

7.1 Why the local upward vector traces a cone

At geodetic latitude phi, the local upward vector does not rotate around the celestial equator. It rotates around the north celestial pole while retaining a constant polar component. In equatorial Cartesian coordinates,

```
u(theta) = [ cos(phi) cos(theta),
             cos(phi) sin(theta),
             sin(phi) ]
```

At the D101 site, phi = 42.129 degrees. The constant polar component is sin(phi) = 0.6708020808, and the radius of the rotating equatorial component is cos(phi) = 0.7416364125. The zenith therefore traces a cone whose half-angle from the north celestial pole is 47.871 degrees.

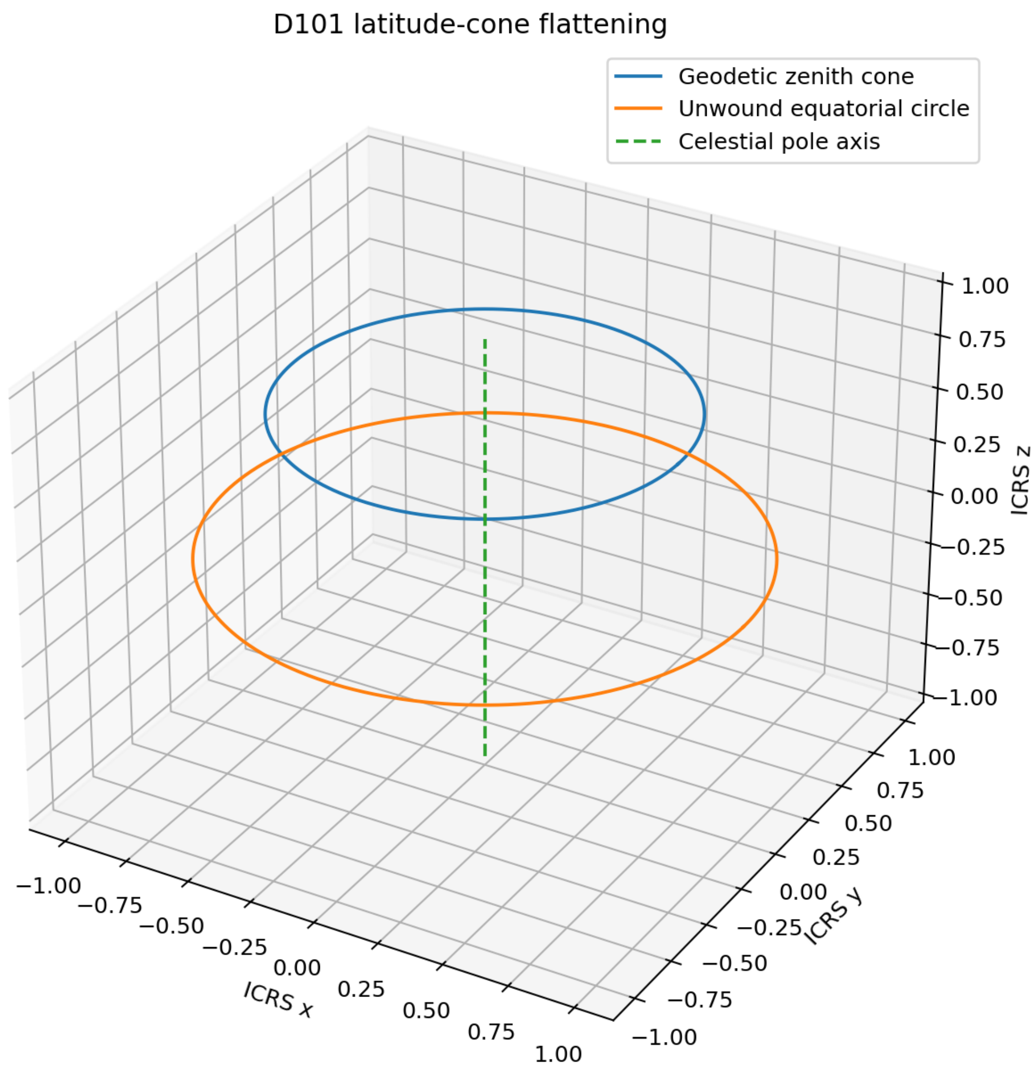

Figure 21. Exact latitude-cone flattening.

Reading guide. The geodetic zenith cone is flattened by removing the constant polar component and restoring the equatorial radius to unity.

7.2 Exact cone flattening and co-rotation

```
q(t) = [u(t) - sin(phi) z_hat] / cos(phi)
     = [cos(theta), sin(theta), 0]

q_fixed(t) = Rz[-theta(t)] q(t) = [1,0,0]
```

The required tangential or path-length stretch is sec(phi) = 1.3483696095. Any wavelength measured along the physical zenith cone maps to its equatorial-plane equivalent by dividing by cos(phi).

```
lambda_equatorial = lambda_cone / cos(phi)
```

The numerical implementation reproduces the constant co-rotated vector with error below 5 x 10^-16. The existing lab position vector is geocentric radial rather than geodetic upward; the deterministic separation at the D101 site is approximately 0.191387 degrees. Device-axis work therefore uses the geodetic normal.

7.3 Target-relative normalization

For a target at right ascension alpha and declination delta, the zenith projection includes a constant latitude-declination term and a rotating hour-angle term. The normalized cosine and sine quadratures are

```
C = [a dot u - sin(phi) sin(delta)] / [cos(phi) cos(delta)] = cos(H)
S = -[a dot east] / cos(delta) = sin(H)
```

The phase atan2(S,C) recovers the unwound hour angle. This phase replaces UTC for event stacking. Temporal chirp frequency is then converted into angular harmonic order and angular wavelength.

```
m(t) = 2 pi f(t) / abs[dH/dt]
lambda_H(t) = 360 deg / m(t)
```

For fixed CMB and Galactic directions, d(alpha)/dt = 0, so the transformation cannot create new scalar coherence; it only recalibrates angular scale and derivatives. For the Sun, Moon, and planets, the changing right ascension is removed. This is why lunar horizon events gain coherence while fixed-axis correlations remain unchanged.

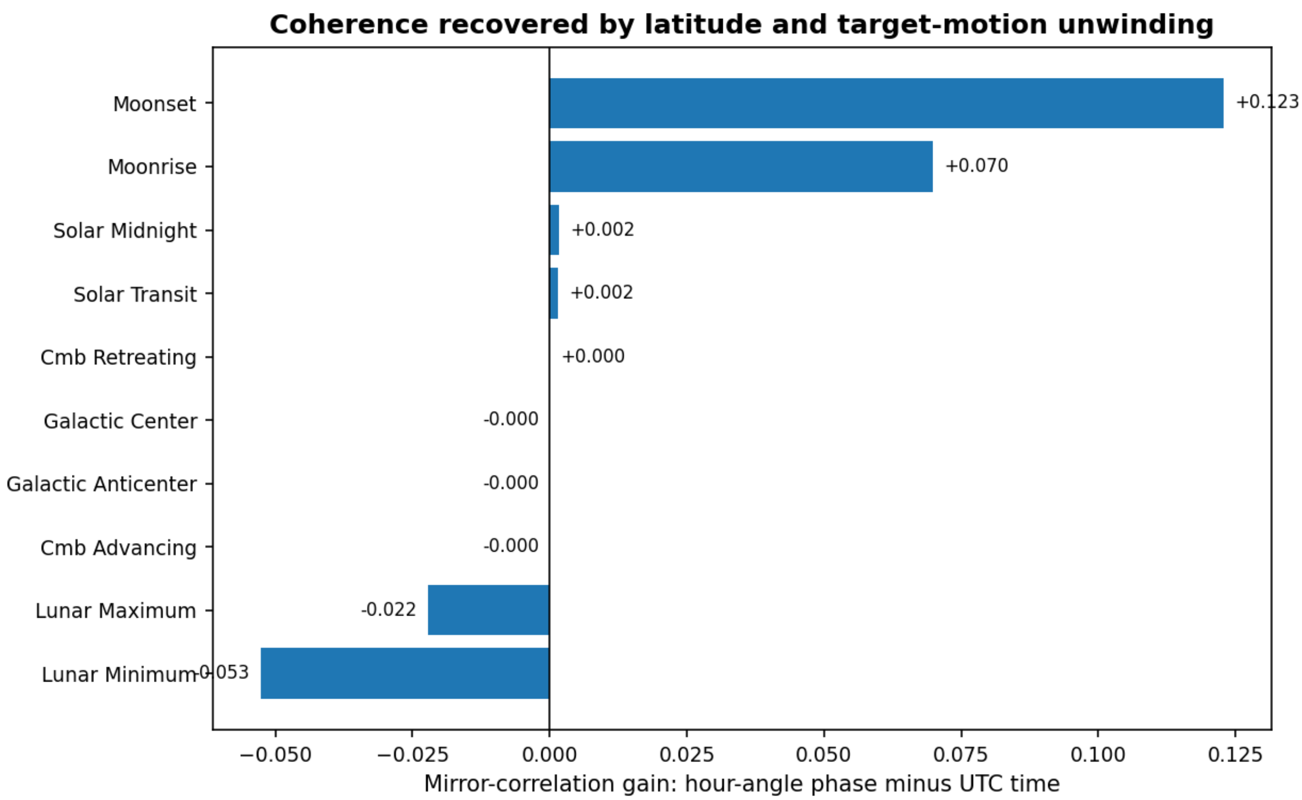

Figure 22. Coherence gained by hour-angle registration.

Reading guide. Moonset and Moonrise receive the largest mirror-correlation gains. Fixed CMB and Galactic directions remain effectively invariant, which is the required analytical behavior.

PART IV. CONNECT SIGNAL RETURNS TO EXPLICIT ICRS GEOMETRY

8. Explicit ICRS Geometry and the Symmetry of Episodic Returns

8.1 The anchor population

The complete geometric reconstruction contains 205 profile pairs with correlation at or above 0.90, 253 unique crossing anchors, 17 event classes, and 21 cross-event clusters. Every anchor is represented by its UTC, target direction, directed antipode where applicable, local zenith, Sun, Moon, CMB relation, Galactic relation, nearest Cell3 plane, nearest projected node ray, and recursive phase closure.

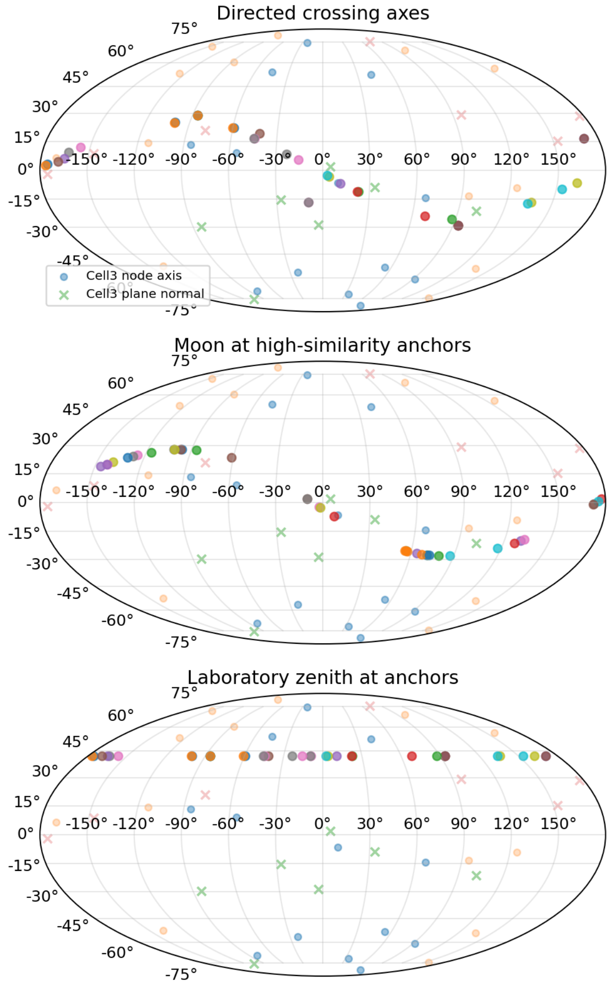

Figure 23. High-similarity anchors on the ICRS sphere.

Reading guide. The anchor population is plotted with fixed K-Field axes and Cell3 directions. The figure allows direct comparison of the returning signal states with their celestial orientation.

8.2 Projected node-ray alignment

For each principal reciprocal plane, a celestial direction is projected into the plane and compared axially with the projected rays of all nonzero Cell3 nodes. The axial comparison identifies a direction and its antipode, matching the centrosymmetric structure of the 27-node cell. Among 19 representative event configurations, nine have both endpoints within one degree of a primary-plane projected node ray and 17 are within two degrees.

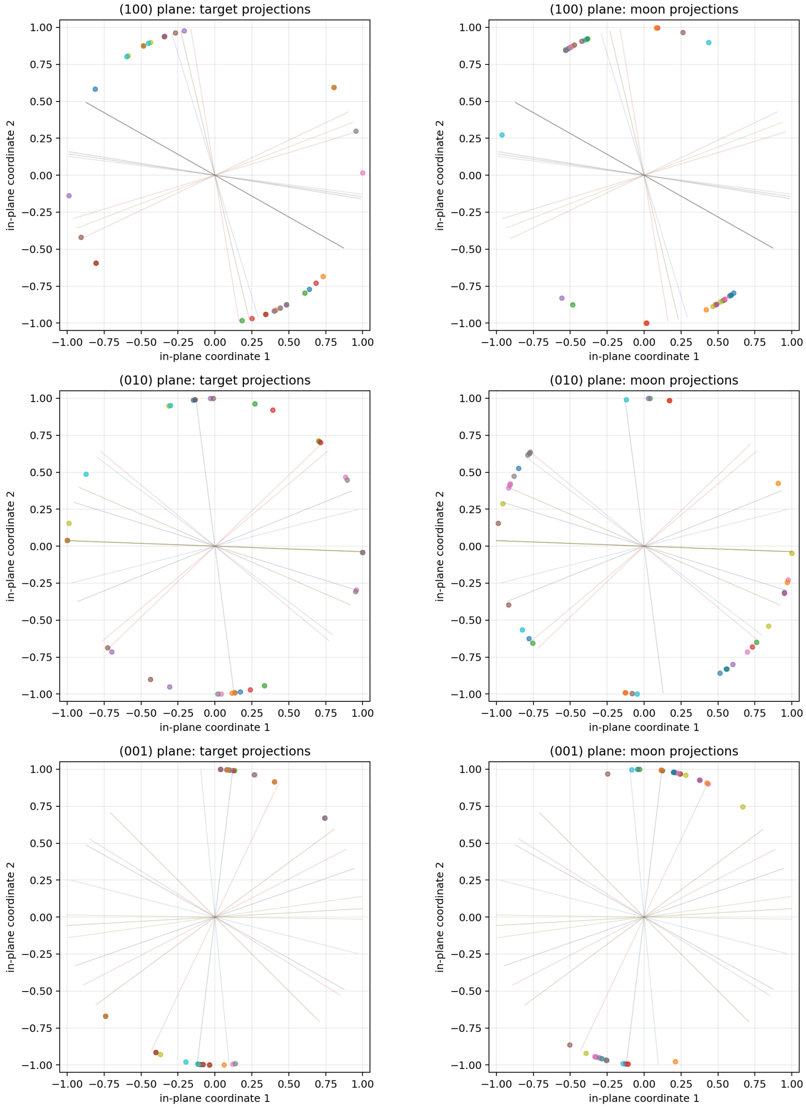

Figure 24. High-similarity anchors projected into the principal Cell3 planes.

Reading guide. Discrete node rays form the projected symmetry registry. The measured event directions cluster near selected rays, including the Galactic-center/anticenter line and several planetary crossings.

8.3 Representative explicit alignments

Table 10. Representative ICRS and projected Cell3 alignments

Event pair	Archived r	Directed target A -> B	Nearest projected ray	Residual A -> B	Recursive closure
Solar midnight	0.949863	RA 165.319, Dec +6.270 -> RA 169.025, Dec +4.718	(001)/(010), N(+1,0,0)	1.992 deg -> 0.778 deg	N=19, (100), 0.012599 cycles
Galactic center	0.983548	Fixed RA 266.405, Dec -28.936	(010), N(0,+1,-1)	0.167936 deg at both anchors	N=20, (110), 0.000205 cycles
Galactic anticenter	0.991085	Fixed RA 86.405, Dec +28.936	(010), N(0,+1,-1)	0.167936 deg	N=21, (100), 0.014642 cycles
Mars advancing	0.992184	RA 99.932,+25.071 -> RA 100.506,+25.024	(010), N(+1,+1,+1)	0.051 deg -> 0.562 deg	N=21, (100), 0.017406 cycles
Mercury retreating	0.955692	RA 156.299,+12.203 -> RA 162.995,+9.464	(001), N(+1,0,0)	0.615 deg -> 0.214 deg	N=19, (101), 0.012075 cycles
Saturn advancing	0.942284	RA 336.973,-11.122 -> RA 337.813,-10.836	(001), N(+1,0,0)	0.220 deg -> 0.219 deg	N=20, (110), 0.015928 cycles

8.4 Angular alignment is a gate, not the complete return condition

The Galactic-center direction attains essentially the same meridian geometry every sidereal day and remains only 0.167936 degrees from the same projected Cell3 ray. Nevertheless, consecutive Galactic-center profiles are usually dissimilar, while isolated pairs reach correlation above 0.98. A fixed angular alignment therefore selects a permitted response sector but does not determine whether the state returns on a particular date.

The missing condition is translational closure. For order N, define q_N = B_N^{-1} r. A complete lattice return is measured by the distance of Delta q_N to the nearest integer triple. A plane-family return is measured by the distance of h^T Delta q_N to the nearest integer. The useful range in this archive is approximately orders 19-22.

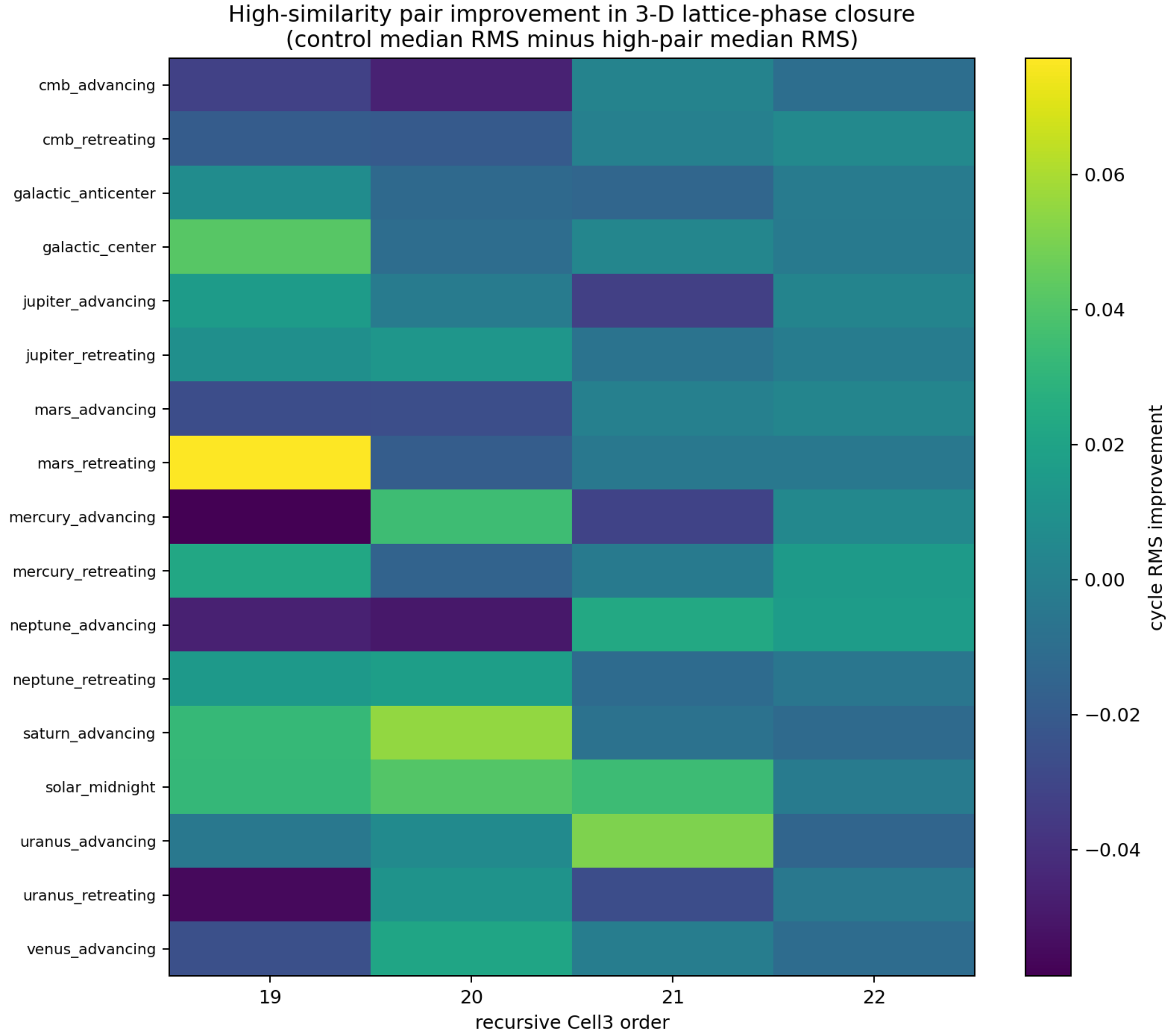

Figure 25. Recursive lattice closure of high-similarity pairs and matched controls.

Reading guide. The high-similarity population improves relative to controls in selected event classes and recursive orders. The result connects episodic waveform return with translational lattice phase.

8.5 The derived symmetry of the system

The Cell3 metric is generic and oblique. Testing signed coordinate permutations against the physical metric leaves only identity and inversion. The exact intrinsic point group is therefore centrosymmetric:

```
C_i = { I, -I },    G_Cell3 = B Z^3 semidirect C_i.
```

The experiment adds an external four-sector Earth-rotation action through direct, +90 degree, antipodal, and -90 degree positions. Together with local mirror reflection H -> -H, the measurement phase has a D4-like organization. The observed system is best represented as a centrosymmetric oblique lattice coupled to a four-sector rotational sampling operation, not as a cubic lattice.

8.6 The complete three-gate return condition

The minimum recurrence model is

```
S_j(t) = F_j[ H_j(t), Phi_EMB(h,N,t), deltaPhi_reflex(h,N,t), deltaPhi_rot(h,N,t), Pi_h v_k(t) ].
```

A high-similarity state requires a compatible angular sector, a compatible EMB lattice phase, and a compatible observer offset. The return may be direct, in which all components recur separately, or compensated, in which the changes sum to an integer plane phase. This model explains why several different celestial labels can return together and why the same nominal transit usually does not reproduce on the following day.

PART V. DEFINE THE SIGNAL OPERATORS AND CONTROLS

9. Signal Operators and Evidence Definitions

9.1 Moving means and moving standard deviations

Moving means isolate slow central tendency, while moving standard deviations convert local fluctuations into a positive variance envelope. The D101 analysis uses raw windows of 31, 121, 301, 601, 1,801, and 3,601 seconds, plus minute-scale windows of 5, 15, 30, 60, 180, 360, and 720 minutes. Each window defines a distinct physical scale. Short windows respond to the 26.84-second carrier and nearby modulation. Long windows respond to broad variance states extending over one or more hours.

No revolution or day averaging is used to erase drift. Event profiles retain the actual chronological structure, and only the minimum smoothing required by each scale is applied.

9.2 Palindrome metric

A crossing-centered profile is palindromic when its pre-event half matches the time-reversed post-event half. For a profile x(H), the primary metric is the correlation between x(-H) and x(+H).

```
r_pal = corr[x(-H), x(+H)]
```

A high value means that the event lies near a mirror center of the variance envelope. It does not require identical peak amplitudes on every day. The analysis therefore also measures aggregate-template recurrence, adjacent-event correlation, archive-half stability, temperature-residualized symmetry, and recurrent peak lags.

Nearby-center optimization is controlled by applying the identical search to displaced event anchors. Smooth signals can produce apparently perfect local symmetry when the center is selected after inspection. Only exact-center or controlled shifted-center results are retained as crossing evidence.

9.3 Chirp metric

A chirp is a moving frequency-selective ridge or notch. Broadband amplitude changes are removed before ridge tracking. The fast tier covers periods of 2 through 512 seconds over a plus or minus two-hour interval. The long tier covers 2 minutes through 4 hours over a plus or minus six-hour interval. A power-notch chirp is a moving suppression band rather than a power enhancement.

The ridge path is fitted in logarithmic frequency or angular wavelength. The fit supplies direction, span in octaves, slope, and R-squared. Each exact event is compared with displaced anchors processed by the same dynamic-programming path search.

9.4 CWT derivatives

The continuous wavelet transform provides localized power as a function of crossing phase H and logarithmic angular wavelength. Three derivative fields are important: the phase derivative, the scale derivative, and the mixed derivative.

```
P = P(H, log2(lambda_H))

dP/dH
dP/dlog2(lambda_H)
d2P/[dH dlog2(lambda_H)]
```

The mixed derivative is especially sensitive to chirps, reversals, ridge branching, and moving notches. Scale-dependent edge regions are removed, and each scale is robustly normalized before derivative comparison.

9.5 FFT derivatives and two-dimensional FFT

Event profiles are assembled into a matrix X(n,H), where n is event number and H is crossing phase. The two-dimensional FFT separates angular structure within one event from recurrence across successive events.

```
F(k_n,k_H) = FFT2{X(n,H)}
```

Nonzero event-axis frequency means that a structure alternates, evolves, or returns every several events. FFT differentiation supplies first and second phase derivatives without finite-difference phase error.

```
FFT{dX/dH} = i k_H F
FFT{d2X/dH2} = -k_H^2 F
```

9.6 Control hierarchy

The evidence hierarchy includes displaced-event controls, temperature residualization, archive-half stability, leave-one-event-out templates, adjacent-event correlation, exact versus shifted centers, UTC versus hour-angle coordinates, and correction for duplicated celestial sectors. A crossing-conditioned profile is reported even when it does not establish unique causal attribution to one astronomical body. The geometric state and the body label are kept distinct when event schedules overlap.

10. Global Recurrence and the Absence of One Invariant Daily Waveform

Archive-wide autocorrelation does not show one complete waveform repeating every solar, sidereal, or lunar day. The largest moving-standard-deviation correlation anywhere from 1,320 to 1,540 minutes is only approximately 0.026. Exact reference-period correlations remain near zero.

Table 11. Global daily-lag recurrence

Reference period	Lag	Observed correlation range
Sidereal day	1,436 min	approximately -0.021 to +0.011
Solar day	1,440 min	approximately -0.023 to +0.010
Lunar day	1,490 min	approximately -0.033 to +0.017

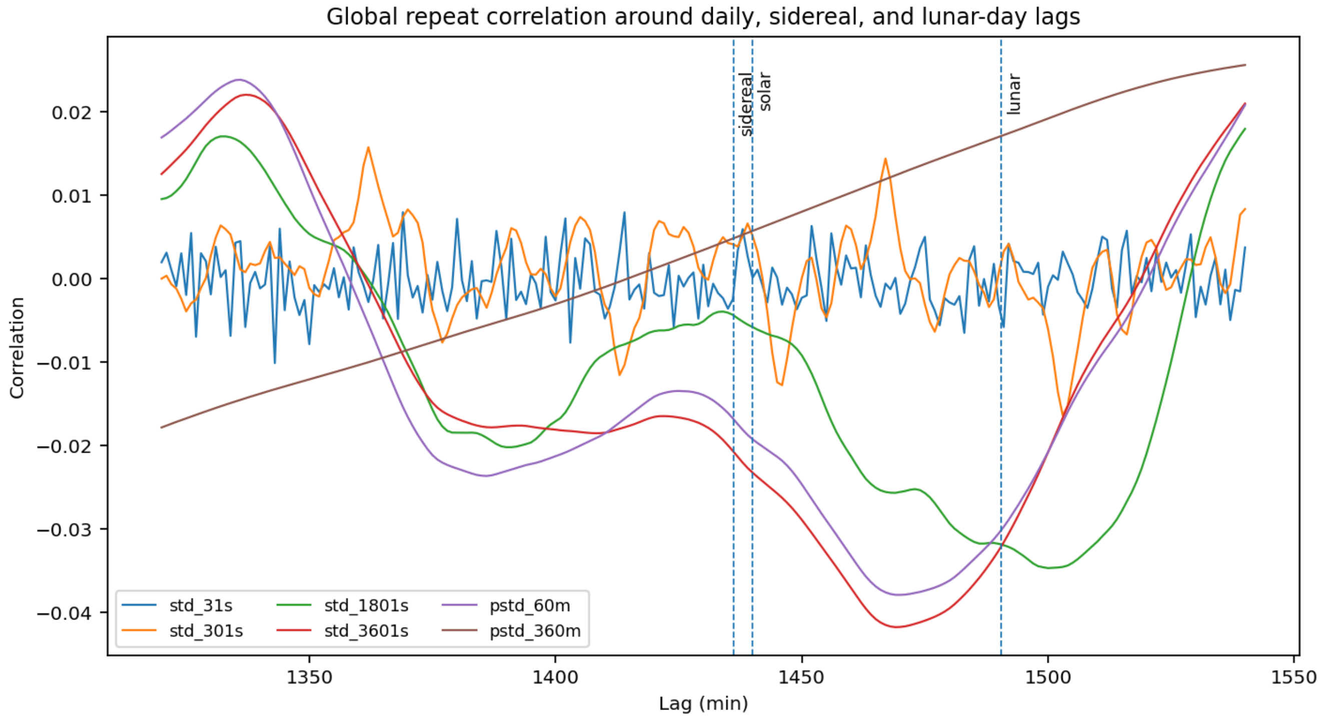

Figure 26. Daily, sidereal, and lunar lag scan.

Reading guide. No moving-standard-deviation channel contains a strong archive-wide repeat at exactly one day. Crossing evidence is therefore event-conditioned rather than a single fixed daily sequence.

This result is essential. The D101 archive repeatedly enters certain symmetry and chirp classes near defined celestial phases, but the detailed peak sequence changes. The two-dimensional FFT shows that dominant recurrence often occurs every two to eight crossings, with alternation, phase reversal, or episodic blocks.

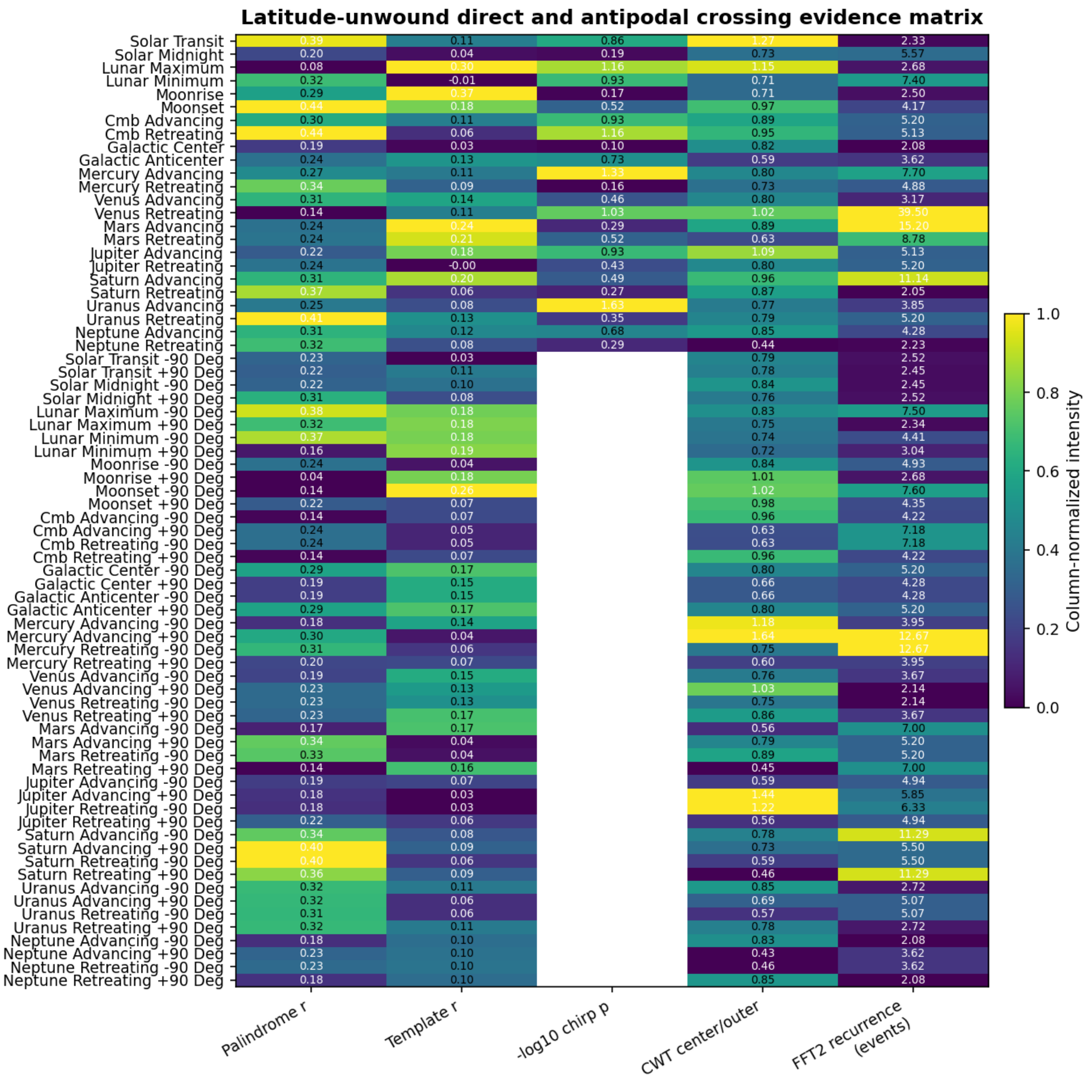

Figure 27. Consolidated direct and antipodal crossing evidence.

Reading guide. The matrix combines palindromic symmetry, template recurrence, controlled chirp evidence, CWT center concentration, and two-dimensional recurrence interval for all 24 direct and antipodal event classes.

PART VI. EXPLAIN THE MEASURED SIGNAL FORMS

11. Palindromic Repeat Sequences and Event-to-Event Morphology

11.1 Symmetry class versus identical sequence

The central palindrome result is statistical but physically specific: individual crossing-centered profiles frequently become mirror-like, while their exact internal peaks change. Solar transit, solar midnight, lunar lower culmination, CMB retreating, Galactic anticenter, Mercury retreating, Saturn transits, Uranus retreating, and Moonset all exhibit this separation between envelope symmetry and weak adjacent-event waveform recurrence.

For solar midnight, the six-hour mirror correlation is 0.4256, while adjacent-event correlation is 0.007. For CMB retreating, the broad palindrome is approximately 0.43, while adjacent-event recurrence is negative or near zero. The event is therefore a recurring symmetry condition, not a digital copy of the preceding day.

11.2 Episodic high-similarity pairs

Several isolated profile pairs correlate above 0.90 at separations of days to months. Examples occur around solar midnight, Galactic center, Galactic anticenter, and planetary events. Because the median consecutive-event correlation remains near zero, these pairs define episodic returns of a signal state rather than uniform daily recurrence.

11.3 Derivative parity

A palindromic envelope does not require palindromic first and second derivatives in the aggregate template. Internal peaks can change sign, split, merge, or shift while the outer variance envelope remains symmetric. Strong negative derivative-mirror correlations at Moonset and Galactic anticenter therefore identify changing internal morphology within a stable event-centered envelope.

12. Raw and Angular Chirp Evidence Across All Crossings

12.1 Directly observable bands

The fast tier covers 2 through 512 seconds. The long tier covers 2 minutes through 4 hours. The high-frequency request extending to 2 Hz cannot be directly satisfied by a one-second archive; only the 0 to 0.5 Hz baseband is uniquely observed. The analysis therefore reports actual measured baseband ridges and explicitly excludes claims that require unaliased content above Nyquist.

12.2 Principal transit-centered chirps

Table 12. Principal crossing-centered chirps

Event	Tier	Structure	Trajectory	Audit
Solar transit	Fast	Segmented power ridge	approximately 46 s toward longer periods	p=0.0256 temporal
Solar midnight	Fast	Power-notch up-chirp	256 s -> 73 s -> 32 s	paired-event p=0.0209
Jupiter advancing	Long	Power-notch up-chirp	4.27 h -> 57 min	p=0.0435 temporal
Uranus advancing	Long	Power-notch up-chirp	4.27 h -> 28 min	p=0.0435 temporal; p=0.0233 angular
Mercury advancing	Long angular	Power-notch up-chirp	64.2 deg -> 24.3 deg -> 14.0 deg	p=0.0465
CMB retreating	Long angular	Frequency-trough trajectory	48.6 deg -> 32.1 deg -> 14.0 deg	p=0.0698
Galactic center	Fast	Frequency trough	20 s -> 64 s -> 57 s	p=0.103

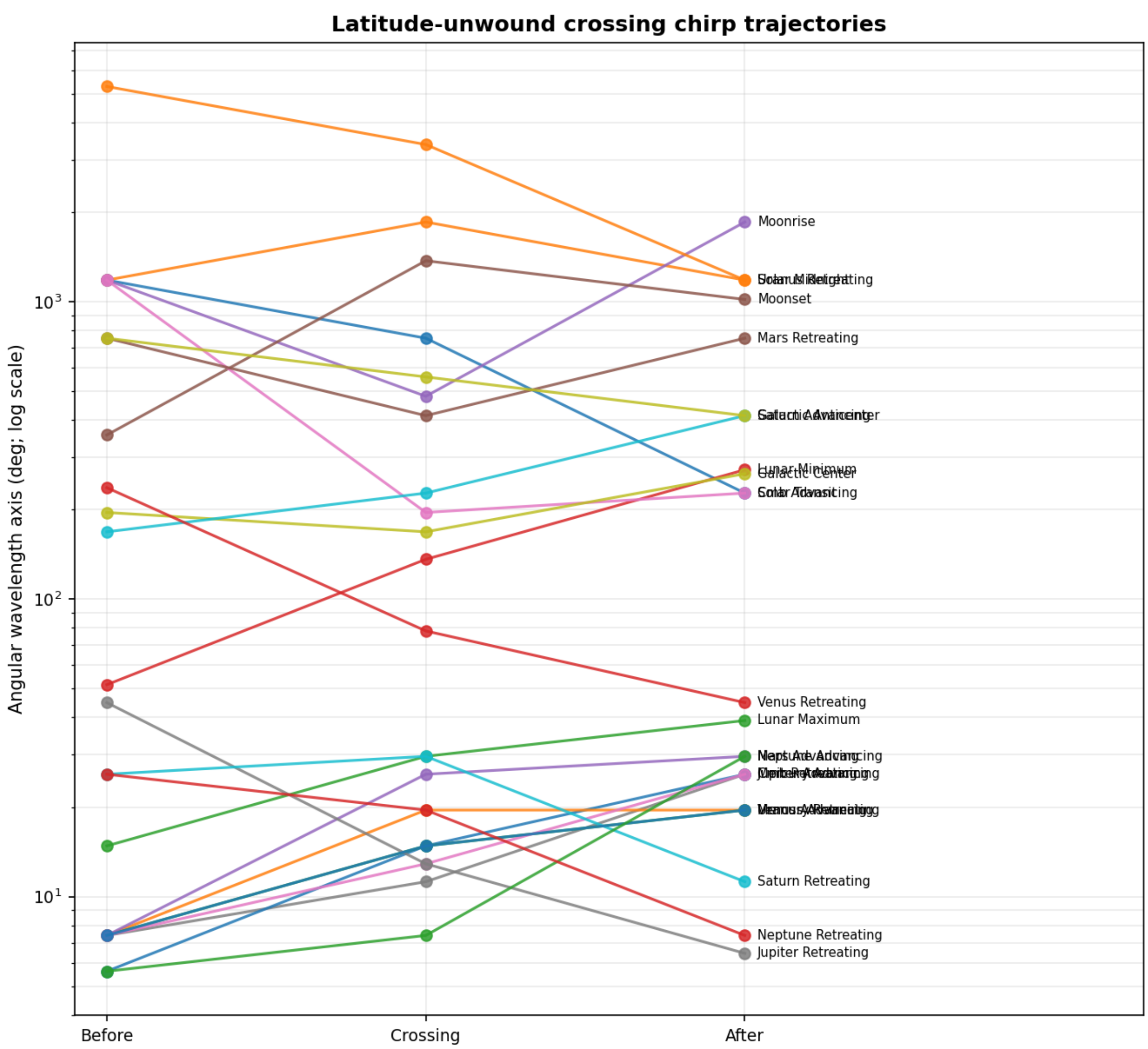

Figure 28. Angular-wavelength chirp atlas.

Reading guide. Temporal chirps are converted to angular wavelength using the target-relative hour-angle rate. The transformation separates intrinsic angular motion from target kinematics.

12.3 Carrier harmonic coherence

Table 13. Events with common carrier-ladder drift direction

Event	Median drift oct/h	1x	2x	4x	8x
Cmb Advancing	0.0244	0.0198	0.0289	0.0427	0.0138

Event	Median drift oct/h	1x	2x	4x	8x
Saturn Advancing	-0.0128	-0.0031	-0.0190	-0.0088	-0.0168
Solar Transit	-0.0186	-0.0090	-0.0027	-0.0282	-0.0571
Uranus Retreating	-0.0203	-0.0266	-0.0464	-0.0141	-0.0019

Common-sign drift across four harmonics is a high-specificity signature. It rejects the interpretation that the observed motion is merely the wandering of one spectral bin. The coherent event classes are CMB advancing, solar transit, Saturn advancing, and Uranus retreating, with additional orthogonal carrier-ladder classes at Galactic anticenter -90 degrees and Venus advancing -90 degrees.

13. Continuous Wavelet Transform and Derivative Evidence

13.1 Why derivatives matter

A conventional scalogram emphasizes power. Crossing events often appear more clearly in derivatives because a moving ridge, a moving notch, and a reversal all create localized gradients in phase and scale. The mixed derivative suppresses stationary bands and emphasizes spectral motion.

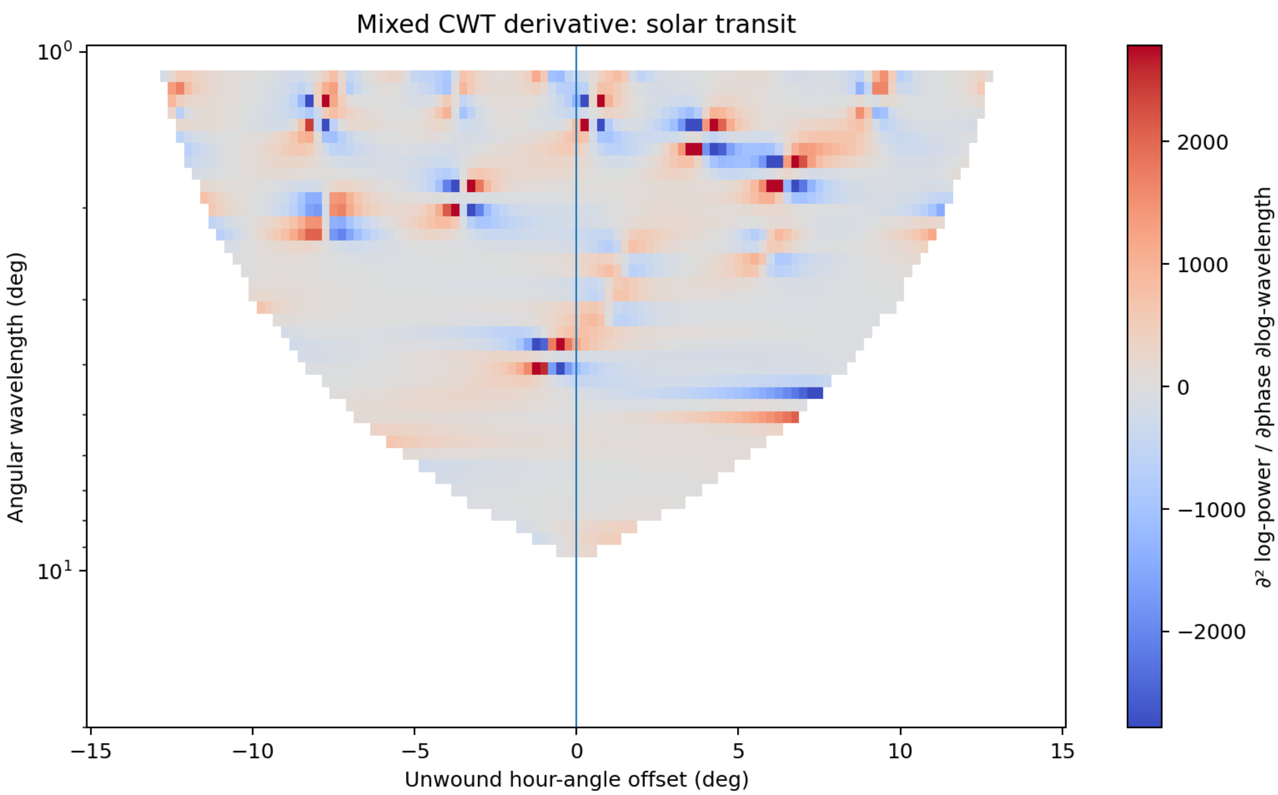

Figure 29. Latitude-unwound CWT mixed derivative.

Reading guide. The derivative field isolates scale-phase transitions that are obscured in the raw power map. Scale edges are removed and each scale is robustly normalized.

13.2 Principal CWT derivative events

Table 14. Strongest CWT mixed-derivative concentrations

Event	Channel	Span deg	Ridge span oct	Ridge R2	Center/outer	Dominant order
Mercury Advancing +90 Deg	std_3601s	15	2.883	0.0005	1.645	35.70
Jupiter Advancing +90 Deg	pstd_180m	15	3.350	0.0393	1.439	47.60

Event	Channel	Span deg	Ridge span oct	Ridge R2	Center/outer	Dominant order
Solar Transit	std_3601s	15	3.428	0.0036	1.269	35.70
Jupiter Retreating -90 Deg	pstd_180m	15	3.350	0.0287	1.222	47.60
Mercury Advancing -90 Deg	std_1801s	15	3.350	8.7771e-06	1.180	59.50
Lunar Maximum	pstd_360m	15	3.584	0.0021	1.153	83.31
Jupiter Advancing	pstd_180m	15	3.350	0.0424	1.088	71.40
Venus Advancing +90 Deg	std_3601s	15	3.350	0.0048	1.034	47.60
Moonset -90 Deg	pstd_60m	15	3.350	0.0460	1.024	23.80
Venus Retreating	std_3601s	15	2.649	0.0434	1.020	47.60
Moonrise +90 Deg	pstd_360m	15	4.285	0.0691	1.011	142.81
Moonset +90 Deg	pstd_360m	45	5.016	0.0040	0.978	27.92

Solar transit has the strongest event-centered derivative concentration, 1.269, but ridge R-squared 0.0036. Lunar upper culmination reaches 1.153 with ridge R-squared 0.0021. Jupiter advancing reaches 1.088 with R-squared 0.0424. These values establish that the visible structures are segmented, branching, or reversal-type chirps, not single straight lines.

13.3 CWT interpretation

A broad variance palindrome and a non-palindromic internal CWT sequence can coexist. The moving-standard-deviation envelope reports the outer energy state; the CWT derivative reports how energy transfers among wavelengths inside that state. This distinction explains why strong event-centered palindromes can have weak or negative derivative parity.

14. FFT Derivatives and Two-Dimensional Event Recurrence

14.1 Event-by-phase matrices

For every event class, standardized profiles are arranged in chronological order. The horizontal axis is unwound crossing phase, and the vertical axis is event number. A two-dimensional FFT then measures angular periodicity within a crossing and recurrence across crossings. This operator directly tests whether a pattern repeats every event, alternates, or returns after several events.

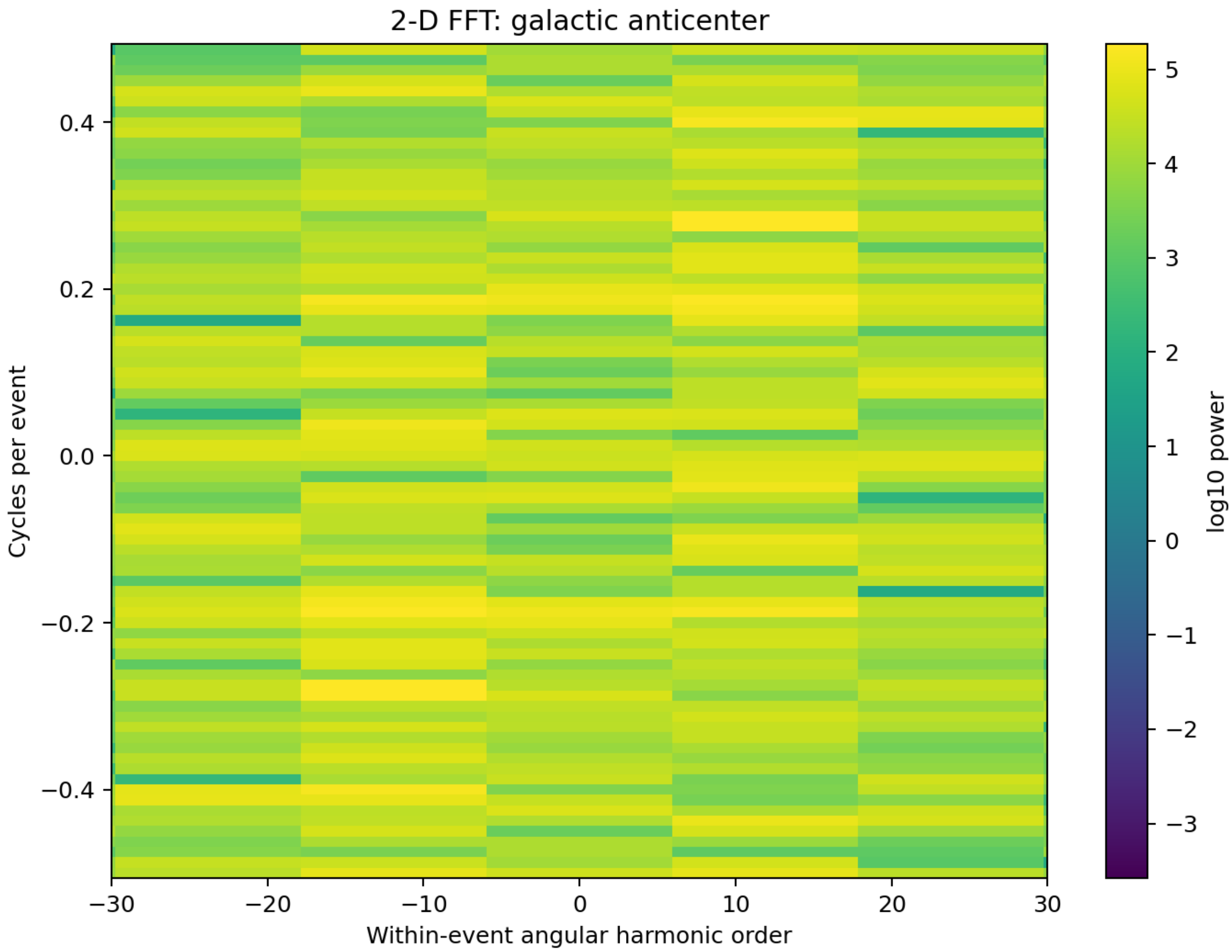

Figure 30. Two-dimensional FFT of a crossing-profile matrix.

Reading guide. Horizontal frequency encodes angular order within the event. Vertical frequency encodes recurrence across event number. Dominant power away from zero vertical frequency demonstrates multi-event structure.

14.2 Representative recurrence modes

Table 15. Representative 2-D FFT recurrence modes

Event	Angular order	Wavelength deg	Cycles/event	Recurrence events	Top10 fraction	Entropy
Solar Transit	11.90	30.25	0.4286	2.33	0.120	0.653
Galactic Anticenter	11.90	30.25	0.2763	3.62	0.139	0.636
Moonset	2.00	180.25	0.2400	4.17	0.114	0.556
Cmb Retreating	3.99	90.25	0.1948	5.13	0.118	0.578
Uranus Retreating	2.00	180.25	0.1923	5.20	0.135	0.549
Mercury Retreating	2.00	180.25	0.2051	4.88	0.117	0.552
Saturn Retreating	11.90	30.25	0.4872	2.05	0.132	0.651

14.3 Recurrence classes

Three recurrence forms dominate. Alternation produces power near one-half cycle per event and corresponds to polarity reversal on successive crossings. Multi-event recurrence produces a nonzero event-axis line corresponding to a return every three to six events. Episodic blocks appear as broader event-axis power and correspond to several similar events followed by a regime change.

Solar transit has a dominant within-event wavelength near 30.25 degrees and recurrence interval 2.33 events. Galactic anticenter returns every 3.62 events. Moonset has a broad 180.25-degree mode and recurrence interval 4.17 events. CMB retreating and Uranus retreating recur near five events. These values explain why archive-wide one-day autocorrelation is weak even when event-centered structure is strong.

14.4 FFT derivative parity

First and second phase derivatives are calculated spectrally. Their mirror correlations are often negative even when the variance envelope is positive and palindromic. This is not a contradiction. A symmetric outer envelope may contain internal peaks whose slopes reverse, split, or move. The derivative field records morphology; the moving standard deviation records energy distribution.

PART VII. EXAMINE DIRECT, ANTIPODAL, AND ORTHOGONAL CROSSINGS

15. Solar and Lunar Crossing Evidence

15.1 Consolidated direct-event table

Table 16. Solar and lunar crossing evidence

Crossing	Variance scale	Mirror r	Chirp evidence	FFT2 wavelength	Recurrence
Solar upper transit	3,601-s movstd; +/-15 deg	0.395	Fast segmented down structure; temporal template p=0.0256	30.25 deg	2.33 events
Solar lower transit	360-min movstd; +/-360 min	0.426	Fast power-notch up-chirp; paired-event p=0.0209	90.25 deg	5.57 events
Lunar upper culmination	360-min movstd; +/-15 deg	0.077	Long up-chirp; 24.3 -> 12.2 -> 9.21 deg	30.25 deg	2.68 events
Lunar lower culmination	180-min movstd; +/-180 min	0.352	Long up-chirp; broad lower-culmination envelope	90.25 deg	7.40 events
Moonrise	360-min movstd; +/-45 deg	0.290	Fast nonlinear frequency trough	90.25 deg	2.50 events
Moonset	720-min movstd; +/-90 deg	0.440	Fast frequency peak; phase-registration gain +0.123	180.25 deg	4.17 events

15.2 Solar upper transit

Solar upper transit contains a one-hour variance-envelope palindrome. The original exact-center result in the 3,601-second moving standard deviation is $r = 0.3969$ with displaced-event $p = 0.0396$. The latitude-unwound value is 0.3945, essentially unchanged from UTC stacking. This invariance is expected because the Sun moves slowly in right ascension over a single event span.

The fast spectrogram contains the strongest aggregate solar-transit chirp contrast in the original temporal analysis, with $p = 0.0256$. The path branches and changes slope. A single linear fit explains only a small part of the full ridge. After angular normalization, the aggregate chirp statistic contracts, showing that part of the temporal sweep is kinematic. The CWT mixed derivative nevertheless becomes more concentrated at the crossing, with center-to-outer ratio 1.269. Thus the event is better described as a crossing-centered branching scale transition than as one straight chirp.

The 26.84-second carrier and its 2x, 4x, and 8x components all drift downward in fractional frequency around solar transit. The median drift is -0.0186 octaves per hour. Coherent motion across the ladder demonstrates modulation of a shared carrier state.

15.3 Solar lower transit

Solar lower culmination, or local apparent solar midnight, contains a broader palindrome than solar noon. The strongest original result uses a 360-minute moving standard deviation over a plus or minus six-hour span: $r = 0.4256$ with $p = 0.0249$. The first and second archive halves are 0.4221 and 0.4659, and temperature residualization increases the value to 0.4395. The broad envelope is therefore stable and is not explained by the measured temperature channel.

The fast-band event is a power-notch chirp. Its characteristic period moves from approximately 256 seconds before midnight toward 73 seconds near the crossing and approximately 32 seconds afterward. Among 76 usable individual crossings, 64.5 percent exceed matched displaced windows, with paired-event $p = 0.0209$. The median peak chirp energy occurs approximately three minutes after lower culmination.

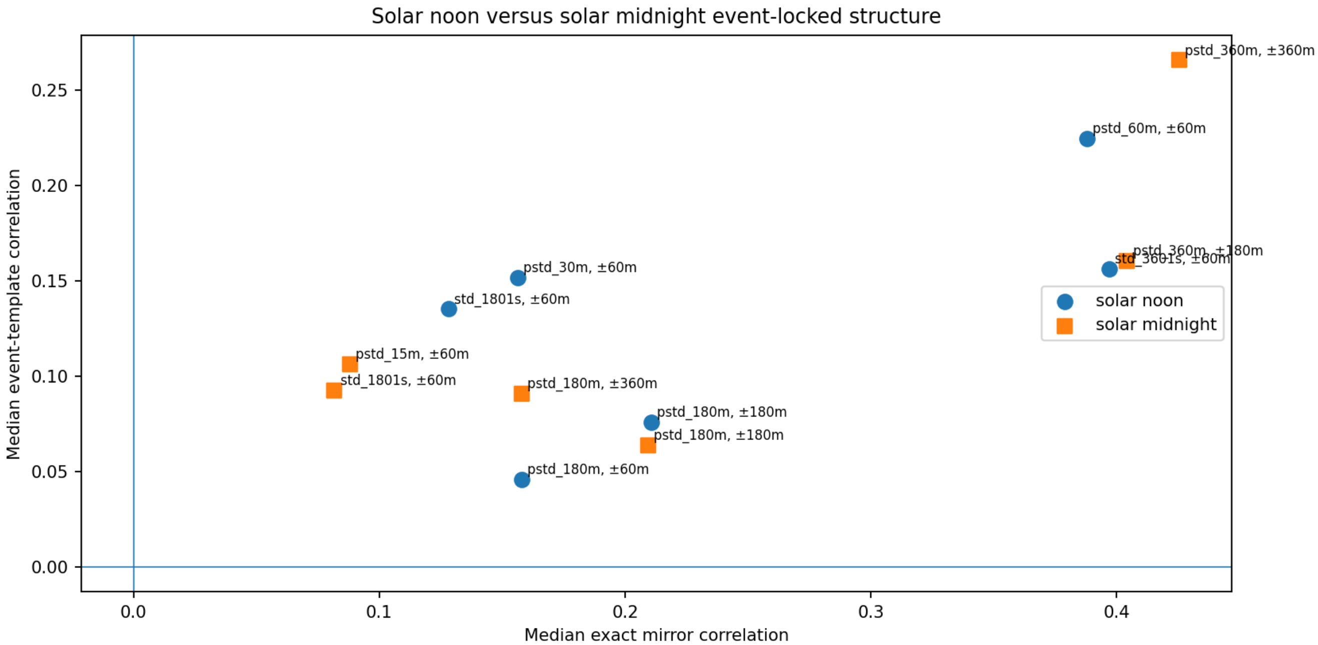

Figure 31. Solar upper and lower culmination variance envelopes.

Reading guide. Solar noon is strongest near a one-hour scale. Solar midnight carries a broader three-to-six-hour mirror envelope. Neither event reproduces an identical daily internal peak train.

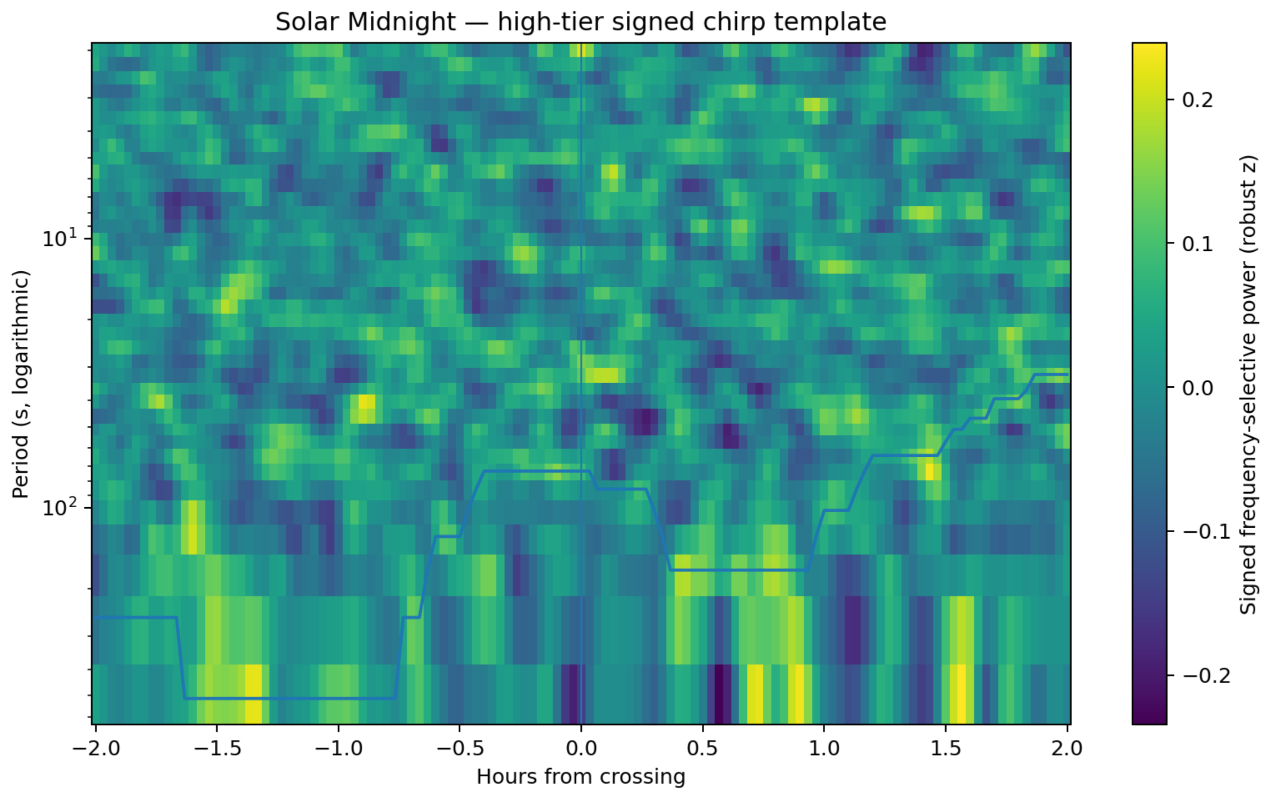

Figure 32. Solar-midnight fast power-notch chirp.

Reading guide. The moving spectral suppression shifts toward shorter periods through the lower solar culmination. The event is distinct from the broad moving-standard-deviation palindrome.

15.4 Lunar upper and lower culmination

The strongest lunar palindrome is centered on lower culmination, the local altitude minimum. The original 180-minute moving-standard-deviation profile over plus or minus 180 minutes gives r = 0.3518 and p = 0.0495. Upper culmination is weaker, with r = 0.1669 and p = 0.158. The asymmetry between the two directed lunar sides is therefore a repeated feature of the archive.

The chirp hierarchy differs from the palindrome hierarchy. Lunar upper culmination contains a near-controlled long-period angular chirp from approximately 24.3 degrees toward 12.2 and 9.21 degrees, with R-squared 0.849 and p = 0.0698. Its CWT mixed derivative center-to-outer ratio is 1.153, while the ridge R-squared is only 0.0021. The event contains a localized multiscale transition with branching rather than one simple ridge.

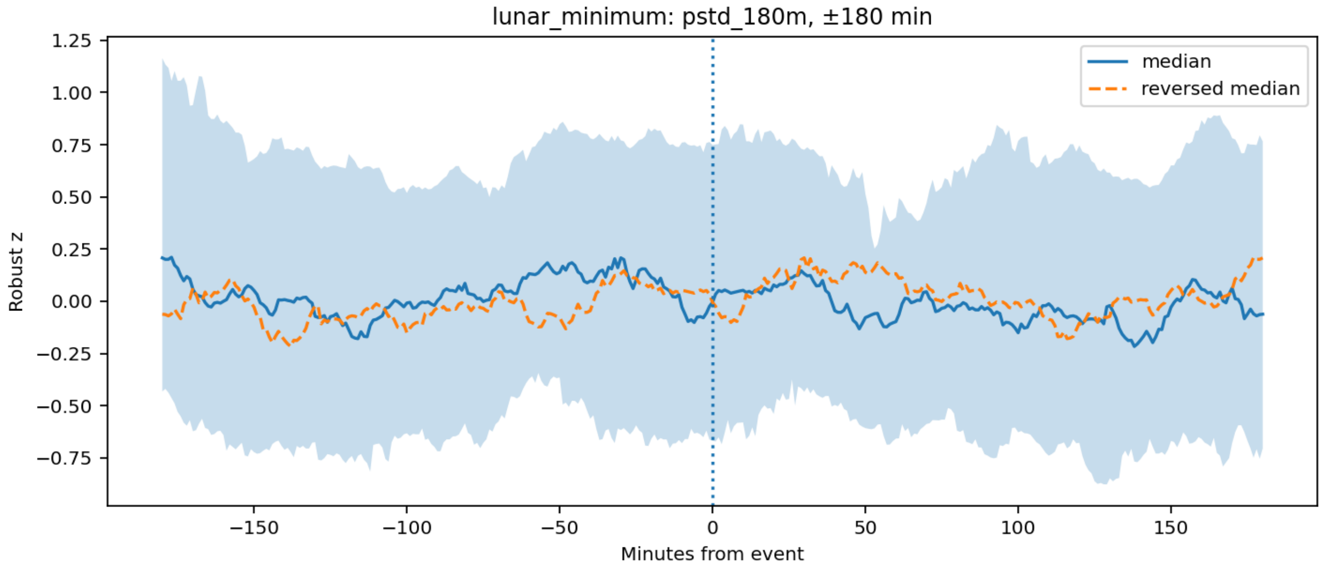

Figure 33. Lunar lower-culmination palindrome.

Reading guide. The broad variance envelope is centered on the Moon altitude minimum. The detailed peaks change from event to event, but the mirror class recurs.

15.5 Lunar horizon crossings and the importance of hour-angle registration

Moonset is the clearest demonstration that the latitude and target-motion unwind recovers physical coherence. Its mirror correlation increases from 0.3176 in UTC coordinates to 0.4403 in lunar hour angle, with p = 0.0465. Moonrise increases from 0.2202 to 0.2901. The improvement follows from removing the Moon's nonuniform right-ascension motion and expressing the event in a common angular coordinate.

Moonset has a dominant two-dimensional FFT wavelength of 180.25 degrees and recurrence interval 4.17 events. It is therefore a broad multi-event state rather than a waveform that repeats every lunar day.

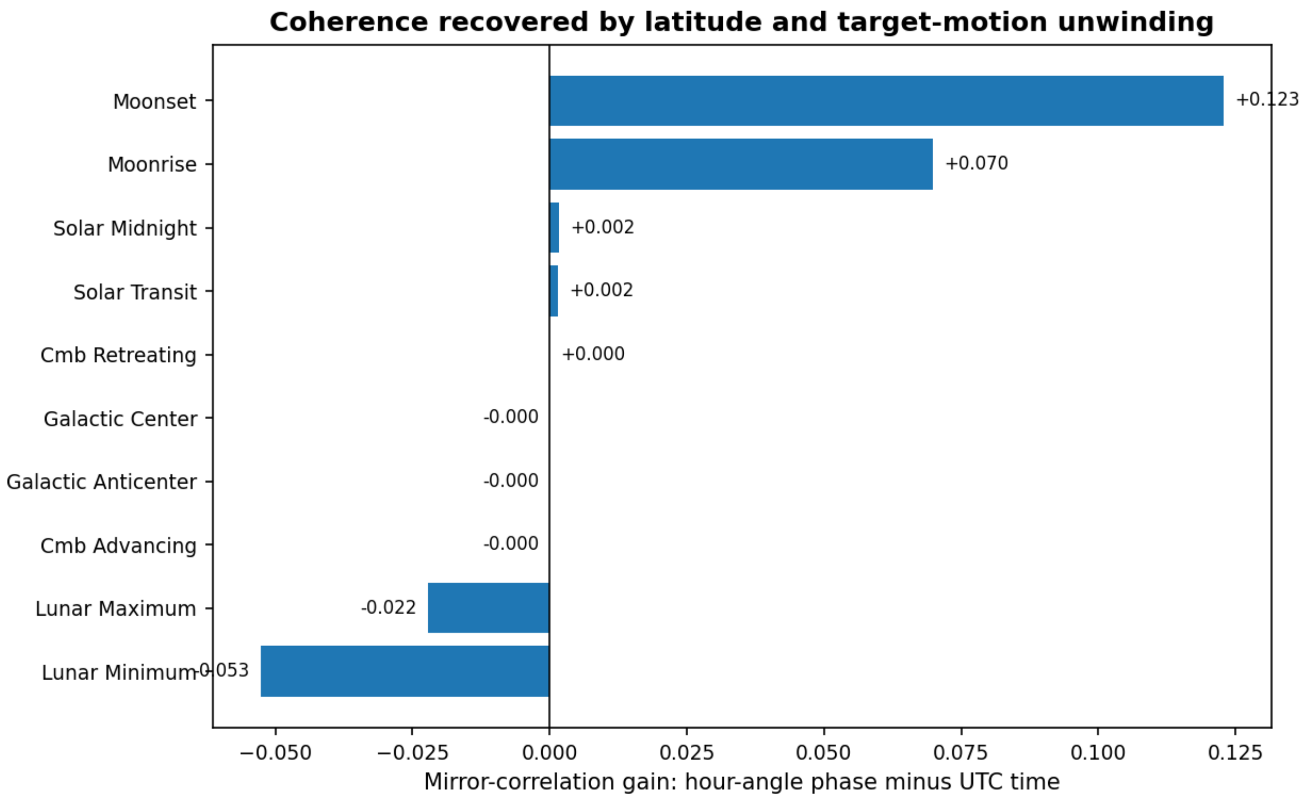

Figure 34. Event registration changes after latitude unwinding.

Reading guide. The large Moonset and Moonrise gains identify target-motion smearing in the UTC-centered profiles. Fixed axes remain invariant.

16. CMB and Galactic-Axis Crossing Evidence

16.1 Consolidated fixed-vector table

Table 17. Fixed-axis evidence summary

Directed crossing	Upper altitude	Primary profile	Control	Chirp/carrier structure	FFT2 recurrence
CMB advancing	64.50 deg	0.353 at one-hour scale	p=0.0579	Carrier ladder up, +0.0244 oct/h	5.20 events
CMB retreating	31.24 deg	0.430 original; 0.439 unwound	p=0.0083 original; 0.0233 unwound	48.6 -> 32.1 -> 14.0 deg trough	5.13 events
Galactic center	18.93 deg	Recurring asymmetric profile	template p=0.0248	Fast frequency trough near crossing	2.08 events
Galactic anticenter	76.81 deg	0.416 broad; 0.379 one-hour	p=0.0496 and 0.0413	Weak fast up-chirp candidate	3.62 events

16.2 CMB advancing

The advancing CMB side contains a weaker one-hour palindrome and an archive-dependent broad envelope. The one-hour moving-standard-deviation result is $r = 0.353$ with $p = 0.0579$ and remains present in both archive halves. The broad 720-minute result reaches $r = 0.421$ but falls from 0.639 in the first half to 0.129 in the second. The shorter scale is therefore the more stable advancing-side signature.

All four tracked carrier harmonics drift upward at this crossing, with median +0.0244 octaves per hour. The long-tier angular chirp statistic increases under coordinate normalization, but the displaced-anchor result is not isolated. This is evidence of phase concentration without a unique event-specific chirp detection.

16.3 CMB retreating

CMB retreating is the strongest fixed-vector palindrome in the archive. The original 360-minute moving-standard-deviation envelope over plus or minus six hours gives r = 0.4301 with p = 0.0083. An independent 180-minute variance scale also exceeds controls, with r = 0.2484 and p = 0.0413. The latitude-unwound standardized result is r = 0.4386 with p = 0.0233. Because the CMB direction is fixed, the near identity before and after unwinding is an important internal validation.

Its angular chirp is a broad frequency-trough trajectory from approximately 48.6 degrees through 32.1 degrees to 14.0 degrees, with R-squared 0.801 and p = 0.0698. The complete CWT spans approximately 5.23 octaves but has very low single-ridge linearity, showing that the principal path is embedded in a more complex branching structure. The two-dimensional FFT recurrence interval is 5.13 events.

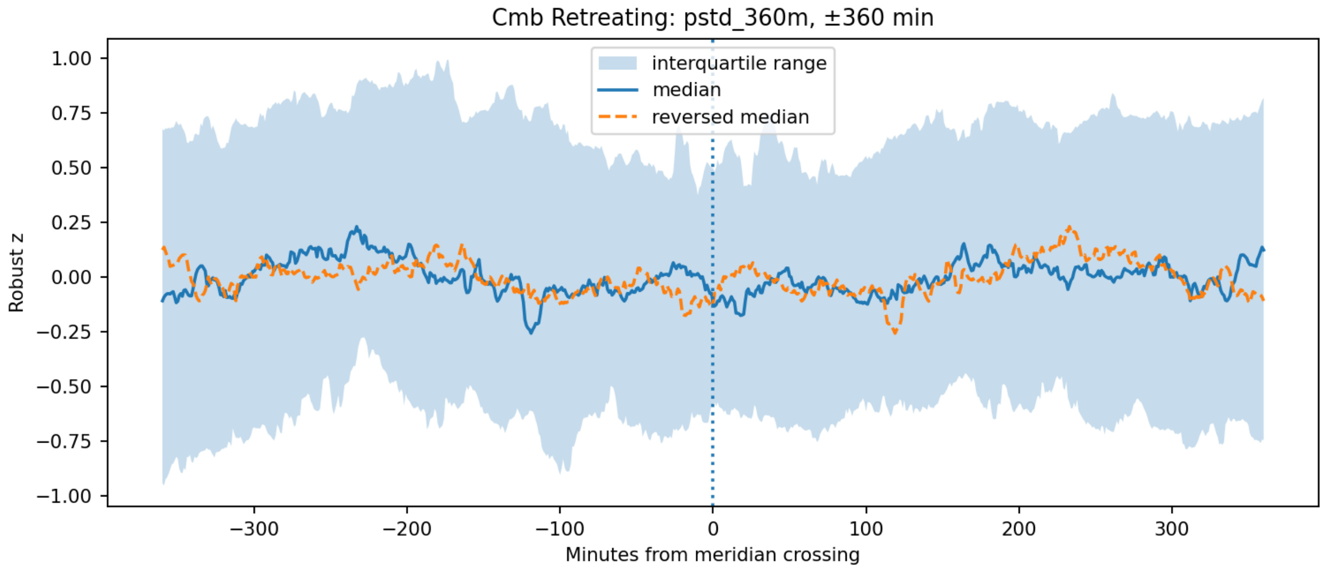

Figure 35. CMB retreating event-centered palindrome.

Reading guide. The broad moving-standard-deviation envelope is stable across the archive and survives temperature residualization and displaced-anchor controls.

16.4 Galactic center

The Galactic-center crossing is not primarily palindromic. Its broad mirror result does not exceed controls, but a different 360-minute channel produces a recurring asymmetric profile with cross-validated template correlation 0.311 and p = 0.0248. The mirror correlation of that profile is negative. Its orientation changes from +0.405 in the first archive half to -0.381 in the second, identifying a phase or polarity reversal rather than loss of structure.

The fast-band spectrogram contains a frequency trough that moves from approximately 20 seconds toward 64 and 57 seconds around the crossing. The path has R-squared 0.703 and p = 0.103. Episodic profile pairs reach correlations above 0.97, but adjacent-event recurrence remains near zero.

16.5 Galactic anticenter

The Galactic anticenter contains both broad and one-hour palindromes. The original broad result is r = 0.416 with p = 0.0496; the one-hour result is r = 0.379 with p = 0.0413. The standardized unwound control selects a more conservative profile with r = 0.2389 and p = 0.0233. Its dominant two-dimensional FFT wavelength is 30.25 degrees with recurrence every 3.62 events.

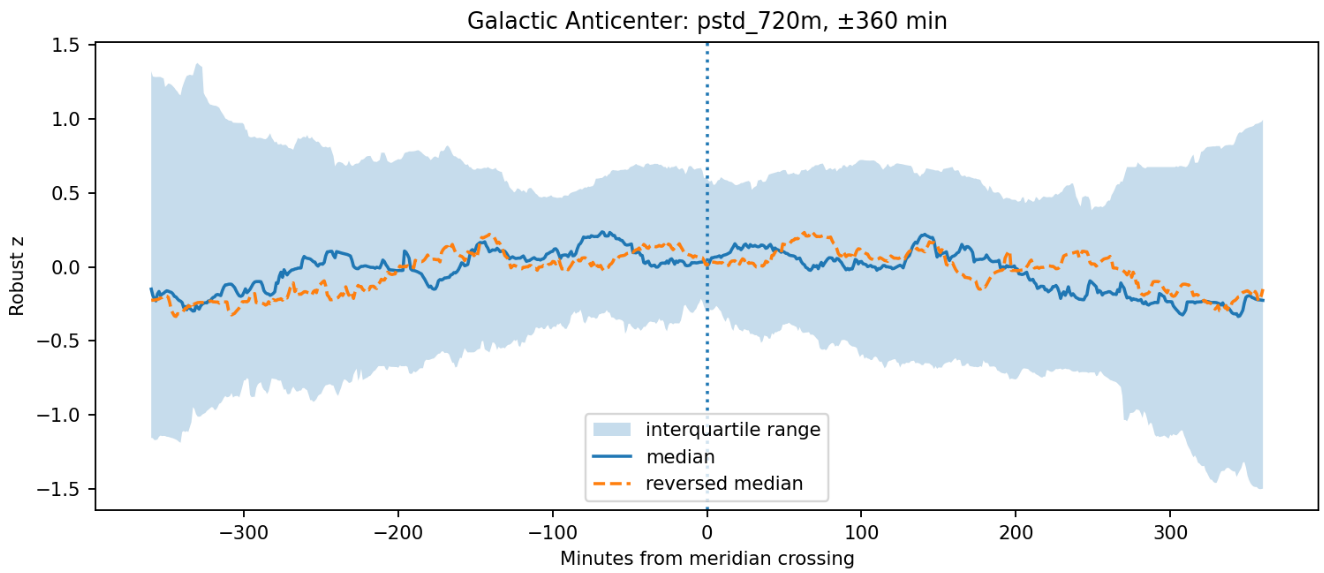

Figure 36. Galactic anticenter palindrome.

Reading guide. The anticenter side contains event-centered symmetry, while the center side preferentially contains a recurring asymmetric sequence.

16.6 Opposite-side reversal geometry

Aggregate advancing and retreating templates often become more similar when the retreating profile is reversed in time. The CMB aggregate reaches approximately 0.720 after reversal, and the Galactic aggregate reaches approximately 0.627. Same-day paired profiles remain much weaker. The relation is therefore a population-level directed geometry whose detailed realization changes between crossings.

17. Planetary Direct and Antipodal Crossing Evidence

17.1 Complete planetary evidence table

Table 18. Planetary direct and antipodal evidence

Planet	Side	Best variance	Span min	Mirror r	Pal p	Template r	Angular chirp trajectory	Chirp p
Jupiter	Advancing	pstd_360m	180	0.288	0.220	0.041	7.40 -> 12.89 -> 25.78 deg	0.116
Jupiter	Retreating	pstd_360m	360	0.356	0.122	0.057	44.88 -> 12.89 -> 6.44 deg	0.372
Mars	Advancing	pstd_720m	360	0.376	0.244	0.460	7.40 -> 25.78 -> 29.61 deg	0.512
Mars	Retreating	pstd_180m	180	0.337	0.098	0.455	753.24 -> 413.60 -> 753.24 deg	0.302
Mercury	Advancing	pstd_60m	60	0.292	0.146	0.171	5.61 -> 14.80 -> 25.78 deg	0.047
Mercury	Retreating	pstd_360m	360	0.519	0.024	0.354	7.40 -> 19.53 -> 19.53 deg	0.698
Neptune	Advancing	pstd_180m	180	0.328	0.122	0.100	5.61 -> 7.40 -> 29.61 deg	0.209
Neptune	Retreating	pstd_720m	360	0.378	0.098	0.090	25.78 -> 19.53 -> 7.40 deg	0.512
Saturn	Advancing	std_3601s	60	0.342	0.073	0.205	753.24 -> 558.16 -> 413.60 deg	0.326
Saturn	Retreating	std_1801s	60	0.387	0.024	0.110	25.78 -> 29.61 -> 11.22 deg	0.535
Uranus	Advancing	std_3601s	60	0.401	0.073	0.098	7.40 -> 14.80 -> 19.53 deg	0.023
Uranus	Retreating	pstd_360m	360	0.389	0.024	0.067	5285.31 -> 3371.38 -> 1180.85 deg	0.442
Venus	Advancing	pstd_720m	360	0.448	0.024	0.614	7.40 -> 14.80 -> 19.53 deg	0.349
Venus	Retreating	pstd_30m	60	0.209	0.171	0.278	236.86 -> 78.14 -> 44.88 deg	0.093

17.2 Mercury

Mercury retreating produces the largest original planetary palindrome: r = 0.5188 with p = 0.0244. It is balanced across archive halves and increases slightly after temperature residualization. The event occurs near solar midnight during this archive, with mean solar hour angle approximately 12.15 hours and concentration 0.978. The observed structure is therefore a real crossing sector, but Mercury-specific attribution is not independent of the solar-midnight geometry.

Mercury advancing contains the second strongest controlled angular chirp in the full direct/antipodal analysis: approximately 64.2 degrees to 24.3 degrees to 14.0 degrees, with R-squared 0.940 and p = 0.0465.

17.3 Venus

Venus advancing combines a broad palindrome with the strongest recurring planetary profile in the original event analysis. Its broad mirror correlation is 0.4479 with p = 0.0244. Its cross-validated recurring template reaches 0.6136 with p = 0.0488, and adjacent-event correlation is 0.2665. The standardized angular analysis is more conservative for the palindrome, but the recurrence evidence remains explicit.

Venus retreating carries a near-controlled angular down-chirp from approximately 1.52 degrees toward 4.61 and 8.02 degrees, with R-squared 0.876 and p = 0.093.

17.4 Mars

Mars is distinguished by recurrent templates rather than controlled palindromes. Advancing and retreating cross-validated template correlations are approximately 0.460 and 0.455, each with original audit p = 0.0488. The advancing adjacent-event correlation is -0.427, demonstrating alternation or polarity reversal rather than direct daily repetition. Aggregate opposite-side templates reach approximately 0.865 after time reversal.

17.5 Jupiter

The original temporal-frequency analysis identifies a controlled Jupiter-advancing long-period power-notch up-chirp from the four-hour analysis boundary toward approximately 57 minutes, with R-squared 0.878 and p = 0.0435. After angular normalization, the path remains coherent but the matched-control audit weakens. This shows that part of the temporal sweep is associated with target-relative phase rate. Jupiter advancing occurs close to solar transit during much of the archive, so the event samples a shared solar-sidereal sector.

17.6 Saturn

Saturn retreating contains a compact one-hour palindrome in the original 1,801-second moving-standard-deviation channel: r = 0.3872 with p = 0.0244. The latitude-unwound standardized analysis finds controlled symmetry on both sides: Saturn advancing r = 0.3118 and Saturn retreating r = 0.3689, each with p = 0.0465. The carrier ladder moves coherently downward at Saturn advancing with median -0.0128 octaves per hour.

17.7 Uranus

Uranus retreating contains a broad palindrome. The original result is r = 0.3892 with p = 0.0244; the unwound result is r = 0.4121 with p = 0.0233. Uranus advancing contains the strongest controlled angular chirp in the complete direct/antipodal study: 48.6 degrees to 24.3 degrees to 18.4 degrees, with R-squared 0.772 and p = 0.0233.

The original temporal analysis also finds a controlled long-period up-chirp from approximately 4.27 hours toward 28 minutes, with R-squared 0.935 and p = 0.0435. Agreement between the temporal and angular analyses makes Uranus advancing the clearest long-period chirp class, while event-overlap analysis still requires separation from neighboring solar and sidereal sectors.

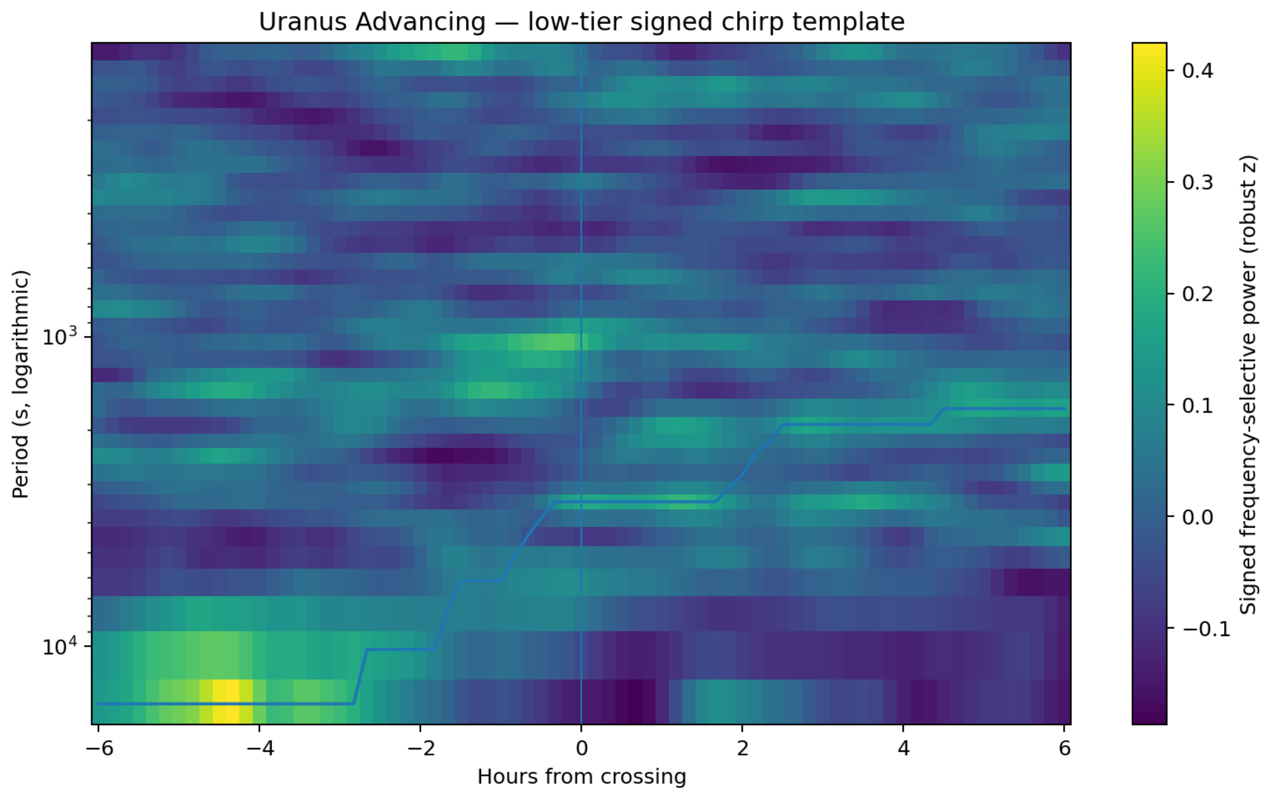

Figure 37. Uranus-advancing long-period chirp.

Reading guide. The moving power-notch ridge progresses from the four-hour limit toward shorter periods. Angular normalization preserves a controlled trajectory.

17.8 Neptune

Neptune does not produce a controlled direct-transit palindrome or chirp after angular normalization, but both sides contain organized trajectories. Neptune differs from the other planets in the aggregate opposite-side comparison: its advancing and retreating templates align more strongly directly, approximately 0.647, whereas Mercury through Uranus generally align more strongly after time reversal.

17.9 Planetary overlap and attribution

Table 19. Planetary crossing schedule overlap

Event pair	Median separation	Overlap statement
Mercury advancing - Uranus advancing	49 min	59.3 percent within 60 min
Mercury advancing - Jupiter advancing	55 min	54.3 percent within 60 min
Saturn - Neptune corresponding sides	82 min	100 percent within 120 min
Jupiter - Neptune corresponding sides	95 min	approximately 90 percent within 120 min

Planetary labels identify real celestial sectors sampled during the archive, but they are not fourteen independent experiments. Unique attribution requires longer records spanning substantially different relative planetary configurations.

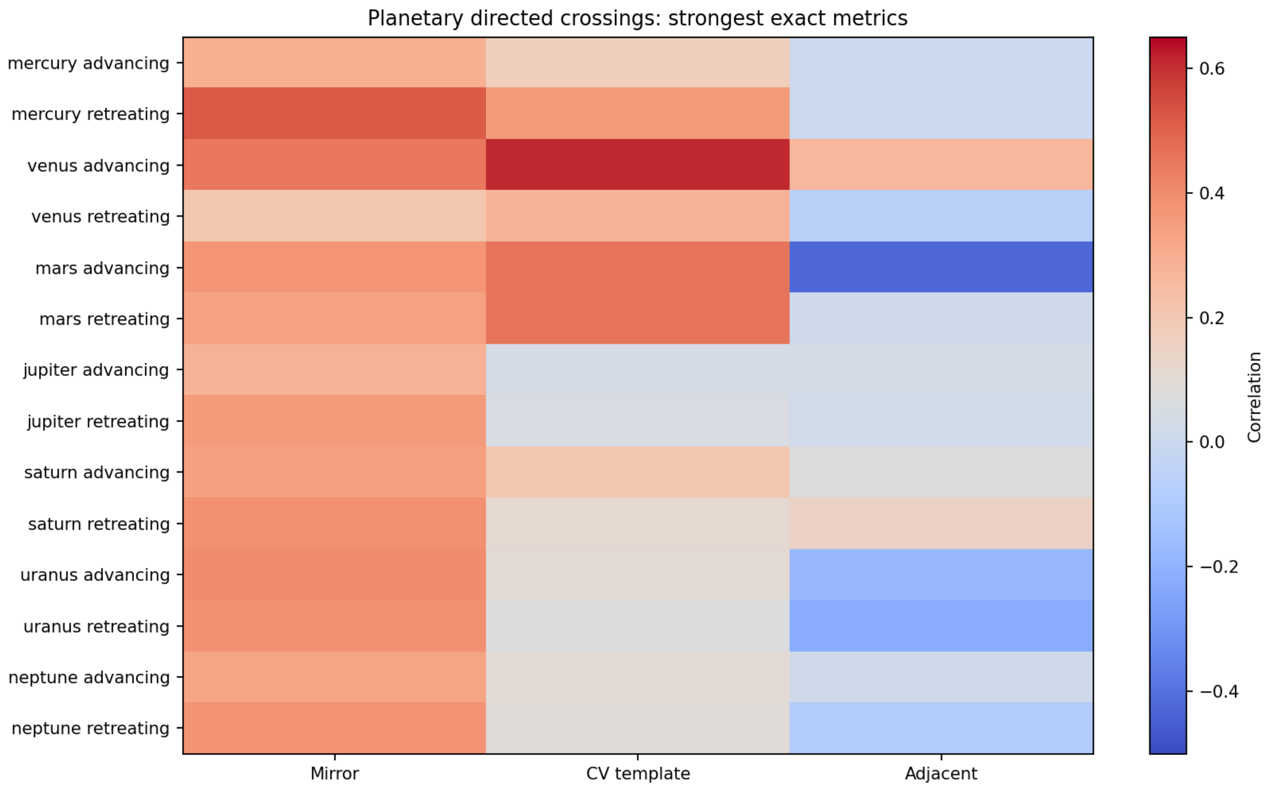

Figure 38. Planetary palindrome and recurrence matrix.

Reading guide. Advancing and retreating results are shown for all seven planets. Strong sectors cluster because several planetary schedules overlap during the 2023 archive.

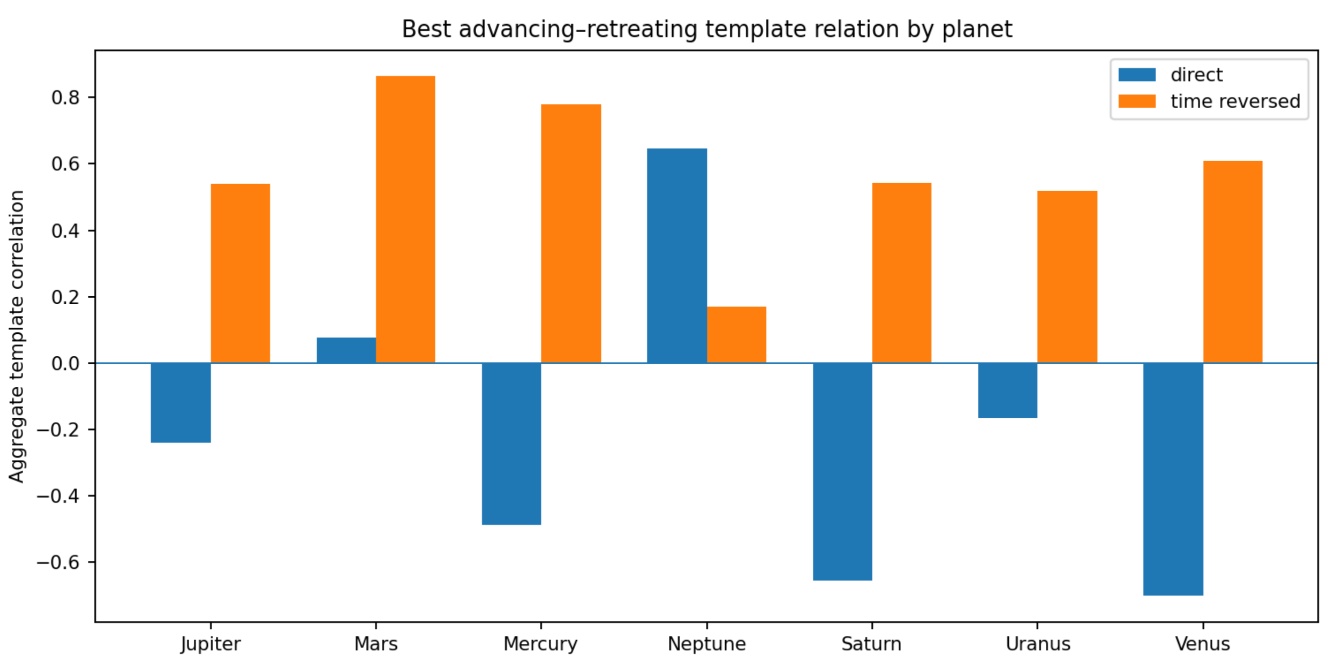

Figure 39. Planetary advancing-retreating template comparison.

Reading guide. Mercury through Uranus generally gain similarity after time reversal. Neptune is the principal direct-alignment exception.

18. The Complete 90-Degree Orthogonal Crossing Analysis

18.1 Universal long-period inversion

At 2-minute to 4-hour periods, every one of the 24 parent event classes has negative direct correlation between its -90-degree and +90-degree templates. The median correlation is -0.275, with interquartile range -0.288 to -0.252. Time reversal does not restore the match; the median remains -0.083. Frequency reversal produces only +0.062. The relation is therefore a spectral-polarity inversion rather than a time-reversed palindrome.

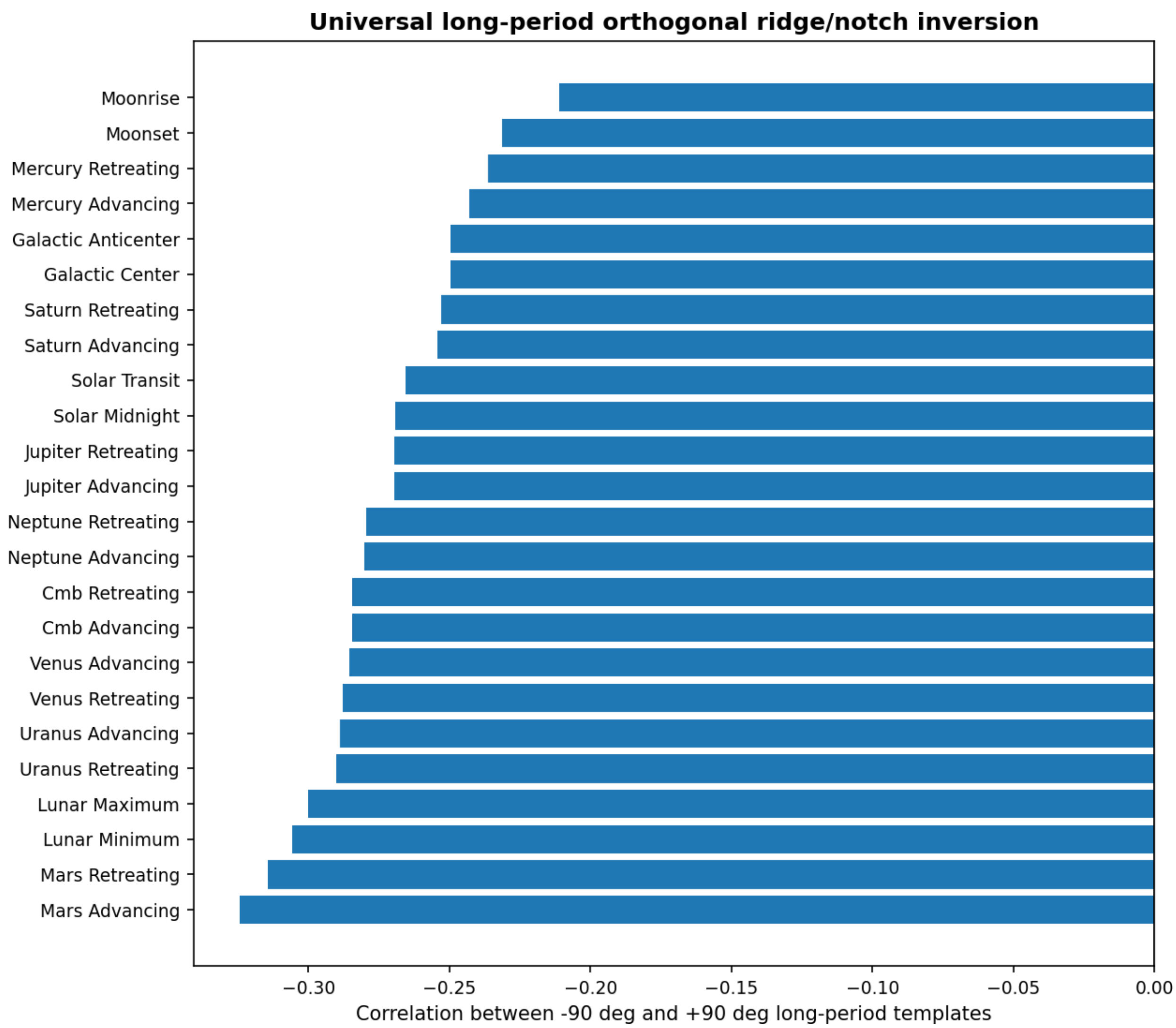

Figure 40. Universal long-period quadrature inversion.

Reading guide. Each bar is the direct correlation between the two opposite orthogonal templates for one parent event. Every bar is negative.

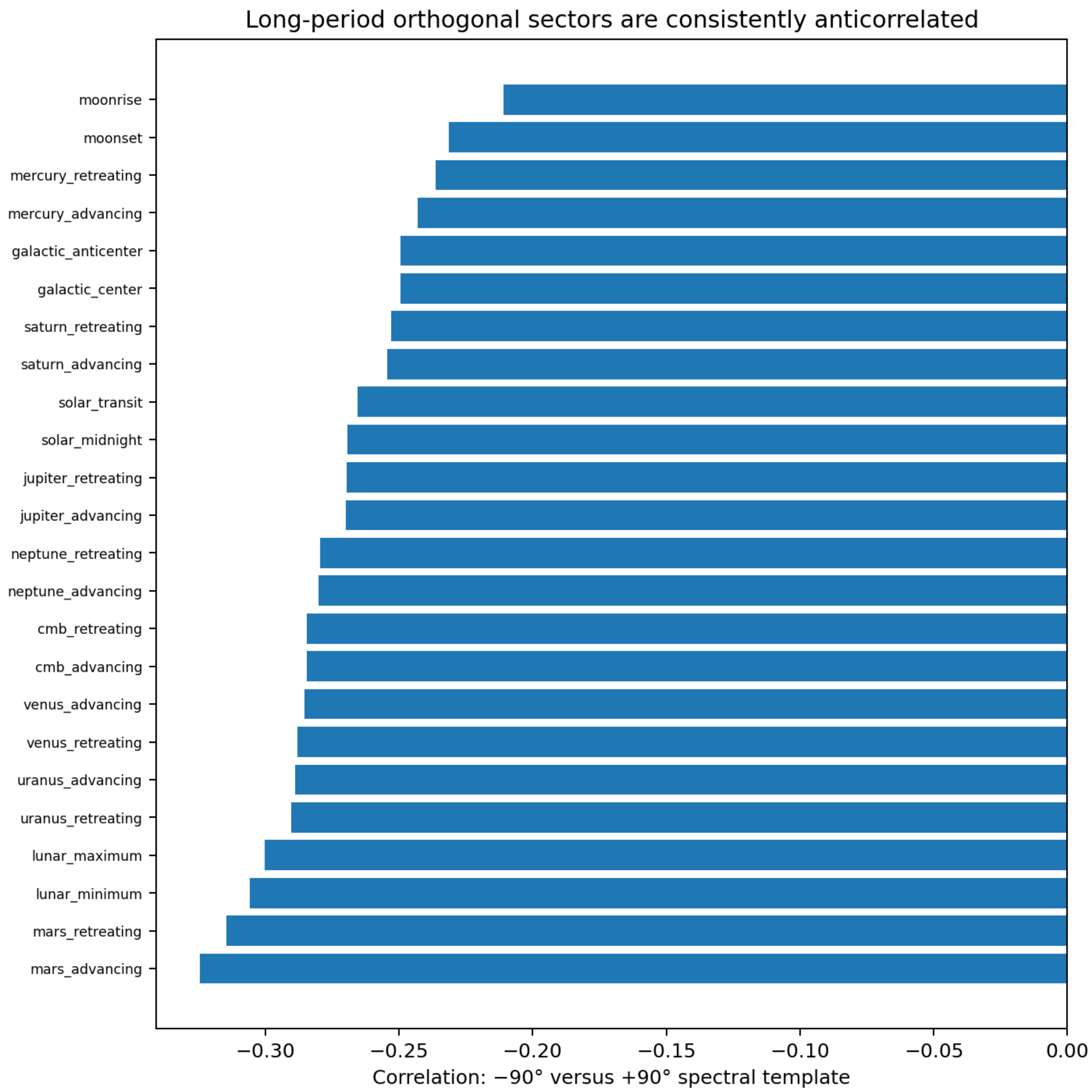

Figure 41. Orthogonal pair correlations from the full spectral templates.

Reading guide. The long-period inversion is present across solar, lunar, fixed-axis, and planetary parent classes. The fast tier does not show a corresponding universal relation.

18.2 Complete orthogonal evidence table

Table 20. Direct-parent orthogonal crossing evidence

Parent crossing	Mirror -90	Mirror +90	Best chirp p -90	Best chirp p +90	Long-tier pair r
Solar Transit	0.226	0.220	0.667	0.043	-0.265
Solar Midnight	0.224	0.308	0.174	0.974	-0.269
Lunar Maximum	0.385	0.321	0.538	0.957	-0.300
Lunar Minimum	0.371	0.158	0.846	0.590	-0.306
Moonrise	0.238	0.040	0.435	1.000	-0.211
Moonset	0.136	0.216	0.087	0.308	-0.231
Cmb Advancing	0.144	0.237	0.795	0.641	-0.284
Cmb Retreating	0.237	0.144	0.641	0.795	-0.284
Galactic Center	0.292	0.189	0.564	0.128	-0.249
Galactic Anticenter	0.189	0.292	0.128	0.564	-0.249
Mercury Advancing	0.177	0.299	1.000	0.744	-0.243

Parent crossing	Mirror -90	Mirror +90	Best chirp p -90	Best chirp p +90	Long-tier pair r
Mercury Retreating	0.315	0.196	0.826	1.000	-0.236
Venus Advancing	0.194	0.231	0.564	1.000	-0.285
Venus Retreating	0.231	0.226	0.957	0.205	-0.288
Mars Advancing	0.166	0.340	0.462	1.000	-0.325
Mars Retreating	0.335	0.143	0.769	0.385	-0.315
Jupiter Advancing	0.188	0.179	0.913	0.718	-0.270
Jupiter Retreating	0.176	0.220	1.000	0.538	-0.269
Saturn Advancing	0.339	0.402	0.652	0.087	-0.254
Saturn Retreating	0.404	0.357	0.043	0.538	-0.253
Uranus Advancing	0.318	0.316	0.103	0.641	-0.289
Uranus Retreating	0.314	0.317	0.256	0.333	-0.290
Neptune Advancing	0.176	0.226	0.077	0.718	-0.280
Neptune Retreating	0.231	0.179	0.282	0.385	-0.279

18.3 Solar evening quadrature

The controlled solar orthogonal chirp is the unique sector represented by solar upper transit +90 degrees and solar lower transit -90 degrees. The two labels differ by only 1.85 minutes and have long-period template correlation 0.994. The sector contains a downward-frequency power-notch sweep whose characteristic period lengthens from approximately 17.1 minutes to 51.2 minutes and then 73.1 minutes. The fitted slope is -0.243 octaves per hour, R-squared 0.856, with displaced-anchor p = 0.0435. The opposite morning quadrature does not contain a comparable controlled chirp.

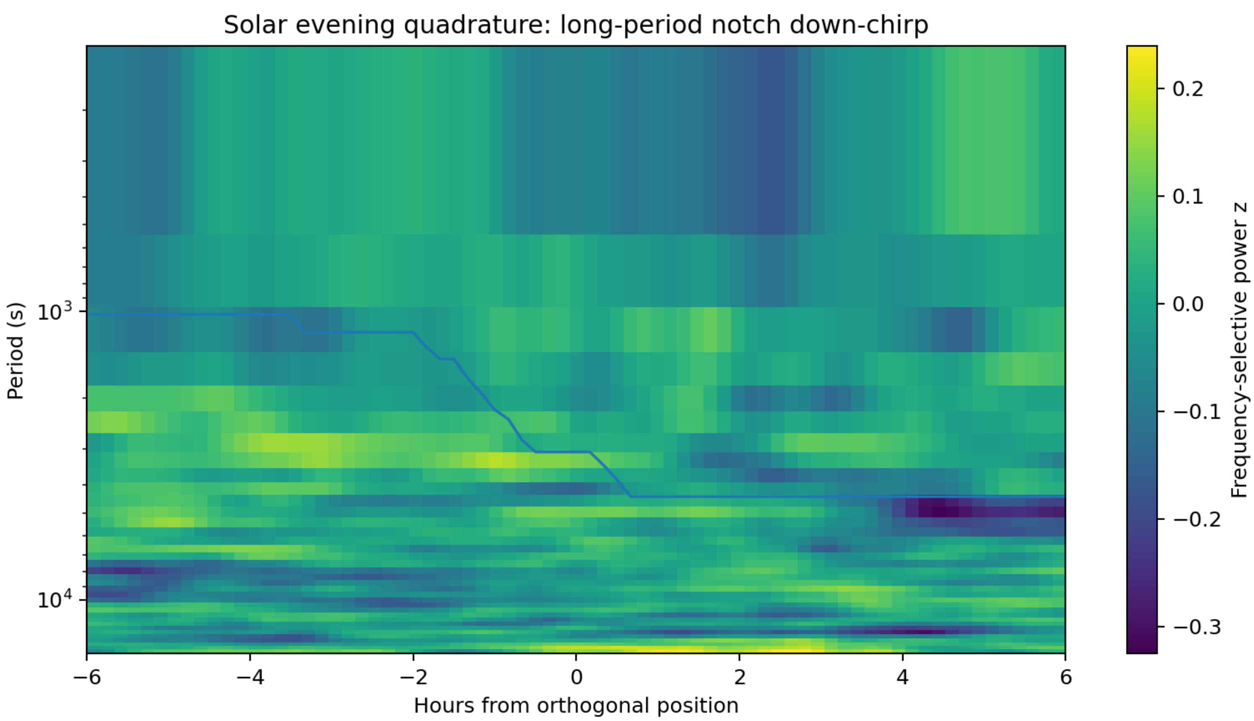

Figure 42. Solar evening orthogonal chirp.

Reading guide. This is a unique Earth-rotation sector represented by two equivalent direct/antipodal labels. The spectral notch moves toward longer periods through the quadrature.

18.4 Saturn quadrature

The Saturn orthogonal sector is represented by Saturn retreating -90 degrees and Saturn advancing +90 degrees. Their event sequences differ by only 0.18 minutes and their long-period templates correlate at 0.999. The event is an upward-frequency power-notch sweep from the four-hour analysis boundary toward approximately 56.9 minutes and then 46.5 minutes. The fitted slope is +0.223 octaves per hour, R-squared 0.779, with p = 0.0435.

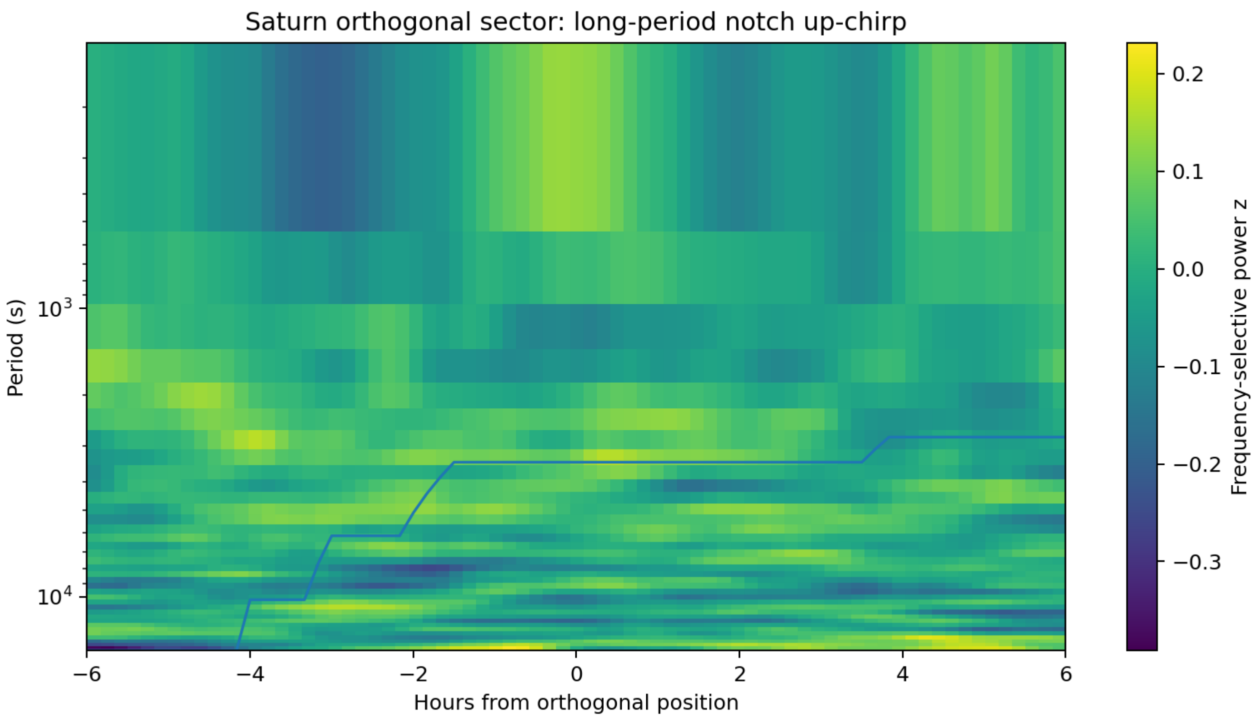

Figure 43. Saturn orthogonal-sector chirp.

Reading guide. The notch advances from the longest resolved periods toward approximately one hour. The opposite Saturn quadrature is weak.

18.5 Neptune near-threshold orthogonal event

Neptune advancing -90 degrees, equivalent to Neptune retreating +90 degrees, contains a fast power-notch up-chirp from approximately 512 seconds through 128 seconds to 64 seconds. The fitted R-squared is 0.760 and the displaced-anchor p value is 0.0769. It is retained as a secondary structured event rather than a controlled detection.

18.6 Orthogonal states are not delayed copies

The median correlation between a direct crossing template and either of its orthogonal templates is only approximately 0.070, and the largest absolute value is approximately 0.291. The orthogonal positions are therefore distinct signal states rather than delayed copies of the corresponding transit.

PART VIII. DERIVE THE SYSTEM SYMMETRY AND UNIFIED MODEL

19. Integrated Evidence and Minimum Sufficient Signal Model

19.1 Observed layers

The D101 crossing evidence is not described by one sinusoid. The minimum sufficient observational model contains five layers: a broad variance envelope, an internal chirp or moving notch, directional spectral polarity, coherent carrier-ladder modulation, and recurrence across multiple events.

Table 21. Minimum sufficient crossing-signal model

Layer	Observable	Principal evidence
Variance envelope	Moving-standard-deviation palindrome	CMB retreating, solar midnight, Moonset, Mercury retreating, Saturn and Uranus sectors
Internal frequency motion	CWT or spectrogram ridge/notch	Uranus advancing, Mercury advancing, solar midnight, Jupiter advancing, orthogonal solar and Saturn sectors
Directional polarity	Direct/antipodal reversal and orthogonal inversion	All 24 opposite 90-degree quadrature pairs are negatively correlated at long periods
Carrier modulation	Common fractional drift of 1x, 2x, 4x, and 8x harmonics	CMB advancing, solar transit, Saturn advancing, Uranus retreating
Multi-event recurrence	Nonzero event-axis frequency in 2-D FFT	Two-to-eight-event recurrence rather than one identical daily waveform

19.2 Direct, antipodal, and orthogonal inequivalence

The directed sides are not interchangeable. CMB retreating is more strongly palindromic than CMB advancing. Galactic anticenter is palindromic while Galactic center carries a recurring asymmetric profile. Planetary opposite sides frequently align only after time reversal. Orthogonal sides exhibit a universal long-period ridge/notch inversion. These observations establish a four-sector directional response around the Earth-rotation cycle.

19.3 Rectilinear-volume interpretation

A rectilinear volume provides a compact geometric interpretation of the combined observations. Translation through plane families can produce repeated crossing envelopes. Rotation changes the projection speed and polarity of the local device axes. A family of plane spacings or nested sheet scales produces multiple chirp wavelengths. Advancing and retreating motion through a polarized structure can produce time-reversed aggregate templates. Orthogonal directions can exchange ridges and notches when the projected coupling changes sign.

This mechanism class is consistent with the observed hierarchy without requiring every day to reproduce the same waveform. The Earth-Moon barycenter follows the smooth trajectory through the volume, while Earth reflex and laboratory rotation modulate the detector around it. Each crossing therefore samples a different angular phase, EMB lattice phase, lunar-opposed reflex offset, daily rotation offset, celestial projection, and internal oscillator state. The event class can recur while the detailed waveform evolves.

The observed waveform is consequently a local encoding of a higher-dimensional state. Palindrome symmetry identifies approximate reflection about a boundary center. Chirp slope identifies changing plane-crossing rate. Orthogonal ridge/notch inversion identifies directional parity. Two-dimensional FFT recurrence identifies return on the event sequence. High-similarity pairs identify near-closure of the full angular and translational state.

19.4 What latitude unwinding changes

Latitude unwinding does not amplify the scalar signal. It changes the coordinate in which signal structure is compared. Fixed-axis coherence remains unchanged, which validates the transformation. Moving-target events gain or lose coherence depending on how much UTC stacking smeared the physical hour-angle trajectory. Angular chirp trajectories become comparable across the Sun, Moon, planets, CMB, and Galactic directions because they are expressed in degrees rather than seconds.

19.5 Evidence hierarchy

Table 22. Crossing-evidence hierarchy

Evidence class	Strongest examples	Interpretive weight
Controlled palindrome	CMB retreating, Galactic anticenter, Moonset, Mercury retreating, Saturn, Uranus retreating	Explicit event-centered variance symmetry
Controlled angular chirp	Uranus advancing, Mercury advancing	Explicit moving angular-wavelength ridge/notch
Controlled temporal chirp	Solar transit, Jupiter advancing, Uranus advancing; solar-midnight individual incidence	Temporal spectral motion before kinematic normalization
Universal orthogonal relation	24 of 24 negative long-tier quadrature correlations	Strongest population-level directional invariant
Recurrent asymmetric profile	Galactic center, Mars, Venus advancing	Crossing-conditioned morphology without mirror symmetry
Carrier-ladder coherence	CMB advancing, solar transit, Saturn advancing, Uranus retreating	Shared oscillator carrier modulation

PART IX. REPLICATION AND FALSIFIABLE PREDICTIONS

20. Replication, Extension, and Falsifiable Predictions

20.1 Required data acquisition

The next D101-class acquisition should retain raw one-second pulse widths and, where possible, add a higher-cadence channel above 4 samples per second so that the 0.5-2 Hz interval is observed without aliasing. Temperature, supply voltage, oscillator oven current, and independent reference frequency should be recorded with synchronized UTC timestamps. Long uninterrupted runs are more valuable than daily fragments because two-dimensional event recurrence spans several crossings.

20.2 Pre-registered analysis

The event definitions, moving-standard-deviation windows, CWT scales, chirp penalties, phase spans, and displaced-anchor controls should be fixed before examining new data. The same target-relative hour-angle transformation should be applied. Direct, antipodal, and orthogonal labels should be collapsed into unique celestial sectors before multiplicity correction.

20.3 Cross-latitude prediction

If the latitude cone is a geometric modulation rather than a site-specific artifact, two instruments at different latitudes should agree after cone flattening and hour-angle registration. Their physical zenith-cone wavelengths should differ by cos(phi), while their equatorial angular wavelengths and event phases should align.

```
lambda_cone,1 / lambda_cone,2 = cos(phi1) / cos(phi2)
for the same equatorial angular structure
```

20.4 Opposite-quadrature prediction

The strongest population-level prediction is that opposite 90-degree sectors retain negative long-period frequency-selective template correlation in a new archive. The exact ridge amplitudes may vary, but the ridge/notch polarity relation should persist if it is a geometric property of the crossing state.

20.5 Planetary decorrelation prediction

A longer record spanning different planetary elongations should separate planet labels that overlap in the 2023 archive. A planet-specific component must follow the planet when its transit moves away from the solar or fixed-vector sector. A purely sector-specific component will remain tied to local solar or sidereal phase instead.

20.6 Cell3 plane prediction

If the crossing signals are associated with rectilinear Cell3 plane families, angular chirp order and recurrence should cluster when the laboratory trajectory projection onto reciprocal normals is used as the independent coordinate. The plane-family assignment should improve coherence without changing the authoritative basis or fitting the basis to the same signal.

20.7 EMB-corrected recurrence prediction

Future high-similarity pairs should concentrate when the sum Delta Phi_EMB + Delta deltaPhi_reflex + Delta deltaPhi_rot approaches an integer for one or more Cell3 plane families. Evaluating only target hour angle should underperform the complete phase model.

20.8 Longitude, latitude, and hemisphere predictions

Two synchronized instruments at different longitudes should share the same EMB phase but encounter different daily laboratory-rotation phases. Instruments at different latitudes should agree after latitude-cone flattening while retaining predictable differences in physical zenith-cone arc. An opposite-hemisphere instrument should reverse selected zenith-projection and palindrome terms while preserving the fixed ICRS lattice.

20.9 Multi-station inversion

Three or more geographically separated D101-class instruments would permit direct inversion of EMB plane phase, Cell3 plane normal, crossing speed, and observer-offset correction. A real spatial boundary should appear at station times related by the stations projected separation along the reconstructed normal.

21. Conclusions

The D101 archive contains explicit evidence that pulse-width variance and spectral structure are organized around celestial crossings. CMB retreating is the strongest fixed-vector palindrome and remains invariant under the exact latitude-cone correction. Galactic anticenter contains controlled palindromic structure, while Galactic center preferentially carries a recurring asymmetric profile. Moonset becomes substantially more coherent after target-relative hour-angle unwinding, demonstrating that moving-target registration matters.

Solar upper and lower culmination produce distinct signal classes. Solar transit carries a segmented fast chirp and a one-hour variance symmetry. Solar midnight carries a broader variance palindrome and a fast power-notch chirp. Mercury retreating, Saturn advancing and retreating, and Uranus retreating contain controlled palindromic envelopes. Uranus advancing and Mercury advancing contain the strongest controlled angular chirps.

Planetary results cluster in overlapping solar and sidereal sectors; they identify real crossing-conditioned states but are not yet uniquely planet-specific. All 24 opposite 90-degree quadrature pairs are anticorrelated at long periods. This universal inversion is one of the most systematic directional results in the archive. The solar-evening and Saturn orthogonal sectors contain controlled stacked chirps, while a Neptune orthogonal chirp occurs at lower confidence.

CWT and FFT derivatives show that the crossing events are usually segmented, branching, or polarity-changing rather than simple linear chirps. Two-dimensional FFTs demonstrate recurrence over multiple crossings rather than one identical waveform repeated every solar, sidereal, or lunar day. The analytical latitude unwind is essential for converting temporal frequency into angular wavelength and recovering coherence in moving-target events, but it cannot manufacture fixed-axis scalar coherence.

The complete body of evidence supports a directional, crossing-conditioned modulation model in which direct, antipodal, and orthogonal celestial phases correspond to distinguishable signal states. The primary observables are variance-envelope symmetry, frequency-ridge motion, notch motion, harmonic carrier drift, quadrature inversion, and recurrence across multi-event sequences.

The explicit high-similarity pairs add a deeper constraint. Projected Cell3 alignment selects permitted angular states, while Earth-Moon barycentric phase and deterministic observer offsets select the dates on which those states recur. The Moon coordinates consequently encode both celestial orientation and the Earth own reflex displacement through the lattice. The resulting model is a centrosymmetric, volume-filling oblique Cell3 structure sampled by a four-sector Earth-rotation phase and modulated by monthly and daily observer motion.

This framework supplies a reproducible foundation for mapping oscillator signals into ICRS direction, EMB lattice phase, and rectilinear spatial geometry. It also defines direct experimental tests across longitude, latitude, hemisphere, cadence, and multi-station baseline.

Appendix A. Explicit High-Similarity Pair and Anchor Ledger

The machine-readable companion contains all 205 pairs and all 253 unique anchors. The table below preserves the representative pair set used throughout the explanatory chapters.

Table 23. Representative explicit ICRS high-similarity pairs

Event	Channel	Span min	r	UTC A	UTC B	Days	Target A	Target B	Projected Cell3 match A
Neptune Retreating	pstd_720m	360	0.992920	2023-03-04 06:27:30+00:00	2023-04-13 03:55:30+00:00	39.8944	RA 175.286, Dec 3.313	RA 176.649, Dec 2.734	(001)/N(+1,+0,+0); 1.174 deg
Mars Advancing	pstd_720m	360	0.992184	2023-04-12 22:50:30+00:00	2023-04-13 22:48:30+00:00	0.9986	RA 99.932, Dec 25.071	RA 100.506, Dec 25.024	(010)/N(+1,+1,+1); 0.051 deg
Galactic Anticenter	pstd_720m	360	0.991085	2023-04-12 21:54:30+00:00	2023-04-13 21:50:30+00:00	0.9972	RA 86.405, Dec 28.936	RA 86.405, Dec 28.936	(010)/N(+0,+1,-1); 0.168 deg
Venus Advancing	pstd_720m	360	0.990685	2023-04-12 20:07:30+00:00	2023-04-13 20:08:30+00:00	1.0007	RA 59.133, Dec 22.051	RA 60.352, Dec 22.335	(001)/N(+1,-1,+1); 1.375 deg
Galactic Center	pstd_360m	180	0.983548	2023-04-12 09:56:30+00:00	2023-04-17 09:37:30+00:00	4.9868	RA 266.405, Dec -28.936	RA 266.405, Dec -28.936	(010)/N(+0,+1,-1); 0.168 deg
Mars Retreating	pstd_180m	180	0.982557	2023-03-30 11:13:30+00:00	2023-05-02 10:19:30+00:00	32.9625	RA 272.348, Dec -25.503	RA 291.342, Dec -23.756	(001)/N(+1,+1,-1); 3.843 deg
Galactic Center	pstd_360m	360	0.980503	2023-04-13 09:52:30+00:00	2023-04-25 09:05:30+00:00	11.9674	RA 266.405, Dec -28.936	RA 266.405, Dec -28.936	(010)/N(+0,+1,-1); 0.168 deg
Uranus Advancing	std_3601s	60	0.973926	2023-04-08 19:24:30+00:00	2023-04-18 18:47:30+00:00	9.9743	RA 44.509, Dec 16.571	RA 45.048, Dec 16.725	(010)/N(+1,-1,+0); 1.193 deg
Neptune Advancing	pstd_180m	180	0.972963	2023-03-21 17:21:30+00:00	2023-05-02 14:41:30+00:00	41.8889	RA 355.892, Dec -3.053	RA 357.223, Dec -2.497	(001)/N(+1,+0,+0); 1.194 deg
Jupiter Advancing	pstd_360m	180	0.964482	2023-03-21 18:39:30+00:00	2023-04-25 16:53:30+00:00	34.9264	RA 15.493, Dec 5.437	RA 23.326, Dec 8.599	(100)/N(+1,+1,-1); 1.186 deg
Uranus Retreating	pstd_360m	360	0.959533	2023-03-29 08:03:30+00:00	2023-05-07 05:38:30+00:00	38.8993	RA 223.982, Dec -16.417	RA 226.102, Dec -17.021	(010)/N(+1,-1,+0); 1.754 deg
Mercury Retreating	pstd_360m	360	0.955692	2023-03-05 05:08:30+00:00	2023-03-09 05:19:30+00:00	4.0076	RA 156.299, Dec 12.203	RA 162.995, Dec 9.464	(001)/N(+1,+0,+0); 0.615 deg
Solar Midnight	pstd_360m	360	0.949863	2023-03-05 05:42:30+00:00	2023-03-09 05:42:30+00:00	4.0000	RA 165.319, Dec 6.270	RA 169.025, Dec 4.718	(001)/N(+1,+0,+0); 1.992 deg
Galactic Anticenter	std_3601s	60	0.946207	2023-03-28 22:53:30+00:00	2023-04-13 21:50:30+00:00	15.9563	RA 86.405, Dec 28.936	RA 86.405, Dec 28.936	(010)/N(+0,+1,-1); 0.168 deg
Saturn Advancing	std_3601s	60	0.942284	2023-04-24 13:52:30+00:00	2023-05-06 13:08:30+00:00	11.9694	RA 336.973, Dec -11.122	RA 337.813, Dec -10.836	(001)/N(+1,+0,+0); 0.220 deg
Cmb Retreating	pstd_360m	360	0.936841	2023-03-03 19:24:30+00:00	2023-04-27 15:48:30+00:00	54.8500	RA 9.285, Dec -16.626	RA 9.285, Dec -16.626	(010)/N(+1,+0,+0); 0.376 deg
Mercury Advancing	pstd_60m	60	0.935879	2023-03-12 17:29:30+00:00	2023-04-18 18:33:30+00:00	37.0444	RA 348.959, Dec -6.794	RA 41.579, Dec 19.168	(001)/N(+1,+0,+0); 0.215 deg
Cmb Advancing	pstd_60m	60	0.935826	2023-04-29 03:42:30+00:00	2023-05-20 02:19:30+00:00	20.9424	RA 189.285, Dec 16.626	RA 189.285, Dec 16.626	(010)/N(+1,+0,+0); 0.376 deg
Jupiter Retreating	pstd_360m	360	0.932332	2023-04-01 06:07:30+00:00	2023-05-09 04:12:30+00:00	37.9201	RA 197.818, Dec -6.400	RA 206.336, Dec -9.745	(100)/N(+0,+1,-1); 1.340 deg

Appendix B. Earth-Moon Barycentric Correction Ledger

Table 24. Representative pair barycentric decomposition

Event	r	Order	Plane	Full observer	EMB + lab	EMB only	Reflex delta km	Lab delta km
Solar Midnight	0.949863	19	(100)	0.012599	0.010705	0.010807	4008.1	325.9
Galactic Center	0.983548	20	(110)	0.000205	0.001106	0.001102	5230.4	13.7
Galactic Anticenter	0.991085	21	(100)	0.014642	0.014615	0.014616	1103.7	1.4
Mars Advancing	0.992184	21	(100)	0.017406	0.017379	0.017378	1105.5	40.0
Venus Advancing	0.990685	21	(100)	0.021739	0.021712	0.021709	1107.2	102.2
Mercury Retreating	0.955692	19	(101)	0.012075	0.015015	0.014715	4014.8	553.7
Saturn Advancing	0.942284	20	(110)	0.015928	0.014446	0.014431	9044.9	66.0
Uranus Advancing	0.973926	21	(001)	0.013692	0.013606	0.013606	8603.9	48.1
Jupiter Advancing	0.964482	19	3D	0.131510	0.124598	0.124986	7125.1	654.8
Cmb Retreating	0.936841	20	(111)	0.031353	0.031390	0.031390	221.6	5.2

Table 25. Complete Earth-reflex and laboratory plane-phase modulation atlas

Order	Plane	Reflex peak-to-peak	Reflex RMS	Lab peak-to-peak	Lab RMS
16	(100)	15.248305	5.384708	15.508006	5.482345
16	(010)	6.209540	2.176944	9.863728	3.487756
16	(001)	8.324676	2.920368	7.889118	2.789724
16	(110)	17.429817	6.156462	18.533749	6.554253
16	(101)	16.355748	5.755033	16.494334	5.829758
16	(011)	2.141426	0.749666	2.188508	0.773645
16	(111)	15.341149	5.414953	14.843293	5.247830
17	(100)	1.694256	0.598301	1.723112	0.609149
17	(010)	0.689949	0.241883	1.095970	0.387528
17	(001)	0.924964	0.324485	0.876569	0.309969
17	(110)	1.936646	0.684051	2.059305	0.728250
17	(101)	1.817305	0.639448	1.832704	0.647751
17	(011)	0.237936	0.083296	0.243168	0.085961
17	(111)	1.704572	0.601661	1.649255	0.583092
18	(100)	0.188251	0.066478	0.191457	0.067683
18	(010)	0.076661	0.026876	0.121774	0.043059
18	(001)	0.102774	0.036054	0.097397	0.034441
18	(110)	0.215183	0.076006	0.228812	0.080917
18	(101)	0.201923	0.071050	0.203634	0.071972
18	(011)	0.026437	0.009255	0.027019	0.009551
18	(111)	0.189397	0.066851	0.183251	0.064788
19	(100)	0.020917	0.007386	0.021273	0.007520
19	(010)	0.008518	0.002986	0.013530	0.004784
19	(001)	0.011419	0.004006	0.010822	0.003827
19	(110)	0.023909	0.008445	0.025424	0.008991
19	(101)	0.022436	0.007894	0.022626	0.007997
19	(011)	0.002937	0.001028	0.003002	0.001061
19	(111)	0.021044	0.007428	0.020361	0.007199
20	(100)	0.002324	0.000821	0.002364	0.000836
20	(010)	0.000946	0.000332	0.001503	0.000532
20	(001)	0.001269	0.000445	0.001202	0.000425
20	(110)	0.002657	0.000938	0.002825	0.000999
20	(101)	0.002493	0.000877	0.002514	0.000889
20	(011)	0.000326	0.000114	0.000334	0.000118
20	(111)	0.002338	0.000825	0.002262	0.000800
21	(100)	0.000258	9.119050e-05	0.000263	9.284399e-05
21	(010)	0.000105	3.686674e-05	0.000167	5.906545e-05
21	(001)	0.000141	4.945669e-05	0.000134	4.724423e-05
21	(110)	0.000295	0.000104	0.000314	0.000111
21	(101)	0.000277	9.746199e-05	0.000279	9.872746e-05
21	(011)	3.626524e-05	1.269566e-05	3.706257e-05	1.310174e-05
21	(111)	0.000260	9.170271e-05	0.000251	8.887247e-05

Appendix C. Controlled Latitude-Unwound Palindrome Ledger

Table 26. Complete controlled phase-palindrome audit

Event	Channel	Span deg	Observed mirror	Control mean	Control q95	p
Cmb Retreating	pstd_360m	45	0.4386	0.1419	0.2948	0.0233
Galactic Anticenter	pstd_360m	15	0.2389	-0.1267	0.0969	0.0233
Uranus Retreating	pstd_720m	90	0.4121	0.1504	0.3031	0.0233

Event	Channel	Span deg	Observed mirror	Control mean	Control q95	p
Mercury Retreating	pstd_720m	90	0.3430	0.1374	0.3258	0.0465
Moonset	pstd_720m	90	0.4403	0.1718	0.3620	0.0465
Saturn Advancing	pstd_180m	15	0.3118	0.0481	0.2287	0.0465
Saturn Retreating	std_3601s	15	0.3689	0.2079	0.3454	0.0465
Lunar Maximum	pstd_360m	15	0.0769	-0.1565	0.0387	0.0698
Moonrise	pstd_360m	45	0.2901	0.1446	0.2763	0.0698
Solar Midnight	pstd_720m	45	0.1972	-0.1452	0.1844	0.0698
Cmb Advancing	pstd_720m	90	0.3016	0.1481	0.3188	0.0930
Neptune Retreating	pstd_60m	15	0.3215	0.1642	0.3307	0.0930
Solar Transit	std_3601s	15	0.3945	0.2001	0.3935	0.0930
Venus Advancing	std_3601s	15	0.3111	0.2073	0.3194	0.0930
Jupiter Advancing	pstd_180m	15	0.2157	0.0539	0.2424	0.1163
Neptune Advancing	pstd_180m	45	0.3106	0.2045	0.3313	0.1628
Lunar Minimum	pstd_180m	45	0.3203	0.2035	0.3385	0.1860
Jupiter Retreating	pstd_720m	90	0.2408	0.1422	0.2820	0.2326
Mercury Advancing	pstd_60m	15	0.2664	0.1798	0.3576	0.2558
Uranus Advancing	pstd_180m	45	0.2471	0.1974	0.3205	0.2558
Mars Retreating	std_3601s	15	0.2416	0.2183	0.3563	0.4186
Mars Advancing	std_3601s	15	0.2383	0.2078	0.3546	0.4419
Galactic Center	std_3601s	15	0.1862	0.2121	0.3563	0.6047
Venus Retreating	std_3601s	15	0.1380	0.2079	0.3773	0.6977

Appendix D. Controlled Latitude-Unwound Chirp Ledger

Table 27. Complete controlled angular-chirp audit

Event	Tier	Chirp stat	Control mean	Control q95	Chirp p	TF mirror	Mirror p
Uranus Advancing	low	0.1281	0.0705	0.0962	0.0233	-0.2955	0.9767
Mercury Advancing	low	0.1396	0.0976	0.1331	0.0465	-0.2130	0.8605
Cmb Retreating	low	0.1403	0.0928	0.1366	0.0698	-0.1952	0.8372
Lunar Maximum	low	0.0993	0.0824	0.0992	0.0698	-0.1003	0.6279
Venus Retreating	low	0.1183	0.0860	0.1218	0.0930	-0.1368	0.7209
Cmb Advancing	high	0.0646	0.0437	0.0780	0.1163	0.0839	0.4419
Jupiter Advancing	low	0.1156	0.0719	0.1224	0.1163	-0.1172	0.8372
Lunar Minimum	low	0.1083	0.0835	0.1089	0.1163	-0.2358	0.9070
Solar Transit	high	0.0452	0.0343	0.0492	0.1395	0.1027	0.3256
Galactic Anticenter	high	0.0635	0.0479	0.0919	0.1860	0.0138	0.6744
Cmb Retreating	high	0.0602	0.0473	0.0748	0.2093	-0.0258	0.8605
Neptune Advancing	low	0.0949	0.0753	0.1223	0.2093	0.0239	0.2326
Jupiter Advancing	high	0.0399	0.0341	0.0483	0.2791	0.0320	0.6279
Mars Retreating	high	0.0337	0.0301	0.0416	0.3023	0.0637	0.5116
Moonset	high	0.0542	0.0451	0.0661	0.3023	0.0189	0.5349
Lunar Maximum	high	0.0435	0.0392	0.0574	0.3256	-0.0448	0.6744
Neptune Advancing	high	0.0357	0.0320	0.0500	0.3256	0.1595	0.0930
Saturn Advancing	high	0.0323	0.0295	0.0396	0.3256	0.0013	0.7209
Venus Advancing	low	0.0841	0.0806	0.1161	0.3488	-0.1133	0.7209
Jupiter Retreating	low	0.0892	0.0790	0.1237	0.3721	-0.0857	0.5349
Mercury Advancing	high	0.0607	0.0508	0.0741	0.3721	0.0940	0.1395
Solar Transit	low	0.0860	0.0792	0.1044	0.3721	-0.1315	0.6977
Uranus Retreating	high	0.0316	0.0312	0.0451	0.4419	0.1023	0.1395
Jupiter Retreating	high	0.0376	0.0362	0.0523	0.4651	0.2356	0.0233
Lunar Minimum	high	0.0335	0.0345	0.0569	0.5116	-0.0155	0.6744

Event	Tier	Chirp stat	Control mean	Control q95	Chirp p	TF mirror	Mirror p
Mars Advancing	low	0.0642	0.0706	0.1206	0.5116	-0.1851	0.8372
Neptune Retreating	low	0.0764	0.0784	0.1248	0.5116	-0.1029	0.7209
Uranus Retreating	low	0.0734	0.0781	0.1261	0.5116	-0.0525	0.4419
Saturn Retreating	low	0.0727	0.0796	0.1199	0.5349	-0.1513	0.7907
Cmb Advancing	low	0.1141	0.1194	0.1872	0.5581	-0.1082	0.6279
Saturn Advancing	low	0.0660	0.0761	0.1214	0.6047	0.0950	0.0465
Mars Advancing	high	0.0266	0.0302	0.0463	0.6279	0.0073	0.6744
Galactic Anticenter	low	0.0794	0.0928	0.1421	0.6512	0.2463	0.0233
Solar Midnight	high	0.0311	0.0341	0.0482	0.6512	0.1152	0.2558
Moonrise	high	0.0369	0.0397	0.0559	0.6744	0.0277	0.4419
Solar Midnight	low	0.0780	0.0822	0.1087	0.6744	-0.0014	0.2093
Mercury Retreating	low	0.0920	0.1066	0.1430	0.6977	-0.0496	0.3488
Moonset	low	0.0722	0.0968	0.1303	0.7674	0.2059	0.0233
Galactic Center	high	0.0313	0.0522	0.0822	0.7907	0.0172	0.6279
Galactic Center	low	0.0676	0.1097	0.1582	0.8605	-0.1628	0.7674
Venus Retreating	high	0.0257	0.0350	0.0480	0.8605	0.0258	0.4651
Uranus Advancing	high	0.0192	0.0315	0.0468	0.8837	0.0603	0.4884
Saturn Retreating	high	0.0213	0.0326	0.0476	0.9302	0.0939	0.1163
Mercury Retreating	high	0.0343	0.0578	0.0866	0.9535	0.0474	0.4651
Neptune Retreating	high	0.0242	0.0371	0.0515	0.9535	0.0425	0.6047
Venus Advancing	high	0.0231	0.0383	0.0531	0.9535	-0.0258	0.9302
Moonrise	low	0.0752	0.1005	0.1235	0.9767	0.0024	0.3721
Mars Retreating	low	0.0359	0.0777	0.1211	1.0000	0.0046	0.1860

Appendix E. Source and Integrity Ledger

Table 28. Source file integrity ledger

Role	Filename	Bytes	SHA-256
archive_zip	d101_b.zip	52,128,215	5132647b4524bb4ab27eb1914b6c8ddc4a4457954aa3c0fc976927a3ce247063
initial_pdf	d101_OCXO_pulse_width_CMB_lunar_analysis_1786014720.pdf	2,443,749	2c8bedfa184c1d58cdbeaa5c2fc3b606442ec883a7c4e83ee394b9f532d5e375
initial_bundle	d101_OCXO_analysis_results_bundle.zip	5,940,611	add95ab3efa69545a5091e4699d269c83832f417c3937d184a5d7714e4d8cf5f
archive_summary	archive_summary.json	1,802	a6d676df7149b00c8b428ceaa54c35c9b18ee88c051eb941d6fb8d33b99fe226
final_summary	d101_final_summary.json	14,313	d834fca6d27fd58d6e7538ddae92981aa24ee70f7cf8e1cf550fd9f32463475e
carrier_summary	carrier_summary.json	7,525	fcea4dcf06c8e6baf57291bf6fbbc72214d4bb70e500ec3044ddcd64b1886e88
fault_summary	fault_burst_summary.json	203	66aec6642b555e04241af461ab4bdacaceee04ff030777c11adb3942271b8bb2
pal_summary	ANALYSIS_SUMMARY.md	8,242	7c4aa9794e784c3a91dbd2a76644bfd46f495c6f0216ba8a33e01d25dc7a8c7c
solar_midnight	SOLAR_MIDNIGHT_EXTENSION.md	9,212	67e8e27b085836710a069618bfb472eb1e9cab34086a7dea9aaa9ca61b1a83cb
fixed_report	FIXED_VECTOR_CROSSING_EXTENSION.md	24,234	b67a536242b4596b96bb422b6d1094eb72af1f992a87b4ed40ab6a34edd83121
planet_report	PLANETARY_CROSSING_EXTENSION.md	40,075	84169c944f937df4e1ec74795b7735f5cae5b180a3c0b391b170de071ce0c0ba
chirp_report	CHIRP_TRANSIT_ANALYSIS.md	9,402	fe97379a94648ce9c37dcb2960b4fd0d2ed5513b7f7f98d449206204fae331f9
ortho_report	ORTHOGONAL90_CHIRP_EXTENSION.md	5,001	50019ff909717015c659ee66df06608f116e521ba1afe4ecc73fb5a1fb3928ff
lat_method	LATITUDE_CONE_UNWIND_METHOD.md	2,019	e827ec9d1296601bea62d74512e1191ecfac21998369a6cb591bf0e6d6734b0c
lat_report	LATITUDE_UNWOUND_SIGNAL_ANALYSIS.md	15,833	c5eea2655775b56bb6f23865d6b2db149d9ac505c05dfa567399bf2262d352a1
cell3_json	K-Field_ICRS_plane_scew_rajectory_analysis_1785606737.json	47,024	03e6d901dbbe0801cada29a559d40b50c3cec5360f8e29f96d80e8e818cae3c3
cell3_compare_json	K-Field_cell3_to_spectrophotometer_planar_comparison_1785937320.json	48,784	3100ea7414fdf104697df366cd299e0f069a7ba8b32ef8f7b917fabbd1f5573b
core_pal	event_symmetry_recurrence_summary.csv	52,247	5edb6aa18d69332d8408bb612caa19cfb852dbbf0193c592bf95e55eb732ad2e
solar_midnight_csv	solar_midnight_symmetry_recurrence.csv	10,924	f02ff498fc8c3cdb74b6dfdbdcbdc66bb3d96c0596aa49dd319f97148ce36c66
fixed_csv	fixed_vector_crossing_symmetry_recurrence.csv	57,988	85171424d3dc3fd94fa41dda77a3612b5d8ebc4cd6b8a046d4841150715e9994
fixed_control	fixed_vector_crossing_jitter_audit.csv	5,265	104852b9bdadf3e8b4ae1fba29e34e2ffbf47fa343c7295e7c92201336f8f549
planet_best	planetary_crossing_best_results.csv	3,850	430a15e9aca8ddc81ac39d211f1a7a3a97f04aae43259b822f2b42041f66951b

Role	Filename	Bytes	SHA-256
planet_full	planetary_crossing_symmetry_recurrence.csv	173,464	a737fe5ab1a3956bdccd92c3e2908ce21e2a9488714a36e592930ec9b98e1a81
planet_opp	planetary_crossing_opposite_side_comparison.csv	35,937	b0a9e94f89f55f48c55e1fdb5be8233d20ec305fa89d4886f848329bb77d4653
planet_overlap	planetary_crossing_event_overlap.csv	5,780	c15f95e535dc04bd4bf2d384a1ebc2d3b28cb5e6eb9e2497281b4e590fecf7ce
chirp_best	transit_chirp_best_by_event.csv	14,157	a353f586fa36875d1c4c5d3fb6555567578e73da43682cb6e0b72c5859adb6e1
chirp_all	transit_chirp_metrics_all.csv	28,088	44b431943770ea7326f0416148fd1834869c94f35a42b661db63bef7206165c6
chirp_individual	individual_transit_chirp_incidence.csv	18,023	4ea8f606bc8f8be6aed9a5dfdff5c28dcec4dc82b117317b095e5620bcc1fad9
carrier_coherence	carrier_harmonic_coherence.csv	3,391	bbc1a31f6cc7b22cde3e7fe673e8a9b69032845fade152fa2226476a985341c0
ortho_metrics	orthogonal_chirp_metrics_all.csv	58,717	26bebbcbf55ab9bc24e0f18f0e4eaa235ac24e6054f21a6c49c992067d316cf8
ortho_pairs	minus_plus90_template_comparison.csv	5,135	9db45cfe2476c311d9be8ec84f7ae05f6667ecd28a6ff1189390e592d051734b
ortho_parent	parent_to_orthogonal_template_comparison.csv	10,893	d2627d5aafc986efb2151800b388361e1ef6dd23d6ca3a679db274c6730282f9
ortho_overlap	orthogonal_event_overlap.csv	106,441	1a7a5d465730b20296cb6300d8a694fc21c840aadd8b9154f714e8d595edeee8
ortho_carrier	orthogonal_carrier_harmonic_metrics.csv	29,267	602e4c3668cb52c033db69247300a1b1b028205fac343acf4663f9bde1a24806
unwind_diag	d101_latitude_unwind_diagnostics.json	1,012	c97de2e22547abcd329ec6efbf916ef95e87954d505b0376f0fc1832ff0ce535
unwind_pal	latitude_unwound_palindrome_best.csv	56,746	cda70f91280c0fca14fd76d126a24f10ec9fe842b6dcd316769f3b66554d9dd6
unwind_pal_control	latitude_unwound_palindrome_controls.csv	2,652	305634e145f460d5d7cd9b355ec0dd282da7f533ddcb0572094a82da0b620c44
unwind_chirp	latitude_unwound_chirp_all.csv	226,490	7916f35e4be373fc72c33c19132df52c2452c53cb14256c0ba063e36703bbb66
unwind_chirp_control	latitude_unwound_chirp_controls.csv	8,780	6b9d7b6049c1f2159cc29509fb3b57faf6fc27ca849ebfc640e3a83e2285e7da
unwind_cwt	latitude_unwound_cwt_derivatives.csv	37,182	792bc9deeeae1603f7a6dbab6c17036e2368573349b2582ce33c09c9f9145b59
minute_features	d101_minute_features.csv.gz	52,034,067	c563a833c9bdda195dd89fe06ba538395e5c3c7aed692c338bb8876502ae6fe3
episodic_top	top_pair_explicit_icrs_geometry.csv	9,367	f5a3329e8d751e19dadaee8515b2ffabd5bdc033a4af409aac1d99179c9994f9
episodic_all	high_similarity_pair_geometry.csv	332,773	2c5b9b9196a17486614f173d025236d5c3140ed14bd2be919de983bddd4332cf
episodic_anchors	high_similarity_anchor_icrs_positions.csv	143,098	26a48bd5902df56351a445dccc1e963f2ef944f8fc59f5bcba2937a0efbb2553
episodic_clusters	cross_event_high_similarity_clusters.csv	7,548	ee45c1152dc80171da18a2a7fbcba54bf1d40e7ee2d8fdde05c18ca737f6e52f
symmetry_axes	cell3_symmetry_axis_atlas.csv	3,056	e6a040ce58b94aeaf0c24e3491f4e6bac4659dab8e39e2aa197af2630d76a223
lattice_closure	high_pair_lattice_closure_by_order.csv	306,112	49edd40558742589e6a62b9ffe0cf7d92a8fcee7d3c181f80f732ffeef6dd1af
closure_controls	lattice_closure_high_vs_control_summary.csv	34,482	49857c6cd506c1e4f5808aaff66799773410bd1ac80f702d53bd819a1cf84e08
episodic_model	episodic_geometry_symmetry_model.json	43,521	ceb7a95d3604777c6d589d9d9ac7c8e457c70b56c36cd8d588f8cc103fd46121
episodic_sky_map	episodic_anchor_icrs_sky_map.png	393,158	e9cd05fd5990c8105e25f9c1c86550847d0542278c4bb1aa379b2c7fd6350848
episodic_projection	cell3_primary_plane_anchor_projections.png	617,824	39eff5a8c10b8a02c359c80c5bdb6c0341ff38f16e40e2a060490b51db8954b3
episodic_closure_fig	recursive_lattice_closure_improvement.png	118,741	b898a2671e5086bd557380508ed71c921091bf5bb0e1c56c5938e4adb0fa8bef
emb_pairs	representative_pair_emb_correction.csv	3,176	b7def21fc74b31896484efd743c3d076047a9b887ef575f10edf06e6204d904a
emb_modulation	emb_reflex_and_lab_phase_modulation_by_order_plane.csv	5,666	b1479f9ca4068e5b09c2fd15d2c7bcd429ff5ba20d7a78f157ccefd50f259c68
emb_event_comparison	event_level_emb_correction_comparison.csv	23,701	91a0f6ce7c471e37348f64e2fbd114c2a59ad4554eef3964688fa6b52fa85cb0
emb_scan	barycentric_reflex_correction_scan.csv	59,646	e79c62cb6828dd176ed4f6e3cc6a88868912da12a49beca15ba07a6a66680b23
emb_summary	emb_barycentric_reanalysis_summary.json	9,780	3250883afaa4978e5806855373a2fced08b50596c407f0b604e77d5c42159af7
emb_modulation_fig	emb_phase_modulation_by_order.png	96,774	ea2ee8859b1ed66fc615ede92b40bd06e2e6dce8c92fe7821076265a0f288853
emb_pair_fig	representative_pair_emb_correction.png	129,182	77b2a04c425dc170a11a9c1670f64c338761c67fc4e8880d10b179662de6e6cd
logo	K-Field_Logo_20260806(1).png	2,464,613	d58ade314a1b3b8edc91952342ec94a42575073fd025223ad6decc3fdfc54370

Appendix F. Machine-Readable Companion and Figure Inventory

The machine-readable companion K-Field_Celestial_Crossing_Evidence_in_the_D101_OCXO_Signal_1786043894.json is the numerical reproduction authority for this paper. It contains the archive census, coordinate model, complete Cell3 basis, all 27 nodes, reciprocal plane families, seven-sheet membership, complete event and control tables, figure manifest, source hashes, and document-control fields.

Table 29. Complete figure manifest

Figure	Filename	Title	Data basis	SHA-256
1	figure_cover_composite_expanded.png	Expanded data-derived title composite	D101 repeated profiles, authoritative Cell3 lattice, and actual EMB ephemeris	b55a443d0f938f02c5e51edc94efe90fd5f2765a7c157ba3c454b82a564fa922
2	figure_repeat_signal_overlays.png	Explicit episodic repeat-signal overlays	D101 minute feature matrix and archived top-pair ledger	8142672ed00d258138ebc4f039303cee6166a7fa5faa69e463a0d0aec8f39f5b
3	figure_repeat_signal_residuals.png	Repeated profile overlays and residuals	D101 minute feature matrix and high-similarity event pairs	f6d04d0a7a33374921cbdbad40e268864ffe3c8855d0020edf6876a0fab1d8e8
4	figure_repeat_correlation_vs_separation.png	Pair correlation versus temporal separation	Complete high-similarity pair geometry ledger	f4931cdd0352fdc902380097d60006da92115cf0afb217babf87bdf7480aa8d9
5	figure_april_12_13_common_state.png	April 12-13 common-state cluster	D101 Venus, Galactic-anticenter, and Mars event profiles	a01a3b81e6c0d0537c1f34bde35e19ea6174d7b3d62502291d2fa7fd13f514fe
6	01_daily_lag_scan.png	01_daily_lag_scan	See caption and machine-readable source ledger	f8b3d9abe479efa18017b296a55feef69f322a02b2454268be93f72fed884b5b
7	01_timeline.png	01_timeline	See caption and machine-readable source ledger	bdaad59718bb20d25a9a929e6470b37eaca87e5619211e61bf167903acd1a026
8	04_high_frequency_psd.png	High-frequency spectral structure and carrier ladder	D101 one-second pulse-width archive	bfe5403288fbc07f14fa2a22074852d7fd627881d04328cdf211df30ce47a283
9	03_moving_std.png	Multiscale moving-standard-deviation envelopes	D101 raw pulse-width archive	3e45b44913b7e8ea80b944cc76f5f8232bd7ff3fea9941bcfa3e09ef634c5aee
10	figure_cell3_base_cell.png	Authoritative Cell3 base cell with all nodes	K-Field ICRS plane screw trajectory machine-readable Cell3 basis	4903814da99d313611ac50d7349ca7cd96f7ad0ebfd42263d11c7f2f75bbafb2
11	figure_cell3_central_planes.png	Cell3 central face-plane system	Authoritative Cell3 basis and 27-node registry	cc76952451cf84739d8fa9a8f1d4d8fd730c92472a0406b430356115b8d51526
12	figure_cell3_seven_sheet_registry.png	Cell3 seven-sheet node registry	Authoritative Cell3 (111) family and node sheet indices	8bedfd2f31fd98aa5ee0551029a118c10f6e68ab22453822d7e013c496269a55
13	figure_volume_filling_cell3_context.png	Volume-filling Cell3 lattice and celestial-direction context	Authoritative Cell3 basis plus measured CMB, Galactic, solar, Mars, and Uranus ICRS directions	530d0a8dc710d05c76011718093221ad7726af88b237b30e57bc78681bd08714
14	figure_actual_boundary_crossing_context.png	Local boundary-crossing geometry at an episodic Galactic-center anchor	Actual D101 lunar vector, Earth reflex offset, geodetic laboratory vector, and authoritative Cell3 (110) normal	717b76902b23c45d14d78ec6a2b40630f8be7fed82396f65d2104a29a5559a52
15	figure_event_geometry.png	Directed crossing taxonomy	Analytical event definitions	da78628b21894423545051235396eb3ef00405412427b6c73b603c53a3d896b4
16	figure_actual_emb_orbit.png	Actual Earth-Moon barycentric paths during the D101 archive	D101 one-minute Swiss-Ephemeris lunar vectors and measured lunar distance	e56c9a38d62014533c9fa383aca82a8b3ef8fda82983d04fc9d19389eb3633a1
17	figure_nested_observer_motion.png	Nested Earth-Moon barycentric and laboratory motion	D101 one-minute lunar ephemeris and geodetic laboratory ICRS position	962b98c16d5ad32fb29d8c6aae4cbc241dba266c17932e80c4e61708e6f569f8
18	emb_phase_modulation_by_order.png	EMB reflex and laboratory phase modulation versus recursive order	D101 EMB modulation ledger	ea2ee8859b1ed66fc615ede92b40bd06e2e6dce8c92fe7821076265a0f288853
19	figure_three_gate_closure_comparison.png	Three-level closure comparison for representative returns	D101 EMB reanalysis representative-pair ledger	24aa90ae6a6ea16df0f2e2c8aa1ec450b119764954d82385c69c37b8e422cb99
20	representative_pair_emb_correction.png	Representative pair closure with and without Earth reflex	D101 EMB representative-pair ledger	77b2a04c425dc170a11a9c1670f64c338761c67fc4e8880d10b179662de6e6cd
21	latitude_cone_unwind_geometry.png	Analytical latitude-cone flattening	Exact laboratory latitude geometry	e2b8335a8dbf1f4c4e5fffaca2b98cf2e9c34617b20d85fc6e163c3f2bae0fd6
22	figure_latitude_registration_gain.png	Latitude-unwinding registration gain	D101 UTC and hour-angle palindrome ledgers	e8403f228f760c1955558174f9dba93b34ae55cfc5d19a9b00a02ab5891aeb04
23	episodic_anchor_icrs_sky_map.png	High-similarity ICRS anchor sky map	D101 episodic anchor ledger and authoritative Cell3 axes	e9cd05fd5990c8105e25f9c1c86550847d0542278c4bb1aa379b2c7fd6350848
24	cell3_primary_plane_anchor_projections.png	Cell3 primary-plane anchor projections	D101 anchor vectors and authoritative projected Cell3 node rays	39eff5a8c10b8a02c359c80c5bdb6c0341ff38f16e40e2a060490b51db8954b3
25	recursive_lattice_closure_improvement.png	Recursive lattice-closure improvement	D101 high-pair and matched-control closure ledgers	b898a2671e5086bd557380508ed71c921091bf5bb0e1c56c5938e4adb0fa8bef
26	01_daily_lag_scan.png	01_daily_lag_scan	See caption and machine-readable source ledger	f8b3d9abe479efa18017b296a55feef69f322a02b2454268be93f72fed884b5b
27	figure_crossing_evidence_matrix.png	Direct and antipodal crossing evidence matrix	D101 latitude-unwound palindrome, chirp, CWT, and FFT2 ledgers	f7f57a2616e2296ee88e4490513a87461f7b469d065dc0139162595c8c04a0b1
28	figure_angular_chirp_summary.png	Angular chirp trajectory atlas	D101 latitude-unwound full-band chirp ledger	40c1cb2d5c6af67cded6a565810a994062660e0770cd4855b5c1dfa25dda6d9b
29	04_cwt_mixed_derivative.png	CWT mixed derivative at a crossing	D101 latitude-unwound CWT product	113f9d47d9a6ec2ac745a2201dfff55f13218fd330b136625ade9659463a97ac
30	03_fft2_strongest_palindrome.png	Two-dimensional FFT of event profiles	D101 event-by-phase matrix	3117d07d5079e55be52281c66ae218c6f061126687e7a99340e9be8ffe2b42db
31	15_solar_noon_midnight_comparison.png	Solar upper and lower culmination comparison	D101 event-centered moving-standard-deviation templates	7933a978a1a2ff7a06c8cc94428c2107b428600ea7c65708a0cab303e8dfd7c9
32	scalogram_solar_midnight_high.png	Solar-midnight fast chirp scalogram	D101 one-second pulse-width spectrogram	e68e85620b24d13a1ba6ccac5feb6929b0cbd383da706359acfa140b4b6c8011

Figure	Filename	Title	Data basis	SHA-256
33	04_lunar_minimum_exact_profile.png	Lunar lower-culmination palindrome	D101 event-centered moving-standard-deviation profiles	ee233acb60902e31744a7610771c8daf896764ed0e4197a5e98442b9c96e83e2
34	figure_latitude_registration_gain.png	Latitude-unwinding registration gain	D101 UTC and hour-angle palindrome ledgers	e8403f228f760c1955558174f9dba93b34ae55cfc5d19a9b00a02ab5891aeb04
35	16_cmb_retreating_exact_profile.png	CMB retreating palindrome	D101 fixed-vector event profiles	c4b07e9718f31827ac185908fd474c988835ce6a77f05b9195885e2984cc33cb
36	16_galactic_anticenter_exact_profile.png	Galactic anticenter palindrome	D101 fixed-vector event profiles	b16dcf0fe44f9a2779e1f52ee3c38c3129cc53e337d14d3e2b42d8fde6504415
37	scalogram_uranus_advancing_low.png	Uranus-advancing long-period chirp scalogram	D101 minute-resampled spectrogram	909b2805ee175b486d678d002eb5f2ed887c770b27ca68aea192845c9beea3e4
38	19_planetary_crossing_metric_matrix.png	Planetary crossing metric matrix	D101 planetary transit ledger	1c22fa42e62c3dc27a815b454d1100a8b5bf47bdc39c480bbe2196862f355f61
39	20_planetary_opposite_side_templates.png	Planetary advancing-retreating template relations	D101 planetary opposite-side ledger	0ffd0d9c04d36663345891cb296a4354cf1a832b3c40133cc9915c403b98ec9e
40	figure_orthogonal_inversion_summary.png	Opposite-quadrature spectral inversion	D101 orthogonal90 template comparison ledger	ebe191d21a80698f50ebfd2679a97ce7bb1b96a6335f131293fed83a23cf283e
41	low_tier_orthogonal_pair_correlations.png	Orthogonal pair correlations across all event classes	D101 orthogonal template ledger	bfd30ee10007b823a60ed16c4cd5faf1a7ddf39e2d9c4b54940fb1a8eb68306d
42	solar_evening_quadrature_low.png	Solar evening quadrature chirp	D101 orthogonal-event spectrogram	8ca46cea855ed792a4f271481521a32df8eecde58bb0a16a62f40049a898cc9a
43	saturn_orthogonal_sector_low.png	Saturn orthogonal-sector chirp	D101 orthogonal-event spectrogram	b78f04d75e9617fe0f217a638eb26b1e6ccf192da84b7a678140413bcb3c1f25

Appendix G. Archival Reproduction Record

Table 30. Archival document record

Field	Value
PDF filename	K-Field_Celestial_Crossing_Evidence_in_the_D101_OCXO_Signal_1786043894.pdf
Machine-readable filename	K-Field_Celestial_Crossing_Evidence_in_the_D101_OCXO_Signal_1786043894.json
Generation epoch	1786043894
Generation UTC	2026-08-06T19:18:14Z
Principal investigator	P. Dwayne Esterline
Credentials	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx, LLC.
Classification	PROPRIETARY CONFIDENTIAL INFORMATION AND TRADE SECRET OF KRYTRONX, LLC.
Rights	ALL RIGHTS RESERVED. DO NOT DISTRIBUTE.
Report engine	Python ReportLab
Core fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier only
Page size	US Letter
Margins	0.68 inch
Figure method	Python numerical computation and raster rendering from exact D101 and Cell3 ledgers
Table contract	Dynamic widths; Paragraph-wrapped cells; repeatRows=1 for long tables

END OF PROPRIETARY CONFIDENTIAL TRADE SECRET ARCHIVAL REFERENCE.